Supplementary Material

> **This document is the Supplementary Material of the manuscript**, submitted together with it.

> **Numbering convention**: the main text uses §n for sections and S n for subsections of this document.

0. Generation and verifiability (read this section first)

This document **was not written by hand**: the body text of every section **embeds verbatim** the content of the following **verified source files**, so "the numbers in this document" and "the numbers in the source files" **cannot be inconsistent**. Every source file records an md5 and a byte count, so anyone can re-check it.

Subsection	Content	Source file	md5	Bytes
S1	S1 · Per-row marking table and mechanical recomputation over 19 public benchmarks × 4 counting units	`split_units_19rows_20260916.md`	`2d542d10226347dfffb69dbab8b61d1a`	5805
S2	S2 · Correction record for the counting convention (old convention → recomputed convention)	`the audit-correction and adjudication record, §2 superseding block`	`94e85a41a6f7c5a5b6788a11b96c2c8e`	24638
S3	S3 · Per-row evidence from the cross-	`xframe_evidence_20260916.txt`	`78b3e251f5d4c8db3b6378ecbbd2b3e0`	13361

	framework configuration audit (5 frameworks / **18** official files printed)			
S4	S4 · 2×2 evaluation matrix over the three-split protocol (41 runs × {best,last} × {val,test})	`xeval_analysis_20260916.txt`	`8b74b2eb6228c078d5a38131d119567b`	8534
S5	S5 · Selection premium on public third-party logs (sample and how to read it)	`selection_premium_public_20260916.md`	`04950c92d1ab4bcfb5bbe616b2dcbec4`	9479

Generation script: `work\gen_supplementary_20260916.py` (purely local and re-runnable; S5 reads the collaborating archive in read-only mode).

Re-check entry point: see §S7 of this document.

Terms this paper uses (one place, so they need not be reconstructed)

term	what it means here
arm	one member of a paired comparison. The **baseline** arm follows the published convention; the **strategy** arm is the intervention under test (e.g. a learning-rate change). "Baseline" does **not** mean unmodified software — both

	arms run the same code
cell	one (corpus pair × arm) unit of the design. A cell name such as `shwd2sf` reads **source→target** (SHWD → SFCHD); the corpora are `dota15`, `aitod`, `visdrone`, `smoke`, `shwd`, `sfchd`, `mende`, `mask`
`best` / `last`	the two checkpoints compared throughout: `best.pt`, written by the training script by **fitness** on the monitored split, and the final epoch
selection premium	`best − last` **within a run**: what selecting the best checkpoint adds, measured on the split the numbers are reported from
Δgap	the paired **difference of the premium between the two arms**, per corpus. Positive means `val` **understates** the strategy arm's effect (§5.1, Tier 2)
realization rate	`prem_test / prem_val`, aggregated over cells (§8.6); the share of the premium selected on `val` that survives on a disjoint `test`
κ	through-origin slope of `ΔT` on `ΔV` over a run's epochs (§8.7): how much of a run's movement on `val` appears on `test`
same directory / alias	a dataset config whose `val:` and `test:` point at the same path, so the checkpoint is selected on the split it is scored on
`shapeiou`	the shape-aware IoU variant used in the registered replication's strategy arm (§9)
`sio_b_results.csv`	the per-run results table of the clean-protocol batch; its unwritten endpoint convention is disclosed as a defect in §10
FROZEN-HASH	the registration's self-referential stamp: the md5 of the registration text above the stamp line (§9, verifiable by re-hashing that text)

S1 · Per-row marking table and mechanical recomputation over 19 public benchmarks × 4 counting units

> **Purpose**: main text §3, §6; this is the **complete per-row evidence** for the claim "how many benchmarks are problematic".

> **Source**: `split_units_19rows_20260916.md` (md5 `2d542d10226347dfffb69dbab8b61d1a`, 5805 bytes); the content below is **embedded verbatim, unmodified**.

19-row × 5-column unit marking table (front matter of the companion paper)

> Generated by `work/recompute_split_units_20260916.py`; all percentages are mechanically recomputed from the markings in this table, with nothing hard-coded.
> Established 2026-09-16. The (h-1) ruling has been applied (D-Fire counts as clean).

The three units (the main text must state which one it uses)

Unit	Definition
release	The benchmark's own release layer: whether the official split artifacts/scripts provide an independent held-out test
protocol	The official protocol layer: the split declared by the official source, and whether that test can be self-evaluated locally
yolo_dist	Generic YOLO distribution-package layer (= the column measured in the §1 table of the audit, i.e. column 4)
reported	The actually-reported layer: which split the numbers reported in the literature come from, and whether they are independent of checkpoint selection

Evidence marks: (A) explicit in the audit｜(D) derived from audit facts｜(?) the audit takes no position

Marking table

#	benchmark	release	protocol	yolo_dist	reported	Evidence category	Confidence
<!-- citations: 1 COCO [34], 2 PASCAL VOC [21], 3 Objects365 [35], 4 Open Images v7 [36], 5 DOTA v1.0 [37],

6 DOTA v2.0 [37], 7 VisDrone-DET [38], 8 AI-TOD [39], 9 UAVDT [40], 10 xView [41], 11 DIOR [42], 12 NWPU VHR-10 [43], 13 SHWD [44], 14 SFCHD [45], 15 MAFA [46], 16 Mendeley face-mask [22], 17 WIDER FACE [47], 18 CrowdHuman [48], 19 D-Fire [49] -->

1	COCO	n_a(A)	independent_test(A)	clean(A)	clean(A)	Official counts + mirror yaml	high
2	PASCAL VOC	n_a(A)	test_gated(A)	alias(A)	alias(A)	mirror VOC.yaml (both keys cited per row)	high
3	Objects365	no_test(A)	no_test(A)	no_test_key(A)	alias(D)	mirror yaml + Ultralytics docs	high
4	Open Images v7	independent_test(A)	independent_test(A)	no_test_key(A)	alias(D)	Official facts page + mirror yaml	high
5	DOTA v1.0	independent_test(A)	independent_test(A)	clean(A)	clean(A)	Official download page + mirror DOTAv1.yaml	high
6	DOTA v2.0	independent_test(A)	test_gated(A)	clean(A)	alias(D)	Official/mirror documentation	high
7	VisDrone-DET	independent_test(A)	independent_test(A)	clean(A)	clean(A)	mirror + converter	high
8	AI-TOD	contradictory(A)	contradictory(A)	unknown(A)	unknown(?)	Official README (contains two mutually contradictory release statements)	low
9	UAVDT	no_val(A)	no_val(A)	unknown(A)	alias(A)	Official paper + two mirrors (the audit calls these unverified beyond these two)	medium
10	xView	no_test(A)	no_test(A)	no_test_key(A)	alias(A)	mirror xView.yaml	low
11	DIOR	independent_test(A)	independent_test(A)	no_test_key(A)	unknown(?)	Official description (no counting) + mirror DIOR-R yaml	low
12	NWPU VHR-10	no_split(A)	no_split(A)	unknown(A)	alias(A)	Official TorchGeo (the split means positive/negative)	high
13	SHWD	no_yaml(A)	no_val(A)	train_val_alias(A)	alias(A)

Official README + mirror Reflective_vests.yaml	medium
14	SFCHD	`alias(A)`	`no_val(A)`	`alias(A)`	`alias(A)`	Official first-party release (machine comparison _ev_sfchd_check2.py)	highest
15	MAFA	`no_val(A)`	`no_val(A)`	`unknown(?)`	`alias(A)`	Official CVPR PDF (extracted with pypdf)	high
16	Mendeley face-mask	`no_split(A)`	`no_split(A)`	`alias(D)`	`unknown(?)`	Official data record + mirror generator	lowest
17	WIDER FACE	`independent_test(A)`	`test_gated(A)`	`no_test_key(A)`	`alias(A)`	Official site (GT not released, verified word by word) + mirror	high
18	CrowdHuman	`test_gated(A)`	`alias(A)`	`no_test_key(A)`	`alias(A)`	Official paper + mirror	high
19	D-Fire (FireSmoke)	`independent_test(A)`	`independent_test(A)`	`clean(A)`	`unknown(?)`	Official README	medium

Recomputation results

Unit	alias/non-independent	Among decidable rows	unknown
release (the benchmark's own release layer)	**10/19 = 52.6%**	10/19 = 52.6%	0
protocol (official protocol layer)	**13/19 = 68.4%**	13/19 = 68.4%	0
yolo (generic YOLO distribution-package layer (yolo_dist))	**4/19 = 21.1%**	4/15 = 26.7%	4
reported (actually-reported layer)	**12/19 = 63.2%**	12/15 = 80.0%	4

Cross-check against the numbers already published in the audit

- Audit 3a "literal `val == test` **2/19**" → this table's "the two keys of the distribution-package config point to the same path" (A evidence source) yields **2/19**, **reproduced** ✓
  - The other 1 alias row is produced by a **generator** (Mendeley face-mask), not by a distribution-package config, and is not counted under the 3a convention
- Audit "**14/19**" → this table's `reported` yields **12/19** (difference -2)

- Audit "**5/19** clean" → this table's `reported=clean` yields **3/19** (difference -2)
- The audit's "12/19" was obtained by manually adjusting two rows and is **not** a recomputation under a consistent unit; this table recomputes by a consistent rule, so the values need not be identical

A self-consistent set of numbers for the main text to adopt

- Literal `val == test` (generic distribution-package config, A evidence source): **2/19 = 11%** — PASCAL VOC, SFCHD
- train/val from the same source, produced by the generator: **1/19** (Mendeley face-mask), **a different convention, which must be stated separately**
- Non-independent at the official protocol layer: **13/19 = 68%**
- Non-independent at the actually-reported layer: **12/19 = 63%** (4 unknown rows) — **13/19** if a paper's own `val` numbers, rather than community `test-dev` numbers, are what the unit judges for COCO (the main text states both readings)
- Clean at the actually-reported layer: **3/19 = 16%**

The four levels of "same directory" (resolving (g0))

- **L1** literal `val==test` in a generic distribution package: 2 rows — PASCAL VOC, SFCHD
- **L2** distribution package with `train==val` (F-C): 1 row — SHWD
- **L3** distribution package clean but reported layer non-independent: 9 rows — Objects365, Open Images v7, DOTA v2.0, UAVDT, xView, NWPU VHR-10, MAFA, WIDER FACE, CrowdHuman
- **L4** all the rest: 7 rows

Corrections already applied

- **(h-1) D-Fire / FireSmoke family**: the "mixed" criterion for the family row is taken from a config of SHWD lineage (A:59 cites Reflective_vests.yaml; A:52 notes that it is based on SHWD). All three copies belonging to FireSmoke/D-

Fire are distinct. Under the "release layer" unit explicitly adopted in 3a, D-Fire goes into the clean group.
- **(d-1) AI-TOD / DOTA asymmetry**: the audit marks F-C for AI-TOD, which contains trainval, but does not mark DOTA v1/v2, which likewise contain trainval. This table uniformly adopts the "release/protocol layer" unit: trainval is a user-protocol caveat and is not used as grounds for a verdict; neither place is re-judged, but AI-TOD is contradictory because its v2 release statement contradicts itself.

S2 · Correction record for the counting conventions (old convention → recomputed convention)

> **Purpose**: main text §6.3; explains why the same fact once had two sets of numbers, and which set has been invalidated.

> **Source**: `the audit-correction and adjudication record, §2 superseding block` (md5 `94e85a41a6f7c5a5b6788a11b96c2c8e`, 24638 bytes); the following content is **embedded verbatim, unmodified**.

> ### The four counts in this section (§2) have been superseded by a mechanical recomputation (note added 2026-09-16)
>
> The **12/19 (release/protocol layer)**, **13/19 (actual reporting layer)**, **6/19 (clean)** and **≤14/19 (upper bound)** in the §2 table
> are inconsistent with the mechanical recomputation completed 4 minutes later, and **must no longer be cited**. The recomputation is the sole counting source: `analysis/eval_validity/split_units_recompute_20260916.txt` and
> `split_units_19rows_20260916.md`
>
> | Unit | Non-independent | Clean | Undecidable |
> |---|---|---|---|
> | Literal `val == test` (the benchmark's own release) | **2/19 ≈ 11%** | — | — |
> | release (release layer) | **10/19 ≈ 53%** | — | — |
> | protocol (official protocol layer) | **13/19 ≈ 68%** | — | — |
> | yolo_dist (generic YOLO distribution package layer) | **4/19 ≈ 21%** | — | 4 |
> | **reported (actual reporting layer)** | **12/19 ≈ 63%** (12/15 = **80%** of the decidable rows) | **3/19 ≈ 16%** | **4** |

>
> The four "undecidable" rows are AI-TOD, DIOR, Mendeley and D-Fire — **the audit expressed no position on their reporting practice**; it does not mean they are clean.
>
> The four concrete discrepancies and their dispositions:
> 1. §2's "release/protocol layer 12/19" does not hold under the recomputed convention — the recomputed release is 10/19 and protocol is 13/19, **12 is neither the one nor the other** (consistent with the ruling in §1(g): it is the result of two rows being hand-adjusted).
> 2. §2's "actual reporting layer 13/19" should be **12/19** (4 rows undecidable).
> 3. §2's "clean 6/19" should be **3/19**; the reason the clean group dropped from 5 to 3 is that those 4 rows **were never addressed by the audit** — this is a finding, not a gap.
> 4. §2's "≤14/19 upper bound" is replaced by a derivable statement: **80% of the decidable rows are non-independent (12/15)**, which is both stronger and more defensible than an unenumerated upper bound.
>
> **"79%" has already been ruled inadmissible in §1(e)**; this is confirmed here as well: the main text uses only the four enumerated points in the table above.

> ⚠️ **This section was revised on 2026-09-16 (second version). The first version (09:56) was written before the 19×5 marking table was built and gave predicted values**
> (release/protocol layer 12/19, actual reporting layer 13/19, clean 6/19). The mechanical recomputation 4 minutes later
> (`eval_validity\split_units_recompute_20260916.txt`, 10:00, with a per-unit 19-row subtotal check) gave different results.
> **The two sets of numbers cannot be cited side by side**. Following the disposition recommendation in §7 #0 of `the asset inventory`,
> **the recomputation is taken as the sole counting source**; the first version's predicted values are retained in the "trail" table below for traceability.

S3 · Line-by-line evidence for the cross-framework configuration audit (5 frameworks / 18 official files printed)

> **Purpose**: main text §6.5; the commit pin and the line-by-line evidence for each framework (the judgements are in the main text, the evidence is here).

> **Source**: `xframe_evidence_20260916.txt` (md5 `78b3e251f5d4c8db3b6378ecbbd2b3e0`, 13361 bytes); the following content is **embedded verbatim, unmodified**.

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Item C: extraction of cross-framework "split and checkpoint-selection" evidence (mechanical extraction, with line numbers; no judgements in this file)

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repo default_branch commit_sha

mmdetection open-mmlab/mmdetection main
cfd5d3a985b0249de009b67d04f37263e11cdf3d

detectron2 facebookresearch/detectron2 main
a2f4a8771ab77e8411c26b27f24f9489a28a2453

yolox Megvii-BaseDetection/YOLOX main
6ddff4824372906469a7fae2dc3206c7aa4bbaee

paddledetection PaddlePaddle/PaddleDetection release/2.9
b25522a0f4bde8c80603f3ba5e3472059972e3b5

detr facebookresearch/detr main
29901c51d7fe8712168b8d0d64351170bc0f83e0

MMDetection :: mmdetection/configs___base___datasets__coco_detection.py (96 lines)

```
  3: data_root = 'data/coco/'
 21: dict(type='LoadImageFromFile', backend_args=backend_args),
 22: dict(type='LoadAnnotations', with_bbox=True),
 23: dict(type='Resize', scale=(1333, 800), keep_ratio=True),
```

```
  24:                                 dict(type='RandomFlip',         prob=0.5),
  25:                                                 dict(type='PackDetInputs')
  28:           dict(type='LoadImageFromFile',   backend_args=backend_args),
  29:             dict(type='Resize',   scale=(1333,   800),   keep_ratio=True),
  31:                     dict(type='LoadAnnotations',      with_bbox=True),
  33:                                                 type='PackDetInputs',
  41:                 sampler=dict(type='DefaultSampler',      shuffle=True),
  42:                 batch_sampler=dict(type='AspectRatioBatchSampler'),
  45:                                                 data_root=data_root,
  46:                         ann_file='annotations/instances_train2017.json',
  47:                                                 data_prefix=dict(img='train2017/'),
  56:                 sampler=dict(type='DefaultSampler',      shuffle=False),
  59:                                                 data_root=data_root,
  60:                         ann_file='annotations/instances_val2017.json',
  61:                                                 data_prefix=dict(img='val2017/'),
  68:                                                 type='CocoMetric',
  69:           ann_file=data_root   +   'annotations/instances_val2017.json',
```

MMDetection :: mmdetection/configs___base___default_runtime.py (25 lines)

(no hits)

MMDetection :: mmdetection/configs__faster_rcnn__faster-rcnn_r50_fpn_1x_coco.py (6 lines)

```
   1:                         _base_                         =                         [
   2:                                         '../_base_/models/faster-rcnn_r50_fpn.py',
   3:                                             '../_base_/datasets/coco_detection.py',
   4:       '../_base_/schedules/schedule_1x.py',  '../_base_/default_runtime.py'
   5:                                                                                 ]
```

MMDetection :: mmdetection/tools__train.py (122 lines)
(no hits)

Detectron2 :: detectron2/configs__COCO-Detection__faster_rcnn_R_50_FPN_1x.yaml (7 lines)
```
1: _BASE_: "../Base-RCNN-FPN.yaml"
2: MODEL:
3: WEIGHTS: "detectron2://ImageNetPretrained/MSRA/R-50.pkl"
4: MASK_ON: False
5: RESNETS:
6: DEPTH: 50
```


Detectron2 :: detectron2/configs__Base-RCNN-FPN.yaml (43 lines)
```
14: PRE_NMS_TOPK_TRAIN: 2000 # Per FPN level
15: PRE_NMS_TOPK_TEST: 1000 # Per FPN level
19: POST_NMS_TOPK_TRAIN: 1000
20: POST_NMS_TOPK_TEST: 1000
32: DATASETS:
33: TRAIN: ("coco_2017_train",)
34: TEST: ("coco_2017_val",)
36: IMS_PER_BATCH: 16
41: MIN_SIZE_TRAIN: (640, 672, 704, 736, 768, 800)
```


Detectron2 :: detectron2/tools__train_net.py (168 lines)
```
105: for name in cfg.DATASETS.TEST
```


Detectron2 :: detectron2/detectron2__data__datasets__builtin.py (260 lines)

```
  11: without having to download the dataset annotations
  36:     "coco_2014_train": ("coco/train2014", "coco/annotations/instances_train2014.json"),
  37:     "coco_2014_val": ("coco/val2014", "coco/annotations/instances_val2014.json"),
  38:     "coco_2014_minival": ("coco/val2014", "coco/annotations/instances_minival2014.json"),
  41:     "coco/annotations/instances_valminusminival2014.json",
  43:     "coco_2017_train": ("coco/train2017", "coco/annotations/instances_train2017.json"),
  44:     "coco_2017_val": ("coco/val2017", "coco/annotations/instances_val2017.json"),
  45:     "coco_2017_test": ("coco/test2017", "coco/annotations/image_info_test2017.json"),
  46:     "coco_2017_test-dev": ("coco/test2017", "coco/annotations/image_info_test-dev2017.json"),
  47:     "coco_2017_val_100": ("coco/val2017", "coco/annotations/instances_val2017_100.json"),
  53:     "coco/annotations/person_keypoints_train2014.json",
  55:     "keypoints_coco_2014_val": ("coco/val2014", "coco/annotations/person_keypoints_val2014.json"),
  58:     "coco/annotations/person_keypoints_minival2014.json",
  62:     "coco/annotations/person_keypoints_valminusminival2014.json",
  64:     "keypoints_coco_2017_train": (
  66:     "coco/annotations/person_keypoints_train2017.json",
  68:     "keypoints_coco_2017_val": ("coco/val2017", "coco/annotations/person_keypoints_val2017.json"),
  69:     "keypoints_coco_2017_val_100": (
  70:     "coco/val2017",
  71:     "coco/annotations/person_keypoints_val2017_100.json",
  77:     "coco_2017_train_panoptic": (
  80:     "coco/annotations/panoptic_train2017.json",
  88:     "coco_2017_val_panoptic": (
```

```
  89:                                                 "coco/panoptic_val2017",
  90:                                 "coco/annotations/panoptic_val2017.json",
  91:                                               "coco/panoptic_stuff_val2017",
  93:                                   "coco_2017_val_100_panoptic":         (
  94:                                               "coco/panoptic_val2017_100",
  95:                                 "coco/annotations/panoptic_val2017_100.json",
  96:                                           "coco/panoptic_stuff_val2017_100",
 235:             gt_dir = os.path.join(root, "annotations_detectron2", dirname)

-------------------------------------------------------------------------------------------
---
### Detectron2 :: detectron2/detectron2__config__defaults.py     (657 lines)
 103:                 _C.DATASETS.TEST                          =                  ()
 607:                 _C.TEST.EVAL_PERIOD                       =                   0

-------------------------------------------------------------------------------------------
---
### YOLOX :: yolox/exps__default__yolox_s.py          (16 lines)
   7:         from yolox.exp import Exp as MyExp
  15:                                                 self.exp_name =
os.path.split(os.path.realpath(__file__))[1].split(".")[0]

-------------------------------------------------------------------------------------------
---
### YOLOX :: yolox/yolox__data__datasets__coco.py         (189 lines)
  42:                                         json_file="instances_train2017.json",
  43:                                                 name="train2017",
  54:             name (str): COCO data name (e.g. 'train2017' or 'val2017')

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---
### YOLOX :: yolox/tools__train.py                (147 lines)
 (no                                                                        hits)

-------------------------------------------------------------------------------------------
```

YOLOX :: yolox/tools__eval.py (221 lines)

```
  93:                                                 "--test",
```

PaddleDetection :: paddledetection/configs__yolov3__yolov3_darknet53_270e_coco.yml (11 lines)

```
   1:                                     _BASE_: [
   2:                                         '../datasets/coco_detection.yml',
   3:                                         '../runtime.yml',
   4:                                         '_base_/optimizer_270e.yml',
   5:                                         '_base_/yolov3_darknet53.yml',
   6:                                         '_base_/yolov3_reader.yml',
   7:                                         ]
   9:                                 snapshot_epoch: 5
  10:                 weights: output/yolov3_darknet53_270e_coco/model_final
```

PaddleDetection :: paddledetection/tools__train.py (220 lines)

```
  51:                                                 "--eval",
  54:                         help="Whether to perform evaluation in train")
  66:                     help="If set True, enable continuous evaluation job."
  94:                         help='Whether to save the evaluation results only')
 100:                         "format \"key1=value1;key2=value2;key3=value3\"."
 161:                                                 trainer.train(FLAGS.eval)
```

DETR :: detr/main.py (249 lines)

```
  16:     from engine import evaluate, train_one_epoch
  95:                         parser.add_argument('--eval', action='store_true')
 143:                 dataset_val = build_dataset(image_set='val', args=args)
```

147: sampler_val = DistributedSampler(dataset_val, shuffle=False)
150: sampler_val = torch.utils.data.SequentialSampler(dataset_val)
157: data_loader_val = DataLoader(dataset_val, args.batch_size, sampler=sampler_val,
162: coco_val = datasets.coco.build("val", args)
163: base_ds = get_coco_api_from_dataset(coco_val)
165: base_ds = get_coco_api_from_dataset(dataset_val)
179: if not args.eval and 'optimizer' in checkpoint and 'lr_scheduler' in checkpoint and 'epoch' in checkpoint:
184: if args.eval:
185: test_stats, coco_evaluator = evaluate(model, criterion, postprocessors,
186: data_loader_val, base_ds, device, args.output_dir)
188: utils.save_on_master(coco_evaluator.coco_eval["bbox"].eval, output_dir / "eval.pth")
214: test_stats, coco_evaluator = evaluate(
215: model, criterion, postprocessors, data_loader_val, base_ds, device, args.output_dir
228: if coco_evaluator is not None:
229: (output_dir / 'eval').mkdir(exist_ok=True)
230: if "bbox" in coco_evaluator.coco_eval:
235: torch.save(coco_evaluator.coco_eval["bbox"].eval,
236: output_dir / "eval" / name)
244: parser = argparse.ArgumentParser('DETR training and evaluation script', parents=[get_args_parser()])

DETR :: detr/datasets__coco.py (159 lines)
151: PATHS = {
152: "train": (root / "train2017", root / "annotations" / f'{mode}_train2017.json'),
153: "val": (root / "val2017", root / "annotations" / f'{mode}_val2017.json'),

156: img_folder, ann_file = PATHS[image_set]

DETR :: detr/engine.py (152 lines)

3: Train and eval functions used in main.py

13: from datasets.coco_eval import CocoEvaluator

14: from datasets.panoptic_eval import PanopticEvaluator

23: metric_logger.add_meter('lr', utils.SmoothedValue(window_size=1, fmt='{value:.6f}'))

24: metric_logger.add_meter('class_error', utils.SmoothedValue(window_size=1, fmt='{value:.2f}'))

43: losses_reduced_scaled = sum(loss_dict_reduced_scaled.values())

45: loss_value = losses_reduced_scaled.item()

47: if not math.isfinite(loss_value):

48: print("Loss is {}, stopping training".format(loss_value))

58: metric_logger.update(loss=loss_value, **loss_dict_reduced_scaled,** loss_dict_reduced_unscaled)

68: def evaluate(model, criterion, postprocessors, data_loader, base_ds, device, output_dir):

69: model.eval()

70: criterion.eval()

73: metric_logger.add_meter('class_error', utils.SmoothedValue(window_size=1, fmt='{value:.2f}'))

77: coco_evaluator = CocoEvaluator(base_ds, iou_types)

80: panoptic_evaluator = None

82: panoptic_evaluator = PanopticEvaluator(

85: output_dir=os.path.join(output_dir, "panoptic_eval"),

102: metric_logger.update(loss=sum(loss_dict_reduced_scaled.values()),

113: if coco_evaluator is not None:

114: coco_evaluator.update(res)

116: if panoptic_evaluator is not None:

124: panoptic_evaluator.update(res_pano)

129: if coco_evaluator is not None:

130: coco_evaluator.synchronize_between_processes()

```
  131:                 if panoptic_evaluator is not None:
  132:                     panoptic_evaluator.synchronize_between_processes()
  135:                 if coco_evaluator is not None:
  136:                     coco_evaluator.accumulate()
  137:                     coco_evaluator.summarize()
  139:                 if panoptic_evaluator is not None:
  140:                     panoptic_res = panoptic_evaluator.summarize()
  142:                 if coco_evaluator is not None:
  144:                     stats['coco_eval_bbox'] = coco_evaluator.coco_eval['bbox'].stats.tolist()
  146:                     stats['coco_eval_masks'] = coco_evaluator.coco_eval['segm'].stats.tolist()
  151:                 return stats, coco_evaluator
```

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Missing files: none

S4 · 2×2 evaluation matrix over the three-split protocol (41 runs × {best,last} × {val,test})

> **Purpose**: main text §8.6 and §5.6; covers archive-reproduction verification, the decomposability of Δgap, and the realization rate of the selection gain.

> **Source**: `xeval_analysis_20260916.txt` (md5 `8b74b2eb6228c078d5a38131d119567b`, 8534 bytes); the content below is **embedded verbatim, unchanged**.

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2×2 evaluation matrix × archive: verification / locally recomputable / decomposability of Δgap

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Matrix entries 164; sio entries 472; (commit/convention: split=test|val, batch=32, imgsz=640, aligned with train_obj.py L75)

Multiple evaluations of the same cell: all consistent

--- per run: archive vs this recomputation (pp)

run	Vmax_a	Vbest_r	Vfin_a	Vfin_r	Tbest_a	Tbest_r	Tfin_r
r10_aitod20_base30_3way_s42n	14.621	14.621	13.835	13.826	10.210	10.211	10.587
r10_shwd2sf_base100_3way_s42n	60.694	60.707	59.055	58.977	43.510	43.513	42.946
r10_shwd2sf_base100_3way_s43n	60.976	60.966	59.761	59.822	43.880	43.885	42.948
r10_shwd2sf_base100_3way_s44n	61.120	61.135	59.230	59.326	43.050	43.053	42.394
r10_shwd2sf_base100_3way_s45n	59.852	59.866	58.473	58.474	43.080	43.079	42.901
r10_shwd2sf_base100_3way_s46n	60.509	60.507	59.293	59.284	43.640	43.641	42.845
r10_shwd2sf_base100_3way_s47n	60.858	60.886	59.766	59.775	43.780	43.781	43.267
r10_shwd2sf_base100_3way_s48n	60.554	60.516	59.575	59.615	43.390	43.390	42.784
r10_shwd2sf_base100_3way_s49n	61.026	60.999	59.169	59.248	43.990	43.992	42.910
r10_shwd2sf_base100_3way_s50n	60.896	60.854	58.967	59.018	43.640	43.641	42.608
r10_shwd2sf_base100_3way_s51n	60.209	60.208	58.713	58.734	43.630	43.626	42.636
r10_shwd2sf_lr005_100ep_3way_s42n	60.860	60.913	59.581	59.586	44.260	44.260	43.569
r10_shwd2sf_lr005_100ep_3way_s43n	60.408	60.491	59.116	59.105	43.950	43.947	43.861
r10_shwd2sf_lr005_100ep_3way_s44n	60.001	60.003	58.952	58.920	44.440	44.442	44.134
r10_shwd2sf_lr005_100ep_3way_s45n	60.795	60.789	60.578	60.579	44.130	44.132	43.934
r10_shwd2sf_lr005_100ep_3way_s46n	60.462	60.456	59.270	59.298	44.350	44.354	44.242

r10_shwd2sf_lr005_100ep_3way_s47n 60.678 60.667 60.354
60.355 43.970 43.966 43.944
r10_shwd2sf_lr005_100ep_3way_s48n 59.722 59.742 58.701
58.711 43.710 43.713 43.795
r10_shwd2sf_lr005_100ep_3way_s49n 60.363 60.375 59.037
59.028 43.800 43.799 44.027
r10_shwd2sf_lr005_100ep_3way_s50n 60.193 60.188 59.614
59.634 43.780 43.785 44.078
r10_shwd2sf_lr005_100ep_3way_s51n 60.491 60.490 60.119
60.153 44.270 44.269 43.560
r10_smoke2sf_base100_3way_s42n 58.724 58.696 57.618 57.689
42.100 42.103 41.661
r10_smoke2sf_base100_3way_s43n 58.766 58.798 58.438 58.449
42.320 42.323 41.726
r10_smoke2sf_base100_3way_s44n 58.864 58.882 58.513 58.526
42.280 42.282 41.367
r10_smoke2sf_base100_3way_s45n 58.731 58.740 57.594 57.592
41.250 41.253 41.054
r10_smoke2sf_base100_3way_s46n 58.652 58.689 57.887 57.862
41.900 41.899 41.258
r10_smoke2sf_base100_3way_s47n 59.271 59.228 58.532 58.557
42.100 42.101 41.437
r10_smoke2sf_base100_3way_s48n 58.697 58.682 57.778 57.755
42.400 42.402 41.441
r10_smoke2sf_base100_3way_s49n 58.601 58.617 58.376 58.393
41.960 41.964 42.082
r10_smoke2sf_base100_3way_s50n 58.449 58.444 57.430 57.410
42.110 42.108 41.923
r10_smoke2sf_base100_3way_s51n 58.374 58.364 58.057 58.065
42.200 42.202 41.599
r10_smoke2sf_lr005_100ep_3way_s42n 59.463 59.423 59.463
59.423 43.730 43.728 43.728
r10_smoke2sf_lr005_100ep_3way_s43n 60.186 60.171 59.090
59.029 43.730 43.731 43.372
r10_smoke2sf_lr005_100ep_3way_s44n 60.151 60.142 59.578

59.620	43.190	43.191	43.608		
r10_smoke2sf_lr005_100ep_3way_s45n	60.097	60.088	60.019	60.031	43.170
43.174	43.502				
r10_smoke2sf_lr005_100ep_3way_s46n	59.114	59.131	58.735	58.744	43.790
43.785	43.363				
r10_smoke2sf_lr005_100ep_3way_s47n	59.793	59.749	58.790	58.852	43.710
43.714	43.871				
r10_smoke2sf_lr005_100ep_3way_s48n	59.165	59.202	58.739	58.682	43.390
43.389	43.730				
r10_smoke2sf_lr005_100ep_3way_s49n	59.876	59.875	58.560	58.577	43.500
43.503	43.721				
r10_smoke2sf_lr005_100ep_3way_s50n	59.144	59.108	58.178	58.267	42.990
42.989	43.841				
r10_smoke2sf_lr005_100ep_3way_s51n	59.866	59.892	59.649	59.583	43.760
43.759	43.800				

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(1) Verification: can this recomputation reproduce the existing archive

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(best,test) recomputed vs sio archive: n=41 max diff 0.0047 median 0.0021 pp

⇒ ✅ digit-for-digit agreement (<0.05 pp): the evaluation chain follows the project convention, and the archive values can be reproduced independently

(best,val) recomputed vs results.csv maximum: n=41 max diff 0.0834 median 0.0148 pp

⇒ **i** One earlier attribution is withdrawn here. We had put this residual down to `best.pt` being selected by **fitness** rather than by mAP50-95; in the pinned version **fitness** ≡ **mAP50-95** on the monitored split, so the selection rule does not explain it. The candidate causes are a difference between the training-time validator and this standalone re-evaluation, or

best.pt sitting at an epoch other than the logged argmax. We do not know which, and record the residual as **unidentified**, not attributed.

(last,val) recomputed vs results.csv final epoch : n=41 max diff 0.0963 median 0.0215 pp

==========

(3) Decomposability of Δgap

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Identity: gap = M_val(ê) − M_test(ê) = (V_final − T_final) + (prem_val − prem_test)

prem_val = V_max − V_final (results.csv); prem_test = T_best − T_final (measured in this run)

cell	arm	n	V_fin	T_fin	prem_val	prem_test	gap	check
aitod20	base	1	13.835	10.587	0.786	-0.377	4.411	4.411
shwd2sf	base	10	59.200	42.824	1.469	0.735	17.110	17.110
shwd2sf	strat	10	59.532	43.915	0.865	0.151	16.331	16.331
smoke2sf	base	10	58.022	41.555	0.691	0.507	16.651	16.651
smoke2sf	strat	10	59.080	43.654	0.605	-0.158	16.189	16.190

cell	n	Δgap	Δdiffgap	Δoptimism	sum(check)	dominant	p(Δgap)
shwd2sf	10	+0.779	+0.759	+0.020	+0.779	Δdiffgap	0.005258
smoke2sf	10	+0.461	+1.041	-0.580	+0.461	Δdiffgap	0.02627

Reading:

shwd2sf Δgap=+0.779 (p=0.005258, n=10) = difficulty gap +0.759 (±1.051) + selection optimism +0.020 (±0.751)

smoke2sf Δgap=+0.461 (p=0.02627, n=10) = difficulty gap +1.041 (±0.626) + selection optimism -0.580 (±0.770)

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(4) The "realization rate" of the selection gain: how much of the score picked on val is left on test

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prem_val = V_max − V_final (book gain obtained by selecting on val)
prem_test = T_best − T_final (gain realized on test by the same selection action)

cell	arm	n	prem_val	prem_test	realized	per-run prem_test				
aitod20	base	10	0.766	-0.218	-28%	-0.38	-0.24	-0.20	-0.27	-0.22
-0.11			+0.14		-0.78			-0.18		+0.06
shwd2sf	base	10	1.469	0.735	50%	+0.56	+0.93	+0.66	+0.18	+0.79
+0.51			+0.61		+1.08			+1.03		+0.99
shwd2sf	strat	10	0.865	0.151	18%	+0.69	+0.09	+0.31	+0.20	+0.11
+0.03			-0.09		-0.23			-0.30		+0.71
smoke2sf	base	10	0.691	0.507	73%	+0.44	+0.59	+0.91	+0.20	+0.64
+0.66			+0.96		-0.12			+0.19		+0.60
smoke2sf	strat	10	0.605	-0.158	-26%	+0.00	+0.36	-0.42	-0.33	+0.43
-0.16			-0.34		-0.22			-0.85		-0.04

Total (**headline convention = run-weighted**, n = 50, all five cells at n = 10): prem_val mean +0.879 pp, prem_test mean +0.205 pp, realization rate **23.3 %** (run-level bootstrap 95% CI [8.4 %, 37.4 %], P(rate ≤ 0) = 0.002, so the interval excludes no-realization; script `work/item2_final_v2_20260925.py`)

The three weightings are **identical** once every cell carries n = 10 (run-weighted ≡ arm-cell-equal ≡ n ≥ 5 only), all three giving **23.3 %**; on the previous 41-run sample (aitod20 at n = 1) they did not, and the ~15 pp spread came from that n = 1 cell — it carried 20 % of the arm-cell weight while contributing a single run to the run-level mean; the values themselves are in §8.6's measurement detail.

Completing that cell removes the spread, not a change of weighting.

The cell-level bootstrap interval of the arm-cell-equal estimator is [−17.3 %, 53.2 %] and **includes zero** (only five clusters), while the run-level interval [8.4 %, 37.4 %] does not;

so "partial realization" is significant only under run weighting (article §8.6).

Note: the total's prem_val is on the **declared** convention `V_max − V_final`.

With the aitod20 cell completed, arm-cell-equal and run-weighted **are the same estimator**, so it is no longer listed as a separate "sensitivity".

Cells with **negative** prem_test: 2/5 —— aitod20/base(-0.218), smoke2sf/strat(-0.157)

⇒ The gain selected on val is only partly realized on an independent split, and can even reverse.

S5 · Selection premium on public third-party logs (sample and reading)

> **Purpose**: main text §5.5; the epoch count, final epoch, best epoch and premium of every log, plus four limitations.

> **Source**: `selection_premium_public_20260916.md` (md5 `04950c92d1ab4bcfb5bbe616b2dcbec4`, 9479 bytes); the content below is **embedded verbatim, unchanged**.

Sample

File	Epochs	Final epoch	Best epoch (position)	Premium max−final	Premium max−mean of last five epochs	Metric column name
`aakashchalla12345678-yolov8n__unstated__yolov8n.csv`	30	0.4861	0.4861 (ep 30)	**+0.00 pp**	**+0.26 pp**	`metrics/mAP50-95(B)`
`adityadgmath070-yolov8-fs-grafik-adamw-results6__unstated__yolov8.csv`	100	0.4284	0.4284 (ep 100)	**+0.00 pp**	**+0.54 pp**	`metrics/mAP50-95(B)`
`adityadgmath070-yolov8-fs-grafik-adamw-results7__unstated__yolov8.csv`	100	0.4403	0.4494 (ep 74)	**+0.91 pp**	**+0.79 pp**	`metrics/mAP50-95(B)`
`adityadgmath070-yolov8-fs-grafik-adamw-results8__unstated__yolov8.csv`	100	0.4275	0.4511 (ep 30)	**+2.36 pp**	**+2.55 pp**	`metrics/mAP50-95(B)`

adityadgmath070-yolov8-fs-grafik-sgd-results1__unstated__yolov8.csv	100	0.3151	0.3167 (ep 97)	**+0.16 pp**	**+0.17 pp**	metrics/mAP50-95(B)
adityadgmath070-yolov8-fs-grafik-sgd-results2__unstated__yolov8.csv	100	0.4402	0.4459 (ep 65)	**+0.57 pp**	**+0.58 pp**	metrics/mAP50-95(B)
adityadgmath070-yolov8-fs-grafik-sgd-results3__unstated__yolov8.csv	100	0.4515	0.4648 (ep 53)	**+1.33 pp**	**+1.33 pp**	metrics/mAP50-95(B)
adityadgmath070-yolov8-fs-grafik-sgd-results4__unstated__yolov8.csv	100	0.4340	0.4458 (ep 84)	**+1.18 pp**	**+1.20 pp**	metrics/mAP50-95(B)
ayyappakorla-yolov3-results__unstated__yolov3.csv	200	0.9011	0.9041 (ep 198)	**+0.31 pp**	**+0.20 pp**	metrics/mAP_0.5:0.95
dhanduramesh-yolov11s__ppe-css-data__yolo11n.csv	1	0.4430	0.4430 (ep 1)	**+0.00 pp**	—	metrics/mAP50-95(B)
ericmorning-pcb-yolo-comparison__pcb-components__yolo11.csv	145	0.4647	0.4863 (ep 115)	**+2.16 pp**	**+2.33 pp**	metrics/mAP50-95(B)
ericmorning-pcb-yolo-comparison__pcb-components__yolo12.csv	150	0.4698	0.4853 (ep 132)	**+1.55 pp**	**+1.57 pp**	metrics/mAP50-95(B)
ericmorning-pcb-yolo-comparison__pcb-components__yolov8.csv	93	0.4467	0.4621 (ep 63)	**+1.54 pp**	**+1.49 pp**	metrics/mAP50-95(B)
ikram0703-yolov8-ema-Model1__unstated__yolov8.csv	123	0.6311	0.6385 (ep 98)	**+0.74 pp**	**+0.62 pp**	metrics/mAP50-95(B)
ikram0703-yolov8-ema-Model2__unstated__yolov8.csv	131	0.6289	0.6301 (ep 106)	**+0.10 pp**	**+0.10 pp**	metrics/mAP50-95(B)
ikram0703-yolov8-ema-Model3__unstated__yolov8.csv	81	0.6293	0.6435 (ep 56)	**+1.42 pp**	**+1.27 pp**	metrics/mAP50-95(B)
ikram0703-yolov8-ema-Model4__unstated__yolov8.csv	182	0.6058	0.6071 (ep 157)	**+0.14 pp**	**+0.15 pp**	metrics/mAP50-95(B)
ikram0703-yolov8-ema-Model5__unstated__yolov8.csv	109

0.6294 | 0.6363 (ep 84) | **+0.70 pp** | **+0.72 pp** | metrics/mAP50-95(B) |
kimmkless-yolodetector__train2-vehicle-5class__yolov8n.csv	100	0.4327	0.4327 (ep 100)	**+0.00 pp**	**+0.83 pp**	metrics/mAP50-95(B)
kimmkless-yolodetector__train4-face-expression__yolov8n.csv	5	0.5803	0.5803 (ep 5)	**+0.00 pp**	**+12.94 pp**	metrics/mAP50-95(B)
kimmkless-yolodetector__train8-plant-disease__yolov8n.csv	100	0.4745	0.5029 (ep 80)	**+2.84 pp**	**+2.82 pp**	metrics/mAP50-95(B)
nhnthanhj-metrics-result-f1-yololight__unstated__yolo-light-seg.csv	47	0.6521	0.7086 (ep 42)	**+5.65 pp**	**+2.42 pp**	metrics/mAP50-95(B)
nhnthanhj-metrics-result-f1-yolov8n__unstated__yolov8n-seg.csv	74	0.7093	0.7300 (ep 54)	**+2.07 pp**	**+2.10 pp**	metrics/mAP50-95(B)
ottogalt-yolo-result__unstated__yolov8.csv	20	0.9595	0.9595 (ep 20)	**+0.00 pp**	**+0.56 pp**	metrics/mAP50-95(B)

> The "best epoch" is read from each log's own **epoch column** (the ep in the table is the value of that column), not from the row number;
> most logs in this directory start their epoch at 1, while the yolov3 of ayyappakorla starts at **0**; both use column values, so they can be compared across rows.

Logs not included in the statistics (too few epochs to speak of "selection"): aakashchalla12345678-yolov8n__unstated__yolov8n.csv (30 epochs), dhanduramesh-yolov11s__ppe-css-data__yolo11n.csv (1 epoch), kimmkless-yolodetector__train4-face-expression__yolov8n.csv (5 epochs), ottogalt-yolo-result__unstated__yolov8.csv (20 epochs)

Reading

- Complete logs (≥40 epochs) number **20** in total, coming from **6** source repositories; the median premium is **+1.04 pp**, the maximum **+5.65 pp**, the

minimum **+0.00 pp**; **18/20** are positive.
- Distribution: +0.00, +0.00, +0.10, +0.14, +0.16, +0.31, +0.57, +0.70, +0.74, +0.91, +1.18, +1.33, +1.42, +1.54, +1.55, +2.07, +2.16, +2.36, +2.84, +5.65
- **The most important point**: the largest single premium **+5.65 pp** appears in an ordinary public release log, whereas the effects reported in this kind of literature are often only of the order of 0.5–2 pp——that is, **the selection component can be of the same order as, or even larger than, the reported effect**. The median +1.04 pp is already not small. This explains why "on which split the reported value was selected" is not a detail.
- But the sample is still small (20 logs, 6 sources, and most sources do not annotate the dataset/protocol), so it **can only** be written as "this component is present and quantifiable in other people's published logs"; it **cannot** provide a field-level distribution, nor **can** benchmark aliasing be inferred from it.

Verifiability

- **Byte-level verifiable**: the **SHA-256 and byte count** of every log are given in `SHA256SUMS.txt` in the same directory (retrieval date 2026-09-16). Anyone can check whether "the bytes analysed" and "the bytes in circulation" agree.
- **Provenance completed**: `PROVENANCE.md` in the same directory has now been written. **All** 24 files are matched to **specific source paths** by content hash or unique byte count (8 Kaggle datasets + 1 GitHub repo, pinned at commit `2417aff0457c`), with 0 untraced entries. The mapping is not guessed from filename similarity: the script hashes **every entry** in `_work\kg\*.zip` and collides it against the sha256 of the local files. Check: `python work/build_provenance.py`.
- Two things that **still** cannot be derived from this directory (see `PROVENANCE.md` §4): ① the **exact downloaded version** on the Kaggle side; ② whether the `val`/`test` of these benchmarks are in the same directory (the logs do not carry a yaml).

What is still missing (to be stated as well when writing the main text)

1. **Sample size**: 20 logs, 6 sources——enough to show that the component exists in other people's logs and to give its order of magnitude, but not enough to give a field-level distribution; discussing a distribution requires several tens of traceable logs.
2. **Protocol unknown**: public logs carry no yaml, so it is impossible to judge whether the `val` and `test` of a run are the same directory; therefore "the premium exists" and "benchmark aliasing" are two independent propositions and cannot stand in for each other.
3. **Endpoint convention**: the logs give only per-epoch validation values; public repositories usually do not record "which epoch was reported", so the premium is an **upper-bound** reading (assuming that what was reported is the best).
4. **Column-name convention**: this table accepts both the new Ultralytics spelling `metrics/mAP50-95(B)` and the old YOLOv5/v3 spelling `metrics/mAP_0.5:0.95` (each row marks which one it uses). The two spellings denote the same quantity, but the **column names differ**—— any script that "filters logs by column name" will miss the other branch; that is the purpose of this item: to make the convention difference visible.

S6 · Process protocols (shared tooling; used by both papers, neither claims originality)

> **Purpose**: main text §2.3 states that "this paper and its companion paper share the same set of process protocols, and neither claims originality".

> This section **grounds that statement in verifiable names and hashes** (this is how cross-citation rule ⑤b is implemented).

> **Form decision (this paper's side)**: these two protocols are **project-internal artifacts**, and **their text is not released with this paper**——they are not public publications,

> and they bear no direct relation to this paper's scientific claims. This paper states only that "**the same set of protocols was used**", and gives the **file names and md5**,

> so that a verifier can request the same files from the authors. **If a reviewer requires it, they can be supplied together with the revision.**

Protocol (internal artifact)	md5	Role in this paper	Claims originality ?

| --- | --- | --- | --- |
| `the multi-model review checklist (shared protocol)` | `e4c73a5ca5992b7af6a5a66c84ba4561` | Stratified review by external models; among the ten self-reported defects in §10, "silent exclusion" and "wrong column taken" were found by that review | **No**, neither paper claims it |
| `the local/cloud working-method guide (shared protocol)` | `10d1b3a3e21f018f3b82e4a293242695` | Local/cloud working method: scripted remote execution, all-or-nothing patches, audit anchors, a local pointer for every number | **No**, neither paper claims it |

The same convention also applies to the audit-anchor system: the 289 anchors of this paper and the 494 anchors of the companion paper are the **same** shared tooling, and neither paper presents it as a methodological contribution.

Why the text is not attached: ① both are **Chinese working documents** with no direct relation to this paper's scientific claims; ② attaching them would make "process" read as "methodological contribution", which is exactly what this rule is meant to avoid; ③ giving the md5 is already enough to confirm that "the two papers use the same one".

S7 · Reproduction entry points, and what each one needs

① Main-text anchors (289 entries; **270 pass from the released copy** — the other 19 read
author-side working-tree documents, and the checker names them)

```
python                                  work\audit_anchors_fupaper.py

# ② 2×2 matrix analysis of S4 (reads the artifacts pulled back in this round)
python                                  work\xeval_analyze_20260916.py

# ③ Mechanism check of §5.6 (reads the D: archive, read-only)
python                                  work\gap_mechanism_20260916.py

# ④ Integrity of the artifacts pulled back in this round (sha256 + byte counts)
python                                  work\verify_pulled.py
python work\verify_pulled.py _xframe
```

Convention statement: the source files of S1–S7 are all different and **must not be mixed**——S1/S2 are a **configuration-level audit** (no datasets are downloaded), S3 is a **targeted sampling of official configs/scripts** (5 frameworks, not a census), while S4/S5 are **empirical** (the former on our own runs, the latter on other people's logs). The strength of every statement in the main text is distinguished accordingly.

S8 · Tables moved from the main text

> **Why they are here**: the venue caps submitted manuscripts at 20–35 double-spaced
> numbered pages, and these evidence tables cost about eight pages in that format. **No
> value has been changed**; every table is reproduced here exactly as it appears, and the
> main text states each table's headline result in prose.
>
> **Four of these tables are also printed in the article** — main text Tables 1–4, which are
> S4 (the Tier 1 test), S6 (the Tier 2 test), S11 (the four counting units) and S20 (the
> per-cell realization) below — because each carries a headline number the reader is asked
> to check; the article's versions print the same values in its own column layout.

> **One further table deliberately stays in the article and is not reproduced here**: the

> defect table of main text §10, because disclosure belongs in the article rather than in the

> Supplementary. (The figure/table inventory has moved into **this §S8**, below the tables.)

Index. (*n* = data rows.)

Supp. table	Moved from	Columns	n
Table S1	2.1 The closest work, and where this paper sits	Work, What it audits, What it measures, Reports a bias direction, Declares a unit	6
Table S2	3. Units: the methodological premise	Unit, Definition, Controlled by	4
Table S3	5. Finding 2: the selection premium — magnitude, direction, and mechanism	Cell, Arm, n, best, last, premium	6
Table S4	5.1 Direction: reported in two tiers, and the second tier's sign varies by corpus	Cell, n, premium baseline, premium strategy, paired diff, sd, t, p, 95 % CI	4
Table S5	5.1 Direction: reported in two tiers, and the second tier's sign varies by corpus	Cell, best (selectable), last, last5	3
Table S6	5.1 Direction: reported in two tiers, and the second tier's sign varies by corpus	Corpus, n, Δgap (pp), p, 95 % CI, Reading	5
Table S7	5.5 The same component in third-party public logs	Quantity, Value	5
Table S8	5.6 Why Δgap's sign varies by corpus: three of five candidate	Cell, n, M_val base / strat, M_test base / strat, Δgap (pp),	13

	mechanisms are refuted by the data	p	
Table S9	5.6 Why Δgap's sign varies by corpus: three of five candidate mechanisms are refuted by the data	Candidate mechanism, Prediction, Verdict, Refutation / evidence	5
Table S10	5.6 Why Δgap's sign varies by corpus: three of five candidate mechanisms are refuted by the data	Corpus, Δgap, final-epoch val diff, Δprem, −test diff, Dominant	7
Table S11	6.1 The result under four units	Unit, non-independent, of decidable rows, undecidable	6
Table S12	6.2 "Same directory" has four levels, and the audit reported only the narrowest	Level, Rows, Meaning	4
Table S13	6.3 We do **not** reproduce the original audit's two numbers, and we say why	Audit's number, Our recomputation, Reason	3
Table S14	6.5 Across frameworks: is the convention specific to YOLO?	Framework (pinned commit), Split the reported metric comes from, Also used for selection?, Status of an independent "test"	6
Table S15	7.2 The two clean-protocol cells clear significance by n = 6–8	Cell, n, Δ, paired t p, permutation p (n = 10 floor 0.001953), seeds positive	2
Table S16	7.3 A cross-archive endpoint conflict on the same run (investigated —	Reading, Δ, p	3

	and itself a disclosure)		
Table S17	8.1 The protocol is the companion's; the measurement under it is this paper's	Cell, clean protocol (n = 10), published (old protocol)	2
Table S18	8.3 ⭐ A direct measurement of selection inflation	Cell, n = 6, n = 9, change	2
Table S19	8.4 The three variance components: the excluded one is the smaller in this sample	Source, What was varied, Magnitude	3
Table S20	8.6 ⭐ The premium's realization rate: **not significant at claim level**; arm-cell median **17.6 %** (five arm-cells, n = 10; run-weighted **23.3 %**)	Cell, Arm, n, `prem_val`, `prem_test`, realization rate	5
Table S21	8.7 The per-epoch view: the realization rate is not a constant	Cell · arm, epoch, V(e), T(e), ΔV(e), ΔT(e)	4
Table S22	8.7 The per-epoch view: the realization rate is not a constant	Group, n, κ, R^2	3
Table S23	8.7 The per-epoch view: the realization rate is not a constant	Cell, checkpoints evaluated, Δgap > 0, Δgap < 0, epochs with Δgap < 0, Δgap range	2
Table S24	9 (the registered replication): the **criterion is not met** (the third row is the one that passes)	Registered pair, n, Δ (pp), 95 % CI, paired t, permutation p, same sign, Δ ≥ +0.30, p < 0.01, ≥ 8/10, unit criterion	3
Table S25	9 (the registered	Question the	3

	replication): the **criterion is not met** (the third row is the one that passes)	frozen criterion asks, Measured	
Table S26	9 (the near-matched control C): the gain does **not** depend on a label-space change	n, Δ (pp), 95 % CI, paired t, permutation p, same sign	1
Table S27	9. The registered multi-target replication: transferred to the companion paper	Registered pair, data YAML, `val`, `test`, Split	5

The tables in this section are printed in the order they were added rather than by number: S23, then the S38–S41 group, then S24–S29, then S37, then S31–S33; S30 is unused and that gap is disclosed rather than filled. The §S8 index above lists them by number, so the two orders differ. Registered pair, data YAML, `val`, `test`, Split | 5 |

The count column gives the number of data rows the table prints (a summary row, where a table has one, is a row): Table S28 prints 11 rows and is counted as 11, Table S29 prints 15 and is counted as 15.

| **Table S28** | 5.1 Direction: reported in two tiers, and the second tier's sign varies by corpus | seed, premium baseline, premium strategy, paired difference | 11 |

| **Table S29** | 5.1 Direction: reported in two tiers, and the second tier's sign varies by corpus | seed, Δgap `p_aitovis`, Δgap `p_vistod15` | 15 |

| **Table S31** | 3.2 Six experiments, five judged criteria, on GPU machines, each with its criterion written down first (G2: the selection split's size) | selection split, arm, n, `prem_val`, `prem_test`, WC, realization rate | 4 |

| **Table S32** | 3.2 Six experiments, five judged criteria, on GPU machines, each with its criterion written down first (G3: the epoch budget) | arm, n, premium at E = 50/100/200/400, ratio 400/100 | 2 |

| **Table S33** | 3.2 Six experiments, five judged criteria, on GPU machines, each with its criterion written down first (G4: a third corpus) | run, `prem_val`, `prem_test`, WC, realization rate | 20 |

| **Table S37** | 3.2 Six experiments, five judged criteria, on GPU machines, each with its criterion written down first | experiment (design), pre-declared criterion, verdict | 6 |

> **Two notes spanning S24–S27 (added 2026-09-18).** ① The experiment contains

> **90 runs**: three registered pairs 3 × 2 × 10 = 60, the saturated control 2 × 5 = 10

> (the registered set is 70), and the control appended by amendment 2 × 10 = 20 — counted

> from the run directories, ten seeds per arm in every cell but the saturated control.

> ② **Val-side alias bias**, `Δ ≈ Δ_reported + d̄` with `d̄ = mean[p_baseline − p_strategy]`

> and `p = max_e − final` from each run's own log: T1-a **+0.051** [0.006, 0.099], T1-c

> **+0.214** [0.086, 0.326], control C **+0.406** [0.313, 0.498], saturated control

> **+0.961** [0.440, 1.674] (Δ + d̄ = −0.529). Its decomposition: baseline arm **2.505 pp**, strategy arm **1.543 pp** — which is a **0.616 reading** of that cell's −1.490 pp (1.543 / 2.505 ≈ three-fifths, **not** two-thirds). T1-b is excluded by construction — its YAML

> separates the keys. Script `work/x10_alias_audit_20260918.py`. **No verdict changes.**

> **`T1-d`'s `val` side, now complete (2026-10-02).** The registered pair `dota->dota15` carried `n = 7` on its

> `val` side because the archived batch had no per-epoch curve for its last three seeds; those runs'

> `results.csv` were still in the training machine's run directories, and the three curves now ship

> (`02_release_data/t1d_valext_20261002/`), with `work/p2_t1d_val_n10_20261002.py` reproducing the seven

> previously published per-run values **bit-exactly** as its control. At `n = 10` the `val`-side arm difference

> is **-0.4056 pp**, 95 % CI **[-0.5175, -0.2937]**, paired t **-8.201**, **p = 1.8x10^-5**, **0+/10-** (at

> n = 7: -0.4043 pp, p = 0.00106): the point estimate is unchanged and the interval tightens. The article's

> Declarations no longer lists this input among the non-recomputable ones. **No verdict changes.**

Table S1. *2.1 The closest work, and where this paper sits* — columns: Work, What it audits, What it measures, Reports a bias direction, Declares a unit.

Work	What it audits	What it measures	Reports a bias direction	Declares a unit
---	---	---	---	---
Reveles-Martínez et al., *Technologies* 2026 [9]	**Image level**: near-duplicate video frames contaminating train/val	Controlled ablation: a random split scores **0.040 mAP50-95 higher** than a sequence-aware one (0.787 vs 0.747)	✗	✗ (0 hits for "same directory"/"alias"/"checkpoint selection" in the full text)
Ruangsang & Pramkeaw, *Drones* 2026 [10]	**Image level**: near-duplicate training images inside the official test split (**46.0 % / 90.7 %**)	Removing contamination costs **2.8 / 12.4 points mAP@0.5**; it **does** report a val-vs-test gap	**Yes, but concludes "no systematic direction"** (−0.41 to +0.49 points)	✗ (0 hits for "same directory"/"alias"/"reported split"/"selection bias")
roboflow/rf-detr PR #1329	**Engineering fix**	none	—	—
ultralytics issue #25650	**Engineering fix** (3 KITTI Eigen test drives were in train)	none	—	—
CVPR 2026 geospatial leakage audit [11]	**Image level** (geospatial)	detection + de-duplication	✗	✗
This paper	**Configuration level**:	**Checkpoint	**Yes, in two	**Yes**: four units, with per-row evidence

the release binds `val:`/`test:` to one path	selection premium** (best − last) **+ paired difference Δgap**	tiers**: the premium is positive in four cells with the baseline arm gaining more; Δgap's sign **varies by corpus** (2 understate / 2 overstate / 9 n.s.)	classes

Table S2. *3. Units: the methodological premise* — columns: Unit, Definition, Controlled by.

Unit	Definition	Controlled by
release	Does the benchmark's **own release artifact** (a split script, an archive, a YAML) provide an independent held-out test split?	the releasing party
protocol	What does the **official protocol** (README, paper, website) declare, and can that test split be evaluated **locally**?	the releasing party
yolo_dist	How do the `val:`/`test:` keys in the **generic YOLO distribution package** (e.g. an Ultralytics-style directory) read?	a third party
reported	Which split does the number **in the literature** come from, and is that split independent of selection?	the user

Table S3. *5. Finding 2: the selection premium — magnitude, direction, and mechanism* — columns: Cell, Arm, n, best, last, premium.

Cell	Arm	n	best	last	**premium**
smoke→SFCHD (strong)	baseline	11	42.156	41.357	**+0.799**
	strategy	11	43.863	43.481	**+0.382**
SHWD→SFCHD (corroborating)	baseline	11	43.864	42.826	**+1.038**
	strategy	11	44.428	43.916	**+0.512**
dota15→aitod (directional probe)	baseline	10	11.802	11.268	**+0.534**
	strategy	10	15.401	15.139	**+0.262**

Table S4. *5.1 Direction: reported in two tiers, and the second tier's sign varies by corpus* — columns: Cell, n, premium baseline, premium strategy, paired diff, sd, t, p, 95 % CI.

Cell	n	premium baseline	premium strategy	**paired diff**	sd	t	p	95 % CI
smoke2sf (strong)	11	+0.799	+0.382	**+0.416**	0.240	5.74	1.9×10^{-4}	[+0.255, +0.578]
shwd2sf (corroborating)	11	+1.038	+0.512	**+0.526**	0.207	8.41	7.6×10^{-6}	[+0.386, +0.665]
a2d15 (directional probe)	10	+0.534	+0.262	**+0.272**	0.183	4.69	1.1×10^{-3}	[+0.141, +0.403]
dota15 (within-domain)	10	+0.941	+0.492	**+0.449**	0.196	7.23	4.9×10^{-5}	[+0.309, +0.589]

Table S5. *5.1 Direction: reported in two tiers, and the second tier's sign varies by corpus* — columns: Cell, best (selectable), last, last5.

Cell	best (selectable)	last	last5
smoke→SFCHD	+1.708	**+2.124**	+2.086
SHWD→SFCHD	+0.564	**+1.090**	+1.079
dota15→aitod	+3.599	**+3.871**	+3.775

Table S6. *5.1 Direction: reported in two tiers, and the second tier's sign varies by corpus* — columns: Corpus, n, Δgap (pp), p, 95 % CI, Reading.

Corpus	n	Δgap (pp)	p	95 % CI	Reading
shwd2sf	10	**+0.779**	0.0053	[+0.297, +1.261]	`val` **understates**
smoke2sf	10	**+0.461**	0.0263	[+0.068, +0.855]	`val` **understates**
p_aitovis	10	**−1.226**	1.3×10^{-8}	[−1.370, −1.081]	`val` **overstates**
p_vistod15	10	**−3.315**	6.5×10^{-9}	[−3.676, −2.953]	`val` **overstates**
other 9 cells	3–10	−2.46 to +13.66	≥ 0.054	all include 0	**not significant**

Table S7. *5.5 The same component in third-party public logs* — columns: Quantity, Value.

Quantity	Value
Complete logs (≥ 40 epochs)	**20 / 6 sources**
Median premium	**+1.04 pp**
Max / min	**+5.65 / +0.00 pp**
Positive	**18/20**
Distribution	+0.00, +0.00, +0.10, +0.14, +0.16, +0.31, +0.57, +0.70, +0.74, +0.91, +1.18, +1.33, +1.42, +1.54, +1.55, +2.07, +2.16, +2.36, +2.84, +5.65

Table S8. *5.6 Why Δgap's sign varies by corpus: three of five candidate mechanisms are refuted by the data* — columns: Cell, n, M_val base / strat, M_test base / strat, Δgap (pp), p.

Cell	n	M_val base / strat	M_test base / strat	Δgap (pp)	p
shwd2sf	10	60.67 / 60.40	43.56 / 44.07	**+0.779**	0.0053
smoke2sf	10	58.71 / 59.69	42.06 / 43.50	**+0.461**	0.0263
p_aitovis	10	11.24 / 12.66	10.21 / 10.40	**−1.226**	1.3×10^{-8}
p_vistod15	10	16.96 / 23.93	11.63 / 15.26	**−3.315**	6.5×10^{-9}
aitod20	3	14.51 / 12.13	10.40 / 9.49	+1.481	0.054
d15d15	3	30.57 / 29.09	15.17 / 14.96	+1.278	0.092
mende20	3	73.40 / 68.03	73.08 / 70.17	+2.472	0.099
fire	3	60.81 / 42.76	39.58 / 35.20	+13.658	0.061
p_d15toai	3	12.83 / 12.31	9.59 / 9.86	+0.781	0.32

mask20	3	34.15 / 31.46	65.35 / 64.42	+1.754	0.66
p_masktomende	3	60.36 / 60.44	69.36 / 66.97	−2.463	0.16
dota	3	30.33 / 29.04	17.17 / 16.24	+0.355	0.56
vis	3	14.19 / 13.39	11.64 / 11.01	+0.169	0.49

Table S9. *5.6 Why Δgap's sign varies by corpus: three of five candidate mechanisms are refuted by the data* — columns: Candidate mechanism, Prediction, Verdict, Refutation / evidence.

Candidate mechanism	Prediction	Verdict	Refutation / evidence
---	---	---	---
M1 multiplicative gain	if the strategy improves by a fixed **ratio**, Δgap has the same sign as the raw gap	**refuted**	3 counterexamples: `mask20` (gap −31.2 but Δgap +1.75), `p_aitovis` (+1.03 but −1.23), `p_visto d15` (+5.34 but −3.32)
M2 ceiling compression	the easier `val` is (larger gap grows	gap\	), the more the strategy's advantage is compressed, so Δgap turns negative as \	gap\	grows	**refuted**	all six cells with gap\	gap\	> 10 have Δgap **positive** (+0.35 to +13.66); overall r = +0.32, p = 0.28
**M3 level determi	cells with higher mAP are	**not supported**	significant cells' baseline						

nes resolvability**	more likely to be significant		val mean **36.83 pp**, non-significant **36.79 pp** (Welch p = 0.998)
M4 whether the strategy truly gains on test	only cells where the strategy gains on test can show `val` understating	**refuted**	2 counterexamples: `p_aitovis` and `p_vistod15` are better on test yet have **negative** Δgap
M5 difference in val-side selection gain	Δgap has the same sign as (premium_baseline − premium_strategy)	**not supported**	only **8 of 13** cells share the sign; r = +0.06, p = 0.84

Table S10. *5.6 Why Δgap's sign varies by corpus: three of five candidate mechanisms are refuted by the data* — columns: Corpus, `n`, Δgap, final-epoch val diff, Δprem, −test diff, Dominant. **The last three columns already carry the sign convention, so Δgap is their sum** (the `−test diff` column is printed with the sign the identity needs). **The `n` column is needed to read this table against the released components file**: `02_release_data/gap_mechanism_20260916.txt` was computed at the **pre-escalation** sample size, so for the two cells later escalated to `n` = 10 its Δgap column still holds the **`n` = 3** reading (`p_vistod15` −3.58, `p_aitovis` −1.11) where this table prints the **`n` = 10** reading (−3.315 / −1.226); the two are not in conflict, they are two sample sizes.

Corpus	`n`	Δgap	final-epoch val diff	Δprem	−test diff	Dominant

| --- | --- | --- | --- | --- | --- | --- |
mask20	3	**+1.75**	**+15.78**	**−13.10**	−0.93	the two large components **nearly cancel**
fire	3	**+13.66**	+15.96	+2.08	−4.39	final-epoch val diff
p_vistod15	**10**	**−3.315**	−7.072	−0.213	+3.971	final-epoch val diff
p_aitovis	**10**	**−1.226**	−1.583	+0.090	+0.267	final-epoch val diff
shwd2sf	**10**	**+0.78**	−0.33	+0.60	+0.51	Δprem
smoke2sf	**10**	**+0.46**	−1.06	+0.09	+1.43	−test diff
dota	3	**+0.35**	+0.75	+0.54	−0.93	−test diff
`aitod20`	3	**+1.48**	+1.95	+0.44	−0.91	final-epoch val diff
`d15d15`	3	**+1.28**	+0.02	+1.46	−0.21	Δprem
`mende20`	3	**+2.47**	+6.23	−0.85	−2.90	final-epoch val diff
`p_d15toai`	3	**+0.78**	+0.41	+0.10	+0.27	final-epoch val diff
`p_masktomende`	3	**−2.46**	−1.06	+0.99	−2.39	−test diff
`vis`	3	**+0.17**	+0.58	+0.22	−0.63	−test diff

The last six rows are derived, not re-measured. They are the six scanned cells the table previously left to the released file; their four columns follow from the identity and the file's own columns as `final-epoch val diff = val diff − Δprem`, with `Δprem` and `−test diff` read directly from it. **The rule was validated before use**: applied to the five rows whose `n` matches the file (the two escalated cells are excluded because the file predates their escalation), it reproduces every printed column to ±0.01, and the identity closes on all thirteen. The `Dominant` label is likewise mechanical — the largest-|·| component, or **nearly cancel** where the two largest both exceed 5 pp and oppose — and reproduces the seven labels this table already carried.

Table S11. *6.1 The result under four units* — columns: Unit, non-independent, of decidable rows, undecidable.

Unit	non-independent	of decidable rows	undecidable
release (the benchmark's own release)	**10/19 ≈ 53 %**	10/19	—
protocol (the official protocol)	**13/19 ≈ 68 %**	13/19	—
yolo_dist (the generic YOLO distribution package)	**4/19 ≈ 21 %**	**4/15 ≈ 27 %**	4 rows
reported (the layer actually reported)	**12/19 ≈ 63 %**	**12/15 = 80 %**	4 rows
↳ of which **literal two-key same path** (= the 2/19 the original audit reported)	**2/19 ≈ 11 %**	2/19	—
↳ of which `reported` is **`clean`** (independent)	**3/19 ≈ 16 %**	3/15	4 rows

Table S12. *6.2 "Same directory" has four levels, and the audit reported only the narrowest* — columns: Level, Rows, Meaning.

Level	Rows	Meaning
L1 distribution package, literal `val == test`	**2**	PASCAL VOC, SFCHD
L2 distribution package, `train == val`	**1**	SHWD
L3 distribution package clean, but the **reported** layer is non-independent	**9**	Objects365, Open Images v7, DOTA v2.0, UAVDT, xView, NWPU VHR-10, MAFA, WIDER FACE, CrowdHuman
L4 the rest	**7**	

Table S13. *6.3 We do not reproduce the original audit's two numbers, and we say why* — columns: Audit's number, Our recomputation, Reason.

Audit's number	Our recomputation	Reason
14/19 (the reported layer)	**12/19**	(h-1) removes D-Fire; DIOR and Mendeley are evidence-insufficient at the reported layer
5/19 clean	**3/19**	the same two rows move to "undecidable" (they were never adjudicated), so the clean group loses two
12/19 (called "release/spec layer")	**release 10/19, protocol 13/19**	**12 is neither**: it was obtained by a **two-row manual adjustment** (removing WIDER FACE and CrowdHuman), not by applying one unit consistently to all 19 rows

Table S14. *6.5 Across frameworks: is the convention specific to YOLO? —* columns: Framework (pinned commit), Split the reported metric comes from, Also used for selection?, Status of an independent "test".

Framework (pinned commit)	Split the reported metric comes from	Also used for selection?	Status of an independent "test"
Ultralytics YOLO [16] (the `yolo_dist` unit of §6.1)	the YAML's `val:`	**yes** (`best.pt`; §5's "selection premium" is defined on exactly this convention)	the config **has** a `test:` key, but **2 of 19 rows bind it to the same path** (§6.1)
Detectron2 [30] `a2f4a8771ab7`	`DATASETS.TEST: ("coco_2017_val",)`	**no** (`TEST.EVAL_PERIOD = 0`, no periodic evaluation by default)	`coco_2017_test` exists, but its annotation file is **`image_info_test2017.json` — image information only, no ground truth**
MMDetection [31] `cfd5d3`	`CocoMetric(ann_file='annotations/instances_val2017.json')`, `val_interval=1`	**no** (`CheckpointHook(interval=1)`, the base config has	**`test_dataloader = val_dataloader` and `test_evaluator =**

a985b0`		**no** `save_best`)	val_evaluator`** (`configs/_base_/datasets/coco_detection.py`, lines 65 and 73) — the `test:` key **exists but is the same object as `val`**, so there is **no independent test split**; an earlier reading of this row said the key was absent, which that file contradicts
YOLOX [29] `6ddff4824372`	`val_ann = "instances_val2017.json"`	**yes** (`trainer.py` tracks `self.best_ap` and saves the best)	`test_ann` is used **only when `testdev=True`**
PaddleDetection [33] `b25522a0f4bd`	`EvalDataset` = `val2017`	**not written by default, and now provable line by line** — `paddledetection/tools__train.py` declares `--eval` as `action='store_true', default=False` and passes it to `trainer.train(FLAGS.eval)`; the assignment `status['save_best_model'] = True` sits inside the validation branch of `paddledetection/ppdet__engine__trainer.py` (`if validate and is_snapshot:`), and `best_model` is written only through `paddledetection/ppdet__engine__callbacks.py`'s gate `if 'save_best_model' in status and status['save_best_m	**`TestDataset` reads an annotation-free `ImageFolder`, with `anno_path` written as `annotations/instances_val2017.json`**

		odel']:` — both files are now in the SHA-256 list	
DETR [32]	`build_dataset(image_set='val')` `29901c51d7fe`	**no** (evaluates every epoch but does not track the best)	the `PATHS` in `datasets/coco.py` have **only train and val, no test key**

> **Correction (2026-10-04, and a defect of our instrument).** The MMDetection row above was first written as "the base config has **no** test split at all". That is **false**: at the pinned commit, `configs/_base_/datasets/coco_detection.py` lines **65** and **73** read `test_dataloader = val_dataloader` and `test_evaluator = val_evaluator`. The extraction behind this table (`work/analyze_xframe_20260916.py`) searched only for `val_*` keys, so the `test_*` keys were **absent from the evidence file rather than absent from the config**. The corrected reading is **stronger for the article's claim**: the key exists but is bound to the validation object — an **alias, not an independent split**. On the same day the other rows were re-checked against the pinned sources and stand (YOLOX: `test_ann` used only when `testdev=True`, `yolox/exp/yolox_base.py` 301/306/307; PaddleDetection: `if validate and is_snapshot:` in `ppdet/engine/trainer.py` 694 with `status['save_best_model'] = True` at 721).

Table S15. *7.2 The two clean-protocol cells clear significance by n = 6–8* — columns: Cell, n, Δ, paired t p, permutation p (n = 10 floor 0.001953), seeds positive.

Cell	n	Δ	paired t p	permutation p (n = 10 floor 0.001953)	seeds positive
smoke→SFCHD	**10**	**+1.434 pp**	8.6×10^{-7}	**0.001953 (= the floor)**	**10/10**
SHWD→SFCHD	**10**	**+0.507 pp**	0.0094	**0.009766**	9/10 (one at −0.19)

Table S16. *7.3 A cross-archive endpoint conflict on the same run (investigated — and itself a disclosure)* — columns: Reading, Δ, p.

Reading	Δ	p
Final-epoch reading (A, 67.12)	**−1.490**	0.0141

Best-epoch reading (B, 70.06), **as released**	**−0.902**	0.2931
Best-epoch reading (B, 70.06), **corrected 2026-10-02** (authoritative)	**−1.490**	0.0141

Table S17. *8.1 The protocol is the companion's; the measurement under it is this paper's* — columns: Cell, clean protocol (n = 10), published (old protocol).

Cell	clean protocol (n = 10)	published (old protocol)
smoke→SFCHD	**+1.434 pp**	+1.78 (3 seeds) / +1.76 (10 seeds)
SHWD→SFCHD	**+0.507 pp**	+0.51 (3 seeds) / +0.54 (10 seeds)

Table S18. *8.3 ⭐ A direct measurement of selection inflation* — columns: Cell, n = 6, n = 9, change.

Cell	n = 6	n = 9	change
smoke→SFCHD	+1.562	+1.420	**−0.14 pp**
SHWD→SFCHD	+0.693	+0.492	**−0.20 pp**

Table S19. *8.4 The three variance components: the excluded one is the smaller in this sample* — columns: Source, What was varied, Magnitude.

Source	What was varied	Magnitude
augmentation RNG	3 augmentation seeds	0.246 pp (strategy) / 0.190 pp (baseline)
initialization (source-side retraining from scratch)	3 → **5** independent initializations	**0.146 → 0.439 pp (gain) / 0.209 → 0.253 pp (absolute endpoint)**
second-stage data order	9–10 seeds	0.403 pp (smoke) / **0.515 pp (SHWD)**

> **The three rows are not in one unit, and we do not compare them as if they were.** The augmentation and

> data-order rows give **standard deviations of the run-to-run change (pp)**; the initialization row is

> reported in **both** of the units its source record carries — **0.146 pp as a gain** and **0.209 pp as an

> absolute endpoint **— because those two are not interchangeable. The claim §8.4 makes is a** ranking**,

> and that ranking holds under either reading; the three magnitudes are **not** comparable across rows in a

> single unit, and **no cross-row arithmetic is performed on them**.

> **The initialization row was extended on 2026-09-30 (five initializations), and the extension weakens
> the reading rather than confirming it.** Two further source-side retrainings (data-order seeds 45 and
> 46, augmentation seeds 404 and 505) give gains of **+0.430** and **+1.390 pp**; with them the gain SD
> moves **0.146 → 0.439 pp** (mean **+0.626**, range **1.110**) and the absolute-endpoint SD
> **0.209 → 0.253 pp** (43.070–43.630). The bootstrap 95 % interval for the five-initialization gain SD is
> **[0.058, 0.580]**, which **overlaps** the data-order row's **0.488 pp** (n = 10; 0.515 pp at n = 9) — so
> "the excluded component is the smaller" holds **in the three-initialization sample only**, and this
> supplement reports the ranking as sample-scoped rather than established. The same run reproduces the
> archived three initializations value by value (43.630/+0.570, 43.590/+0.280, 43.250/+0.460, max
> |Δ| = **0.005 pp**). Script: `work/G3_sigma_n5_20260930.py`.

Table S20. *8.6 ⭐ The premium's realization rate: not significant at claim level; arm-cell median 17.6 % (five arm-cells, n = 10; run-weighted 23.3 %)* — columns: Cell, Arm, n, `prem_val`, `prem_test`, realization rate.

> **Which cells are these five, and why exactly five — the rule, stated as a rule. An arm-cell enters iff its `test` side reached n ≥ 5**; that threshold is the analyser's gate (`work/item2_final_v2_20260925.py`), applied to every cell alike, not chosen per cell. **At the 2026-09-25 freeze exactly five arm-cells met it**, and at that date all three weightings coincide (**23.26 %**); equivalently: **every arm-cell whose `test` side had reached n = 10 at the 2026-09-25 freeze** — both arms of `shwd2sf` and of `smoke2sf`, plus `aitod20` **baseline** (50 runs); `work/item2_final_v2_20260925.py` applies it and prints the pool, the per-cell means and the **23.26 %** aggregate. The `aitod20` **strategy** arm finished after the freeze: it is **the sixth row of Table 4, printed as Table S20c below**, and it **qualifies under the same n ≥ 5 rule** — it is kept out of *this* pool only because the pool is dated at the freeze, so **the primary estimate is the frozen**

one while the six-cell reading is reported alongside it (feeding it in gives **8.7 %**). **The three scopes agree in verdict**: five-cell **[−17.3 %, 53.2 %]**, six-cell **[−4.9 %, +21.3 %]**, eleven-cluster **[19.3 %, 60.4 %]** — the two cell-level intervals cover zero, so **no reported conclusion turns on the pool boundary**; what moves is the point estimate. **Table S20 and Table 4 therefore print five and six cells on purpose.**

Table S20-lite — the pool ledger, and its status (the one thing it is not). Every cell that could enter the §8.6 pool, its completion state at the two dates that matter, and whether it is in:

Cell · arm	`test` side at the **2026-09-25 freeze**	completed	in the five-cell pool	run-weighted rate
shwd2sf · baseline	n = 10	2026-09-25	**yes**	+50 %
shwd2sf · strategy	n = 10	2026-09-25	**yes**	+18 %
smoke2sf · baseline	n = 10	2026-09-25	**yes**	+74 %
smoke2sf · strategy	n = 10	2026-09-25	**yes**	−26 %
aitod20 · **baseline**	n = 10	2026-09-25	**yes**	−28 %
aitod20 · **strategy**	n = 1	**after** the freeze	**no** (Table S20c)	−49 %

> **Status, stated plainly: this is a *mechanical* rule applied at a recorded date, not a *pre-registered* one.** The
> rule is "every arm-cell whose `test` side had reached n = 10 at the 2026-09-25 freeze", and `work/item2_final_v2_20260925.py`
> is what applies it; the date and the completion states above are what make it re-derivable. But the rule was **written
> down when the pool was formed**, not before the runs existed, so it cannot carry the authority of a pre-registration.
> **What carries that authority is that it does not need to**: the §8.6 claim is interval-level and **both pools cover
> zero **(five-cell [−17.3 %, 53.2 %], six-cell [−4.9 %, +21.3 %]), so** no reported conclusion depends on pool membership**.
> The remaining freedom is over the **point estimate**, and §8.6 prints both.

Cell	Arm	n	`prem_val`	**`prem_test`**	**realization rate**
shwd2sf	baseline	10	+1.469	**+0.736**	**50 %**

shwd2sf	strategy	10	+0.865	**+0.152**	**18 %**
smoke2sf	baseline	10	+0.691	**+0.509**	**74 %**
smoke2sf	strategy	10	+0.605	**−0.157**	**−26 %**
aitod20	baseline	10	+0.766	**−0.218**	**−28 %**

> **The same five cells at one decimal place**, so the median can be re-derived rather than taken on

> trust: 50.1 %, 17.6 %, 73.7 %, -26.0 %, -28.5 % **⇒ median 17.6 %** -- the figure the article,

> Table 4 and this table print. The whole-number rates above are the same ratios rounded for display;

> recomputing the median from *those* is what makes it look inconsistent.

> **One earlier reading of this same cell is left standing, on purpose.** The verbatim analysis dump

> in §S4 still shows `aitod20 base 1 … -0.377` — that is **this cell before it was completed**, at

> n = 1, and §S4 is that sample's record, not a competing summary; the row above is the completed

> n = 10 measurement. §S4 keeps its original bytes and its recorded hash because it is an embedded

> dated artifact.

> **A theory-driven prediction, tested on the same runs (added later).** §3.1 derives

> realization rate = 1 − WC / `prem_val` with WC := E[ε_sel − ε_final] ≥ 0, the winner's-curse term.

> On the archive: ① WC = `prem_val` − `prem_test` = **+0.463 pp** per run ($t = 8.34$, $p = 1.8\times10^{-12}$;

> the realized premium is smaller in **68/81** runs); ② sorting runs by the noise scale $\hat{\sigma}$ of their

> `val` curve gives realization-rate terciles of **65.6 % / 49.8 % / 13.8 %**, **$\rho = -0.466$** ($p = 1.3\times10^{-5}$, $n = 80$). **Both readings are re-taken with arm-cell clustering, as the kappa fit already is: WC = +0.463 pp with a cluster-robust 95 % interval [+0.204, +0.721] and p = 0.0033, and ρ = −0.466 with a cluster-bootstrap interval [−0.651, −0.131] and p = 0.0115 — which at the cluster level (n = 9) is −0.433 with p = 0.244 — so the clustered reading is the one this paper reports (Supplementary S12, section 8.6, script**

work/r102_cluster_20260927.py). The stronger form (premium ≈ σ√(2 ln E)) was **withdrawn**:

> slope 0.026 ± 0.222, p = 0.82, n = 101. Scripts: `work/theory_p4_test_20260919.py`,

> `work/theory_meta_and_tests_20260919.py`.

Table S20b — the `aitod20` baseline arm-cell (Table 4's fifth row) per run, so its rate can be re-derived

Table S20b. *8.6 (Table 4's fifth row, the `aitod20` baseline)* — the per-run readings behind that cell; its companion for the **sixth** row is **Table S20c** below. Source: `02_release_data/aitod20_n10_20260924/item2_aitod20_n10_20260925.csv` (the completed n = 10 measurement, released); the n = 1 predecessor is left standing verbatim in §S4.

run	seed	`prem_val`	`prem_test`	per-run realization rate
`r10_aitod20_base30_3way_s42n`	42	+0.7952	-0.3762	-47.3 %
`r10_aitod20_base30_3way_s43n`	43	+0.4240	-0.2404	-56.7 %
`r10_aitod20_base30_3way_s44n`	44	+0.8800	-0.1963	-22.3 %
`r10_aitod20_base30_3way_s45n`	45	+0.3800	-0.2663	-70.1 %
`r10_aitod20_base30_3way_s46n`	46	+0.5750	-0.2225	-38.7 %
`r10_aitod20_base30_3way_s47n`	47	+1.4780	-0.1080	-7.3 %
`r10_aitod20_base30_3way_s48n`	48	+0.7230	+0.1424	+19.7 %
`r10_aitod20_base30_3way_s49n`	49	+0.7800	-0.7821	-100.3 %
`r10_aitod20_base30_3way_s50n`	50	+1.3660	-0.1808	-13.2 %
`r10_aitod20_base30_3way_s51n`	51	+0.2690	+0.0552	+20.5 %

> **Reproduction.** These ten rows give **per-run means** `prem_val` **+0.7670 pp** and `prem_test` **-0.2175 pp**, i.e. the cell as **Table 4** prints it (**+0.766 / -0.218**) to **0.001 pp**; the run-weighted rate is **-28.36 %** (the printed **−28 %** is that ratio rounded). **8 of the 10 runs are negative** and the per-run rate spans **-100.3 % to +20.5 %** — this cell is the reason the article calls the rate *not a constant*, and the

reason it is **outside the five-cell headline pool** (whose exclusion §8.6 states as post-hoc) whose median and run-weighted rate §8.6 reports.

Table S20c — the sixth arm-cell (`aitod20` strategy) per run

Table S20c. *8.6* — the per-run readings behind the **sixth** row of Table 4 (the `aitod20` **strategy** arm), the cell whose pool membership is discussed in §8.6. Source: `02_release_data/g6_prem_per_run.csv` (the released 140-run per-run file, `prem_test` = `T_best − T_final` as Table 4 defines it). Rows are the ten seeds 42–51.

run	seed	`prem_val`	`prem_test`	per-run realization rate
`r10_aitod20_lr005_30ep_3way_s42n`	42	+0.4500	-0.8953	-199.0 %
`r10_aitod20_lr005_30ep_3way_s43n`	43	+0.3410	-0.4488	-131.6 %
`r10_aitod20_lr005_30ep_3way_s44n`	44	+0.0000	+0.0000	— †
`r10_aitod20_lr005_30ep_3way_s45n`	45	+0.3760	-1.2383	-329.3 %
`r10_aitod20_lr005_30ep_3way_s46n`	46	+1.9340	+0.2805	+14.5 %
`r10_aitod20_lr005_30ep_3way_s47n`	47	+1.4560	-0.8829	-60.6 %
`r10_aitod20_lr005_30ep_3way_s48n`	48	+0.8280	-0.0041	-0.5 %
`r10_aitod20_lr005_30ep_3way_s49n`	49	+1.2450	-0.9036	-72.6 %
`r10_aitod20_lr005_30ep_3way_s50n`	50	+3.1230	-0.5097	-16.3 %
`r10_aitod20_lr005_30ep_3way_s51n`	51	+1.3560	-0.8301	-61.2 %

> **Reproduction.** These ten rows give per-run means `prem_val` **+1.1109 pp** and `prem_test` **-0.5432 pp** ⇒ run-weighted rate **-48.90 %** (the **−49 %** of Table 4 is that ratio rounded). † **Zero-denominator rule**: seed 44 has `prem_val = 0` exactly, so its per-run rate is **undefined** and it is **excluded from the per-run rates but kept in the run-weighted ratio** — that is the rule for every cell, stated here once. Adding this cell to the five-cell pool is the **stated sensitivity** of §8.6 (run-weighted **23.3 % → 8.7 %**, median **+17.6 % → −4.2 %**), and the six-cell cluster interval **[−4.9 %, +21.3 %]** still covers zero.

Table S21. *8.7 The per-epoch view: the realization rate is not a constant* — columns: Cell · arm, epoch, V(e), T(e), ΔV(e), ΔT(e).

Cell · arm	epoch	V(e)	T(e)	ΔV(e)	ΔT(e)
shwd2sf · baseline	20 / 40 / 60 / 80	57.62 / 58.57 / 59.37 / 59.79	41.58 / 42.73 / 43.14 / 43.46	−1.58 / −0.63 / +0.17 / +0.59	−1.24 / −0.10 / +0.32 / +0.63
shwd2sf · strategy	20 / 40 / 60 / 80	52.34 / 56.17 / 58.51 / 59.38	37.98 / 41.21 / 42.69 / 43.91	−7.20 / −3.36 / −1.02 / −0.15	−5.94 / −2.71 / −1.22 / −0.00
smoke2sf · baseline	20 / 40 / 60 / 80	53.69 / 56.25 / 57.20 / 58.08	38.91 / 41.04 / 41.68 / 42.10	−4.33 / −1.77 / −0.82 / +0.05	−2.64 / −0.52 / +0.13 / +0.55
smoke2sf · strategy	20 / 40 / 60 / 80	50.99 / 55.15 / 57.38 / 58.46	36.91 / 41.03 / 42.73 / 43.55	−8.09 / −3.94 / −1.70 / −0.62	−6.74 / −2.63 / −0.92 / −0.11

> **Complete curve (added 2026-09-19).** This table keeps the five-epoch sample. The complete

> evaluation — every retained checkpoint of the same 41 runs, 806 rows — is archived as

> `xeval_perepoch_20260918/matrix_perepoch.csv` (SHA-256 listed) rather than printed, and the two

> agree on all **202** overlapping (run, epoch) points to **0.0000 pp**.

Table S22. *8.7 The per-epoch view: the realization rate is not a constant* — columns: Group, n, κ, R^2.

Group	n	**κ**	R^2
All four arm-cells	765	**0.708**	0.937
Baseline arm only	385	**0.600**	0.946
Strategy arm only	380	**0.767**	0.951

> **Complete curve (added 2026-09-19).** These are the values over **every retained checkpoint**

> (765 (seed, epoch) points, 41 runs, 806 evaluations, 0 failures). The five-epoch grid of the

> first version gave **0.734 / 0.532 / 0.769** over n = 161 / 81 / 80; the overall estimate therefore

> moves **−0.026** with overlapping intervals, and the arm difference narrows from +0.237 to

> **+0.168**. Script: `work/x3b_perepoch_corrected_20260918.py`.

> **Uncertainty (added 2026-09-18) — scoped to the five-epoch grid (n = 161), not to the complete-curve

> value the article prints.** The epochs within a run are not independent

> observations, so the fit is re-done with run-clustered inference (41 runs):

> all four arm-cells **κ = 0.734, 95 % CI [0.683, 0.784]** (sandwich SE 0.025; a run-level

> bootstrap gives [0.685, 0.784]); baseline arm **[0.456, 0.608]**; strategy arm **[0.712, 0.827]**.

> Resampling runs *within* each arm (the two arms are never pooled) puts the arm difference at

> **+0.237, 95 % CI [0.145, 0.323]**; a within-run fit, which removes run-level differences

> entirely, gives 0.774. Scripts: `work/x2_kappa_clustered_20260918.py`,

> `work/x2b_kappa_arms_20260918.py` (both reproduce the published 0.734 / 0.902 / n = 161 first).

Table S23. *8.7 The per-epoch view: the realization rate is not a constant* — columns: Cell, checkpoints evaluated, Δgap > 0, Δgap < 0, epochs with Δgap < 0, Δgap range.

Cell	checkpoints	**Δgap > 0**	**Δgap < 0**	epochs with Δgap < 0	Δgap range (pp)
shwd2sf	19	**19**	0	—	+0.404 to +4.474
smoke2sf	19	**16**	**3**	**epochs 5, 10, 15**	−4.454 to +1.501

> **Why this table has changed shape (2026-09-19).** The first version sampled epochs {20, 40, 60, 80}

> and reported "positive at all four sampled epochs" for both cells. With **every** checkpoint

> evaluated, that holds for `shwd2sf` (all 19) but **not** for `smoke2sf`, whose Δgap is negative at

> epochs 5, 10 and 15 (−4.454 / −1.089 / −0.058 pp) and positive at every checkpoint from epoch 20

> on. The sampled grid began at epoch 20. The four sampled values the old table printed

> (+1.678 / +0.886 / +0.404 / +0.860 and +0.702 / +1.092 / +0.875 / +1.062) are unchanged and remain

> correct for those four epochs. Script: `work/x3b_perepoch_corrected_20260918.py`.

Table S38 — the selection term, cell by cell: `prem_test` and WC for all thirteen scanned corpora

Table S38. *5.6 / 8.6* — the selection contribution proper, `prem_val − prem_test` = WC, formed here for **every** scanned corpus and both arms. **Two scopes, two counts, one table**: this table covers **26 arm-cells / 190 runs** — the five arm-cells printed in Table S20 (their values repeated here, **50 runs**) plus **21 arm-cells / 140 runs** closed on **2026-09-30** by scoring **every archived 5-epoch checkpoint** on each run's own held-out `test` split: **1,568 grid checkpoints over 140 runs** — the two figures are a **decomposition, not two counts of the same thing**: **1,848 evaluations in total = 1,568 grid checkpoints + 140 runs × 2 endpoints (`best`/`last`)**, on one machine, with the pipeline's registered call (`YOLO(w) → val(split="test", batch=32, imgsz=640, plots=False, save_json=False)`, no loss re-import), the data YAML read from each run's own `args.yaml` and asserted to be three-way with a `test:` split. `n` is the number of seeds carrying a `test` curve for that arm (3 where the scan ran at n = 3, 10 where the cell was later escalated, and 10 for the five arm-cells of Table S20); `prem_val` and `prem_test` are means over those seeds, in pp, with `prem_test = T_best − T_final`; realization rate = `prem_test / prem_val`. **The same run reproduces the values printed earlier in this supplement**: over the 21 arm-cells carrying both readings, the val maximum, `prem_val` and the archived `T_best` agree to **max |Δ| = 0.005 pp**, which is the printed rounding (`work/g6_repro_vs_printed_20260930.py`); the coverage audit reports **140/140 runs, 1,848 rows, 0 anomalies** (`g6_prem_per_run.csv` and `g6_perepoch_test_curves.csv`). Per-run readings and the curves themselves: release files `g6_prem_per_run.csv` and `g6_perepoch_test_curves.csv`.

Cell	Arm	n	`prem_val`	`prem_test`	WC	realization rate

shwd2sf	baseline	-10	+1.469	**+0.736***	+0.733	**50 %**
shwd2sf	strategy	10	+0.865	**+0.152***	+0.713	**18 %**
smoke2sf	baseline	10	+0.691	**+0.509***	+0.182	**74 %**
smoke2sf	strategy	10	+0.605	**−0.157***	+0.762	**−26 %**
aitod20	baseline	10	+0.766	**−0.218***	+0.984	**−28 %**
aitod20	strategy	10	+1.111	**−0.543***	+1.654	**−49 %**
d15d15	baseline	10	+1.963	**+0.332***	+1.631	**17 %**
d15d15	strategy	10	+0.270	**−0.295** (−0.294 before the correction)	+0.564	**−109 %**
dota	baseline	10	+1.271	**+0.334***	+0.937	**26 %**
dota	strategy	10	+0.547	**−0.811***	+1.357	**−148 %**
fire	baseline	3	+10.953	**+2.820***	+8.133	**26 %**
fire	strategy	3	+8.869	**−0.264***	+9.134	**−3 %**
mask20	baseline	3	+16.071	**+0.325***	+15.746	**2 %**
mask20	strategy	3	+29.172	**+0.445***	+28.727	**2 %**
mende20	baseline	3	+4.828	**+3.277***	+1.551	**68 %**
mende20	strategy	3	+5.682	**+1.512***	+4.170	**27 %**
p_aitovis	baseline	10	+0.438	**+0.357***	+0.080	**82 %**
p_aitovis	strategy	10	+0.268	**+0.278***	−0.010	**104 %**
p_d15toai	baseline	3	+0.488	**−0.154***	+0.642	**−32 %**
p_d15toai	strategy	3	+0.392	**−0.170***	+0.561	**−43 %**
p_masktomende	baseline	3	+2.335	**+2.022***	+0.313	**87 %**
p_masktomende	strategy	3	+1.342	**−0.144***	+1.486	**−11 %**
p_vistod15	baseline	10	+1.080	**+0.601***	+0.478	**56 %**
p_vistod15	strategy	10	+0.972	**+0.280***	+0.692	**29 %**

vis	baseline	10	+0.484	**+0.358**	+0.126	**74 %**
vis	strategy	10	+0.180	**−0.107**	+0.287	**−59 %**

> **What this table does, and what it does not do.** It closes a **verifiability** gap: the ten corpora
> and the `aitod20` strategy arm previously carried no `test` curve at all, so `prem_test`, and hence
> WC, could not be formed for them — any reader can now recompute every WC above from the released
> per-run file. It does **not** upgrade a claim, and **§7.1's power argument is unchanged**. Read descriptively, the asymmetry the five arm-cells of §8.6 show is the
> general case here: **WC > 0 in 25 of the 26 arm-cells**, the **arm-cell median realization rate is
> 17.2 % **(mean 8.9 %, ten of the 26 negative), and the realized premium is** negative in 8 of the 13
> strategy arms** (and in 2 of the 13 baseline arms). Scripts: `work/g6_prem_analyze_20260930.py`,
> `work/g6_repro_vs_printed_20260930.py`; the evaluation itself: `work/g6_perepoch_eval.py`,
> `work/g6_coverage_audit.py`.
>
> **Ten of the arm-cells above that stood at n = 3 were escalated to n = 10 on 2026-10-06** (seeds 45–51 trained on the two pods; the three earlier seeds are reused, not replaced, so each cell
> pools seeds 42–51 with no duplicate). Their pooled values, side by side with the n = 3 readings
> printed in the table above, are **Table S38b**. Read descriptively with the escalation folded in,
> the three quantities quoted just above move as follows: **WC > 0 holds in 25 of the 26 arm-cells (unchanged)**, the **arm-cell median realization rate rises from 17.2 % to 21.6 %** (mean 9.0 % →
> 11.6 %, the negative cells go from ten to **nine**), and the realized premium is still **negative in

> 8 of the 13 strategy arms **(the baseline arms go from two negative to** one**). The direction of the

> section's descriptive claim therefore survives the escalation; what changes is the magnitude of the

> median.

Table S38b — the ten arm-cells escalated from n = 3 to n = 10 (2026-10-06)

Table S38b. *5.6 / 8.6, supplementing Table S38* — the ten arm-cells of Table S38 that stood at n = 3 when the 2026-09-30 scan closed, re-read at **n = 10** after seeds 45–51 were trained on 2026-10-06 (42 runs on pod A at 30 epochs for the three small corpora, 28 runs on pod B at 100 epochs for the two `p_` cells). **The three earlier seeds are reused, not replaced**, so each cell pools seeds 42–51 and no run is counted twice. `prem_val` is from each run's own `results.csv` (`V_max - V_final`); `prem_test` is `T_best - T_final`, with `T_best` the archive's own end-of-training `test` reading on `best.pt` and `T_final` obtained here by scoring `last.pt` on the same held-out `test` split with the pipeline registered call (`val(split="test", batch=32, imgsz=640, plots=False, save_json=False)`, data YAML read from each run's own `args.yaml`); the two collections agree with the released per-run file to its printed rounding on the shared seeds, and all 56 transferred checkpoints load cleanly. WC = `prem_val - prem_test`; realization rate = `prem_test / prem_val` (means of the per-run values, in pp).

Cell	Arm	n	`prem_val`	`prem_test`	WC	realization rate	was (n = 3)
fire	baseline	**10**	+11.678	**+2.511**	+9.167	**22 %**	+10.953 / +2.820 / 26 %
fire	strategy	**10**	+12.218	**-0.842**	+13.061	**-7 %**	+8.869 / -0.264 / -3 %
mask20	baseline	**10**	+20.523	**+0.801**	+19.722	**4 %**	+16.071 / +0.325 / 2 %
mask20	strategy	**10**	+24.311	**-0.777**	+25.089	**-3 %**	+29.172 /

mende20	baseline	**10**	+4.243	**+2.533**	+1.710	**60 %**	+0.445 / 2 %
mende20	strategy	**10**	+6.006	**+1.302**	+4.704	**22 %**	+4.828 / +3.277 / 68 %
p_d15toai	baseline	**10**	+2.100	**+0.684**	+1.416	**33 %**	+5.682 / +1.512 / 27 %
p_d15toai	strategy	**10**	+1.547	**-0.234**	+1.781	**-15 %**	+0.488 / -0.154 / -32 %
p_masktomende	baseline	**10**	+6.301	**+2.775**	+3.526	**44 %**	+0.392 / -0.170 / -43 %
p_masktomende	strategy	**10**	+6.552	**+2.192**	+4.361	**33 %**	+2.335 / +2.022 / 87 %

> **What this table does, and what it does not do.** It **adds runs, not a new metric**: the same

> pipeline and the same `test` side as Table S38, on seven further seeds per cell, so that no cell in

> this table rests on n = 3. It does **not** re-define the pool, the confirmatory family or the claim

> level, and it does **not** revise Table S38's printed rows — those are what the 2026-09-30 scan

> measured at the *n* it had, and they are left standing. Scripts: `work/collect_test_final.py` (the

> `last.pt` test readings) and `work/_archive_results_csv.py` (the val curves); per-run file:

> `g6_prem_per_run_210_20261006.csv` (the released 140 rows plus these 70).

>

> **Availability of the three names just given (author-side, added 2026-10-10).** None of the three —

> `work/collect_test_final.py`, `work/_archive_results_csv.py`, `g6_prem_per_run_210_20261006.csv` —

> is in the released package: they are **author-side**, and this is stated here rather than left to be

> discovered, because this table's escalated readings cannot be recomputed from the released copy

> alone. What **is** released for this family is the earlier scan's per-run file

> `02_release_data/g6_prem_per_run.csv` (140 runs / 11 corpora, named in the manifest), from which the

> **pre-**escalation rows of Table S38 are recomputable.

Table S39 — the second intervention axis (`weight_decay` 1e-4 → 1e-3)

Table S39. *the second (weight-decay) axis, cell by cell* — both levels re-run fresh on one machine (A), `lr0` fixed at 0.001; `reference` = wd 1e-4 (the archived value), `treatment` = wd 1e-3

Cell	seed	reference	treatment	gain
smoke2sf	s42	+42.310	+41.880	**-0.430**
smoke2sf	s43	+41.930	+42.400	**+0.470**
smoke2sf	s44	+41.930	+41.750	**-0.180**
smoke2sf	s45	+42.060	+42.090	**+0.030**
smoke2sf	s46	+42.060	+42.570	**+0.510**
smoke2sf	s47	+42.670	+42.460	**-0.210**
smoke2sf	s48	+42.660	+42.740	**+0.080**
smoke2sf	s49	+42.470	+42.450	**-0.020**
smoke2sf	s50	+42.130	+42.220	**+0.090**
smoke2sf	s51	+41.860	+41.720	**-0.140**
shwd2sf	s42	+43.510	+42.950	**-0.560**
shwd2sf	s43	+42.770	+43.200	**+0.430**
shwd2sf	s44	+43.530	+43.510	**-0.020**
shwd2sf	s45	+42.870	+43.490	**+0.620**
shwd2sf	s46	+42.830	+43.700	**+0.870**
shwd2sf	s47	+43.140	+43.300	**+0.160**
shwd2sf	s48	+43.600	+43.290	**-0.310**
shwd2sf	s49	+43.180	+43.540	**+0.360**
shwd2sf	s50	+43.550	+42.890	**-0.660**
shwd2sf	s51	+43.690	+43.700	**+0.010**

smoke2sf: n = 10, sign 5+/5−, mean +0.020 pp, SD 0.293, sign-flip permutation p = 0.8535; the axis moves the absolute endpoint by SD 0.321 (range 1.020); second-axis Δgap mean -0.009 pp (SD 0.537, 6+/4−, permutation p = 0.9648); archive anchor mean |Δ| 0.362, max 0.810 pp.*

shwd2sf: n = 10, sign 6+/4−, mean +0.090 pp, SD 0.500, sign-flip permutation p = 0.582; the axis moves the absolute endpoint by SD 0.315 (range 0.930); second-axis Δgap mean +0.190 pp (SD 0.567, 7+/3−, permutation p = 0.3066); archive anchor mean |Δ| 0.442, max 1.110 pp.*

Table S40 — the third intervention axis (`mosaic` 1.0 → 0.0)

Table S40. *the third (augmentation) axis, cell by cell* — both levels re-run fresh on one machine per cell (`smoke2sf` on B, `shwd2sf` on A), `lr0` fixed at 0.001; `reference` = mosaic 1.0 (the archived default), `treatment` = 0.0

Cell	seed	reference	treatment	gain
smoke2sf	s42	+42.310	+41.200	**-1.110**
smoke2sf	s43	+41.930	+41.400	**-0.530**
smoke2sf	s44	+41.930	+41.270	**-0.660**
smoke2sf	s45	+42.060	+41.180	**-0.880**
smoke2sf	s46	+42.060	+40.580	**-1.480**
smoke2sf	s47	+42.670	+41.470	**-1.200**
smoke2sf	s48	+42.660	+40.410	**-2.250**
smoke2sf	s49	+42.470	+41.170	**-1.300**
smoke2sf	s50	+42.130	+40.820	**-1.310**
smoke2sf	s51	+41.860	+41.230	**-0.630**
shwd2sf	s42	+43.510	+41.810	**-1.700**
shwd2sf	s43	+42.770	+42.680	**-0.090**
shwd2sf	s44	+43.530	+42.660	**-0.870**
shwd2sf	s45	+42.870	+42.580	**-0.290**
shwd2sf	s46	+42.830	+42.140	**-0.690**
shwd2sf	s47	+43.140	+42.770	**-0.370**
shwd2sf	s48	+43.600	+42.400	**-1.200**
shwd2sf	s49	+43.180	+42.950	**-0.230**
shwd2sf	s50	+43.550	+43.020	**-0.530**
shwd2sf	s51	+43.690	+42.670	**-1.020**

smoke2sf: n = 10, sign 0+/10−, mean -1.135 pp, SD 0.510, sign-flip permutation p = 0.001953; the axis moves the absolute endpoint by SD 0.664 (range 2.260); second-axis Δgap mean +0.035 pp (SD 0.521, 6+/4−, permutation p = 0.8242); archive anchor mean |Δ| 0.362, max 0.810 pp.*

shwd2sf: n = 10, sign 0+/10−, mean -0.699 pp, SD 0.503, sign-flip permutation p = 0.001953; the axis moves the absolute endpoint by SD 0.501 (range 1.880); second-axis Δgap mean -0.520 pp (SD 0.677, 2+/8−, permutation p = 0.0293); archive anchor mean |Δ| 0.442, max 1.110 pp.*

Table S40b — the three intervention axes side by side

Table S40b. *the three intervention axes, cell by cell* — the estimand is the article's Table 2 one, `Δgap = (M_val(reference) − M_val(treatment)) − (M_test(reference) − M_test(treatment))`, paired by seed inside a cell and computed on one machine per cell; the `lr0` rows are article Table 2's, the other two axes are Tables S39 and S40. `p` is that axis' sign-flip permutation *p* of Δgap, except on the `lr0` rows, where it is the paired *t p* article Table 2 prints (that table's own convention); every *p* is raw and uncorrected. "Arm difference created" is the axis' own paired gain on the endpoint: an axis that moves the endpoint by no more than seed noise cannot say anything about whether the shift reproduces.

Axis (reference → treatment)	Cell	Δgap (pp)	*p*	direction	arm difference created	reproduces the corpus-dependent direction
`lr0` 0.001 → 0.005 (headline axis)	shwd2sf	**+0.779**	0.0053	`val` understates	**yes** (article Table 1)	**yes** — the four cells split 2 understate / 2 overstate
	smoke2sf	**+0.461**	0.0263	`val` understates	**yes**	
	p_aitovis	**−1.226**	1.3×10^{-8}	`val` overstates	**yes**	
	p_vistod15	**−3.315**	6.5×10^{-9}	`val` overstates	**yes**	
`weight_decay` 1e-4 → 1e-3	smoke2sf	−0.009	0.9648	6+/4−	**no** (+0.020 pp, *p* = 0.8535)	**uninformative**
	shwd2sf	+0.190	0.3066	7+/3−	**no** (+0.090 pp, *p* = 0.582)	**uninformative**
`mosaic` 1.0 → 0.0	smoke2sf	+0.035	0.8242	6+/4−	**yes** (−1.13	no

				5 pp, *p* = 0.001953)	
shwd2sf	**−0.520**	**0.0293**	**2+/8−**	**yes** (−0.699 pp, *p* = 0.001953)	**yes** — one cell at zero, one significantly negative

Which axis reproduces, and which one cannot say. Of the two intervention axes run besides the headline `lr0` one, `weight_decay` **never produced a comparable arm difference** — its own paired gain is +0.020 and +0.090 pp, the same order as seed noise — so its non-reproduction is **uninformative**, and the paper does not read "the phenomenon fails to cross axes" off it. `mosaic` did produce one (**0+/10−** in both cells, −1.135 and −0.699 pp) and its Δgap has the **same form as the `lr0` axis'**: one cell at zero, one significantly negative. The systematic shift is therefore **not specific to the learning-rate axis**, which is what section 12 of the article states and what this table now lets a reader check in one place.

Table S41 — the `p_vistod15` clean re-run, and what it does to the memory-pressure bound

Table S41. *10 (defect 10)* — the cell defect 10 flags, re-run on one machine with ample VRAM (one training lane per GPU), with the **archived script and the recorded configuration**: four `(arm, seed)` pairs, each **twice** (the clean re-run `clean1` and an independent replicate `clean2`). Test endpoints are mAP50-95 (pp). `|archive − clean1|` is the quantity a bound would be about (the article now reports it as **0.000 / 0.020 / 0.150 / 0.000 pp** and writes the event effect **unidentified** where there is no no-event control); `|clean2 − clean1|` is what an identical configuration reproduces at, on the same machine, the same day — the **observation-level reproduction residual**. Source: `work/G2_jitter_20260930.py`; per-run logs screened for the two defect-10 fingerprints (`work/g2_screen_20260930.py`) — **no hits**.

arm	seed	archive	clean1	clean2	\	archive − clean1\	* *\	clean2 − clean1\	*
---	---	---	---	---	---	---			

base100	s42	11.450	11.450	11.450	+0.000	**+0.000**
base100	s43	11.680	11.700	11.700	+0.020	**+0.000**
lr005_10ep	s42	15.190	15.040	15.040	+0.150	**+0.000**
lr005_10ep	s43	15.280	15.280	15.280	+0.000	**+0.000**

Table S24. *§9: the registered replication (the criterion is not met — the third row is the one that passes)* — columns: Registered pair, n, Δ (pp), 95 % CI, paired t, permutation p, same sign, Δ ≥ +0.30, p < 0.01, ≥ 8/10, unit criterion.

Registered pair	n	Δ (pp)	95 % CI	paired t	permutation p	same sign	Δ ≥ +0.30	p < 0.01	≥ 8/10	**unit criterion**
T1-a dota15→aitod	10	**+0.147**	[−0.036, +0.330]	1.816 (p = 0.1027)	0.123	6+/4−	✗	✗	✗	**FAIL**
T1-b aitod→visdrone	10	**+0.168**	[+0.083, +0.253]	4.455 (p = 0.0016)	**0.001953**	**10+/0−**	**✗**	✓	✓	**FAIL**
T1-c visdrone→dota15	10	**+3.727**	[+3.534, +3.920]	43.774 (p = 8.5×10⁻¹²)	**0.001953**	**10+/0−**	✓	✓	✓	**PASS**

Table S25. *§9: the registered replication (the criterion is not met — the third row is the one that passes)* — columns: Question the frozen criterion asks, Measured.

Question the frozen criterion asks	Measured		
p > 0.1? (should	Δ\ **no** (paired t	< 0.15 pp? (a saturated domain should be ≈ 0)	**no** (T2 mask→mendeley **−1.490 pp**)

be non-significant)	= −4.164, **p = 0.0141**, 0+/5−)	
A **positive** gain ≥ +0.30 pp with p < 0.01? (if present, the narrative collapses)	**no**	

Endpoint and floor (added 2026-10-01; corrected 2026-10-04). Every value in the Measured column above is the

`final`-epoch reading of the archive-A row (§7.3, Table S16). The frozen endpoint convention of §5 is the **`best`** reading, and the **released** `test_best_epoch` entry for this cell read **Δ = −0.902 pp** (**p = 0.2931**); that cell was corrected on **2026-10-02** (§S12): the entry had used one run's **`last`** reading in place of its `best` reading, and the corrected value is **Δ = −1.490 pp, p = 0.0141**. **So all three questions get the same answer under both endpoints** (|−1.490| ≫ 0.15 pp; p ≪ 0.1; a negative reading cannot be a positive gain): the table is **endpoint-independent** and the criterion **not met** either way. The p values here come from the paired **t** test; at n = 5 the permutation floor is **$2/2^5$ = 0.0625**, which is *above* the printed p = 0.0141 and is printed here as §11 recommendation ③ requires.

Table S26. *§9: the near-matched control C (the gain does not depend on a label-space change)* — columns: n, Δ (pp), 95 % CI, paired t, permutation p, same sign.

n	Δ (pp)	95 % CI	paired t	permutation p	same sign
10	**+0.727**	[+0.640, +0.814]	18.854 (p = **1.53×10^{-8}**)	**0.001953** (= the n = 10 floor)	**10+/0−**

Table S27. *9. The registered multi-target replication: transferred to the companion paper* — columns: Registered pair, data YAML, `val`, `test`, Split.

Registered pair	data YAML	`val`	`test`	Split
T1-b aitod→visdrone	`visdrone_20p.yaml`	`images/val`	`images/test`	**separated** ✓
T1-a dota15→aitod	`aitod_20p.yaml`	`images/val`	`images/val`	**same directory** ✗
T1-c visdrone→dota15	`dota15_20p.yaml`	`images/val`	`images/val`	**same directory** ✗

T2 mask→mendeley	`mende_20p.yaml`	`images/val`	`images/val`	**same directory** ✗
C dota→dota15 (control)	`dota15_20p.yaml`	`images/val`	`images/val`	**same directory** ✗

Where these configurations live. The four per-pair configs named above are archived under `eval_validity/yamls/`, each copy named after the path it had on its pod (for example `A__root__datasets__AI-TOD_yolo__aitod_20p.yaml`); that directory also carries a `MANIFEST.md` listing all 28 archived configs and, for each, whether `val` and `test` point at the same directory. The table above is a recomputation from those files, not a restatement.

Table S28. *5.1 (the dota15 escalation, X8)* — columns: seed, premium baseline, premium strategy, paired difference. Ten seeds, from n = 3 to n = 10. Premium = best-epoch minus final-epoch mAP50-95 on the run's own monitored split; source: the twenty run directories in `x8_dota15_20260918/runs/` (10 seeds x 2 arms; **16 of them trained for this round**, the seeds-43/44 pairs pre-existing).

seed	premium baseline	premium strategy	**paired difference**
42	+0.997	+0.370	**+0.627**
43	+1.004	+0.183	**+0.821**
44	+0.841	+0.317	**+0.524**
45	+0.775	+0.585	**+0.190**
46	+0.997	+0.740	**+0.257**
47	+1.027	+0.658	**+0.369**
48	+0.986	+0.492	**+0.494**
49	+0.952	+0.407	**+0.545**
50	+0.941	+0.715	**+0.226**
51	+0.891	+0.454	**+0.437**
mean (n = 10)	**+0.941** (sd 0.082)	**+0.492** (sd 0.182)	**+0.449** (sd 0.196, t = 7.23, p = 4.924e-05)

> **The criterion was fixed before the runs existed**: a ten-seed paired difference failing to be

> positive with **≥ 8/10 same-sign** and a **95 % CI excluding zero** would remove `dota15` from the

> four-cell headline and force the abstract and Tier 1 to be rewritten. Measured: **+0.449 pp**,

> **10+/0−**, CI **[+0.309, +0.589]** — met as stated.
>
> **Recipe identity.** All twenty `args.yaml` files of the cell (the sixteen new runs plus the two
> pre-existing seeds, both arms) are identical in every recipe field: same pretrain
> (`dota15_pretrain.pt`), same data YAML (`dota15_20p.yaml`), 100 epochs, `patience` 100, batch 32,
> imgsz 640, `save_period` 5, SGD, `lr0` 0.001 (baseline) versus 0.005 (strategy). Old and new runs
> are therefore poolable, and the seed is recorded in every directory name.
>
> **GPU-memory fallback, and its sensitivity.** 10 of the 16 newly trained runs printed
> `CUDA OutOfMemoryError in TaskAlignedAssigner, using CPU` (92 events; 79 of them in
> `dota15_lr005_100ep_s51n`), while the per-epoch sweep of §8.7 shared the GPU. This is an
> implementation-level fallback inside one batch, not a change of objective, and it is reported
> rather than smoothed: dropping each seed pair in turn leaves the criterion satisfied in all ten
> subsets (worst lower bound **+0.277**), and dropping the two most affected pairs (s51: 79 events,
> s42: 7) gives **+0.428 pp**, CI **[+0.252, +0.605]**, 8+/0−. Scripts:
> `work/x8_dota15_analysis_20260918.py`, `work/x8_oom_counts_20260918.py`,
> `work/x8_oom_sensitivity_20260918.py`.

Table S29. *5.1 (the two `val`-overstating cells, X1)* — columns: seed, Δgap for `p_aitovis`, Δgap for `p_vistod15`. Ten seeds per cell, from n = 3 to n = 10. Δgap = (arm difference on `val`, at each run's own val-curve maximum) − (arm difference on `test`, at `best.pt`); paired within seed. Sources: `x1_tier2_20260918/runs/` (40 run records) and `teval.csv` (80 test evaluations, SHA-256 verified against the machine that produced them).

seed	Δgap `p_aitovis`	Δgap `p_vistod15`

42	-1.127	-3.131
43	-0.843	-4.563
44	-1.358	-3.058
45	-1.495	-2.819
46	-1.267	-3.468
47	-0.960	-3.453
48	-1.280	-2.952
49	-1.341	-3.590
50	-1.404	-2.995
51	-1.182	-3.118
mean (n = 10)	**-1.226**	**-3.315**
95 % CI	[-1.370, -1.081]	[-3.676, -2.953]
paired t / p	-19.17 / 1.3e-08	-20.76 / 6.5e-09
sign	0+/10−	0+/10−
sign-flip permutation p	**0.001953**	**0.001953**

The `paired t / p` row is **raw** (no multiplicity adjustment); the permutation p is exact at this cell size and is printed as its attainable floor. The two are separate tests and are not combined.

> **What the comparison is, exactly.** The plan fixed before the runs: 40 training runs

> (seeds 42–51 × 2 arms × 2 cells), paired *and* sign-flip tests, `{best, last} × {val, test}`, and a

> prespecified correction; the strong branch of the pre-declared outcome rule was *"both remaining

> negative and significant with permutation p ≤ 0.002"*, the weak branch being that the abstract's

> "direction is not single-valued" must be withdrawn or limited. All ten seeds of each cell are

> negative and the permutation p is $2/2^{10}$ = **0.001953**, the smallest value attainable at n = 10

> (§7.1), so the strong branch applies. **Holm across the two targets** (the prespecified

> correction) multiplies the larger p by 2 and the smaller by 1: 2.6×10^{-8} and 6.5×10^{-9}, both ≤ 0.05.

>

> **The endpoint was identified, not assumed.** The archived `sio_b_results.csv` that the published

> Table 2 used for the n = 3 values carries no endpoint convention (§7.3, disclosed defect 3). Our

> own evaluation of `best.pt` on `test` reproduces its values for the twelve runs it covers to

> **0.0046 pp** (median 0.0023) — so that archive records **the best checkpoint evaluated on `test`**,

> and the n = 3 → n = 10 comparison is like-for-like rather than a change of measurement. Under the

> same construction restricted to seeds 42–44 the two cells give −1.110 and −3.584 against the

> published −1.114 and −3.580.

>

> **Both arms and all four quantities.** Per cell (baseline / strategy means, n = 10): `val`-curve

> maximum 11.242 / 12.657 and 16.963 / 23.927; `best.pt` on `test` 10.210 / 10.396 and 11.626 / 15.263.

> The cells' realization rates (§8.6's quantity, computed the same way) are **82 % / 104 %** and

> **56 % / 29 %**, pooled **55 %** — the opposite end from §8.6's aggregate realization rate, which is why the article says

> the realization rate is not a constant across corpora. The exact three-component decomposition of

> each Δgap is in Tables S10's conventions: for `p_aitovis` −1.226 = (−1.583) + (+0.090) + (+0.267)

> and for `p_vistod15` −3.315 = (−7.072) + (−0.213) + (+3.971); both identities check to the printed

> digit. Scripts: `work/x1_gap_analysis_20260918.py`, `work/x1_aux_20260918.py`.

Table S37. *3.2* — the six pre-registered experiments and the criterion each was judged against. **G1 is void** (design confounded), so no verdict is entered for it; **two of the five judged experiments are negative — G1′ and G2** — and one design had to be redone, all reported as such. Counts are over experiments, not criteria. The values behind every verdict are in Tables S31–S33 (the G1 family in S12).

Experiment (design)	Pre-declared criterion	Verdict
G1 aliasing	none entered: design	**void** — the pair

calibration, disjoint split	confounded	differed by 14.7 pp in difficulty; redone as G1′ (§S12)
G1″ three-way carve levels	the paired levels differ by ≤ 3 pp	**met** (−0.122 pp; §S12)
G1′ aliasing, level-matched pair	WC > 0, paired t, p < 0.05	**not met** (p = 0.83, n = 10; §S12)
G2 selection-split size, 30 vs 90	the rate rises with size; `WC_small/WC_large` ∈ [1.4, 2.1]	**direction met, magnitude not met** (pooled ratio 1.19; Table S31)
G3 epoch budget `E`	C1 rise in `E`; C2 the σ√(2 ln E) form in [0.86, 1.43]; C3 shape; C4 WC on a disjoint readout	**revised** (§3.1, S32 · S34b, **clean re-run**): C1 met, C2 out of band, so the scaling stays withdrawn; C3 passes on the mean only; C4 not met · cross-lineage direction also probed on a **second detector family** (RetinaNet, two corpora): S34e
G4 third corpus, `mende20_3way`	identity value by value; WC > 0	**met** (WC = +5.21 pp, CI [+4.82, +5.66], 20/20; Table S33)

Table S31. *3.2 (G2)* — the selection split's **size** is the only thing that changes: two nested validation splits drawn from one verified-disjoint pool (30 and 90 images), the same training set, and an identical test side. Columns: selection split, arm, n, `prem_val`, `prem_test`, WC, realization rate. Source: `g2_valsize_20260919_runs/`, `g2t_g1p_20260919/g2_test.csv`.

selection split	arm	n	`prem_val`	`prem_test`	WC	realization rate
small (30)	base100	5	+2.747	+0.646	+2.101	**23.5 %**
small (30)	lr005_100ep	5	+1.258	+0.064	+1.194	**5.1 %**
large (90)	base100	5	+2.549	+0.688	+1.861	**27.0 %**
large (90)	lr005_100ep	5	+1.062	+0.165	+0.897	**15.6 %**

> **Pre-declared criteria.** Direction: the realization rate rises with the selection split's size — **met** (the two arms move the same way). Magnitude: `WC_small / WC_large` within [1.4, 2.1] (√3 ≈ 1.73) — **not met** (measured 1.19, pooled over the two arms), so the 1/√n step is a scale estimate, not a law.

Table S32. *3.2 (G3)* — the epoch budget is the only thing that changes (four budgets, E = 50–400, same corpus, recipe and validation split). Columns: arm, n, premium at E = 50, at E = 100, at E = 200, at E = 400, ratio 400/100. Source: `g3_epochs_runs/`. **Two bases appear in this table and its note, and they are labelled:** the E = 400 and ratio columns report the **six full runs** at that budget, while the C2 ratio **1.329** and the all-runs C1 verdict use the **ten runs** at E = 400 (arm-equal mean **+0.907 pp**); four of the ten stopped early under the protocol's own `patience`. **n = 5 seeds per arm at E = 50–200**; the E = 400 and ratio columns report the **six full runs** of the ten at that budget, which is why the table row and the note's all-ten arithmetic are both stated.

arm	n	premium (E = 50)	premium (E = 100)	premium (E = 200)	premium (E = 400; **six full runs**)	ratio 400/100 (**six full runs**)
base	5	+0.724	+0.897	+1.187	+1.315	×1.47
lr005	5	+0.163	+0.468	+0.756	+1.241	×2.65

> **Pre-declared criteria (C1–C3, fixed before the runs). C1 — monotone rise of the premium with E — is not met**: **on all ten runs** the four-budget series peaks at E = 200 and falls at E = 400 (four of the ten E = 400 runs stopped early at effective budgets 388/318/362/375; on the six full runs the E = 400 mean is +1.278 pp, which is the column this table prints). The ln E regression over E = 50–200 gives slope **+0.3990** (se 0.0592), **p = 9.466e-08**. **C2 — the σ√(2 ln E) form — is not rejected**: premium(400)/premium(100) = **1.329** (the E = 400 arm-equal mean over all ten runs is **+0.907 pp**) against the predicted 1.141, **inside** the pre-declared band [0.86, 1.43]. **C3 — the shape control — passes** (ê/E 0.833 → 0.815). The extension (E = 50 and 400) was added for this revision; it uses the same corpus, recipe, seeds and arms as the E = 100/200 runs, and only `--epochs` differs. **The early-stopped runs decide the outcome**: the four E = 400 runs that stopped under the protocol's own `patience` (effective budgets 388/318/362/375) are what makes the series non-monotone; dropping them flips **both** criteria — the six full E = 400 runs run to a mean of +1.278 pp on a monotone series (C1 met) and move the ratio outside the band (C2 rejected). Both readings are conditional on that restriction and are reported as such; the pre-declared verdict above is the one computed on **all** runs. **This note**

supersedes an earlier reading of the two-budget series, which reported only three budgets and mislabelled them as monotone; the four-budget verdict is in S11 and §3.2. The table below is the with-early-stopping reading; the same four budgets re-run on one machine **with early stopping disabled** satisfy C1**, put** C2 outside its band **and** pass C3 on the pre-declared mean reading **(only the lr005 arm exceeds it) — see Table S34b.**

Secondary criterion C4 (WC on a disjoint readout) — not met, and systematically so. With ê selected on `val` and read out on the disjoint third C3 (100 images; `C3 ∩ val = 0` and `C3 ∩ train = 0` verified programmatically), WC = `prem_val` − `prem_C3` is **negative in all 16 readouts** (mean **-0.616 pp**): the peak-to-endpoint gap is *larger* on the small disjoint split than on `val`. That is consistent with the shape reading above — the premium is largely the split's own curve shape, and a 100-image split has a steeper curve — and it means Proposition 4's inequality **cannot be tested by reading out on a small disjoint carve**. Two limitations are stated with the result: the readout split has 100 images only, and the four early-stopped E = 400 runs are excluded, leaving n = 6 at that budget.

Table S33. *3.2 (G4)* — replication on a third corpus (`mende20_3way`, 2 arms × 10 seeds, test side evaluated separately, 40 evaluations). Columns: run, `prem_val`, `prem_test`, WC, realization rate. Source: `g4_mende3way_runs/`, `g4_mende3way_20260919_g4_test.csv`.

run	`prem_val`	`prem_test`	WC	realization rate
base100_s42n	+8.873	+3.384	+5.489	38.1 %
base100_s43n	+8.309	+2.701	+5.608	32.5 %
base100_s44n	+7.731	+3.783	+3.948	48.9 %
base100_s45n	+7.818	+2.796	+5.022	35.8 %
base100_s46n	+7.257	+2.481	+4.776	34.2 %
base100_s47n	+7.608	+3.666	+3.942	48.2 %
base100_s48n	+7.534	+3.046	+4.488	40.4 %
base100_s49n	+8.185	+4.312	+3.873	52.7 %
base100_s50n	+8.456	+3.834	+4.622	45.3 %
base100_s51n	+7.671	+2.930	+4.741	38.2 %
lr005_100ep_s42n	+9.613	+4.893	+4.720	50.9 %
lr005_100ep_s43n	+9.450	+3.434	+6.016	36.3 %
lr005_100ep_s44n	+6.249	+0.416	+5.833	6.7 %
lr005_100ep_s45n	+8.592	+2.205	+6.387	25.7 %
lr005_100ep_s46n	+8.528	+1.960	+6.568	23.0 %
lr005_100ep_s47n	+9.957	+5.044	+4.913	50.7 %
lr005_100ep_s48n	+7.300	+1.684	+5.616	23.1 %

lr005_100ep_s49n	+12.210	+4.314	+7.896	35.3 %
lr005_100ep_s50n	+7.976	+2.983	+4.993	37.4 %
lr005_100ep_s51n	+7.355	+2.563	+4.792	34.8 %

> **Result.** WC = **+5.212 pp**, 95 % bootstrap CI [+4.822, +5.655], positive in **20 of 20** runs — the same direction as the 81 runs of §8.6 and the 40 of §5.1. No pass/fail is set for the rate itself (Proposition 4 bounds only its expectation).

Table S34 — cross-lineage realization test (non-YOLO detector)

Table S34. *6.5* — the mechanism outside the YOLO lineage. A **Faster R-CNN (ResNet-50 FPN)** detector (torchvision `fasterrcnn_resnet50_fpn`), a family that **reports the final epoch by default** (the cross-framework audit above), trained from COCO-initialised weights on the same **three-way protocol** as the rest of this paper: train = a 20 % subset (1,294 images), selection (`val`) = a **verified-disjoint carve** (500), readout (`test`) = the corpus's held-out split (1,610). 40 epochs, batch 8, SGD lr 0.005 with a 1,000-iteration warmup, 640-px input, **six seeds**. `best` is chosen by the **`val`** curve's argmax; `test` is inferred **once** for `best` and once for `last`, and evaluated with the **same** COCO implementation used on the YOLO side (IoU 0.50:0.95, 101-point interpolation). Source: `nonyolo_realization.json` (the original run; **author-side, not part of the released package**) and `nonyolo_settle_20260925/` (`settle_all.json`, `nonyolo_n6_aggregate.json`; the five added seeds are aggregated by `work/aggregate_nonyolo_n6_20260925.py`). The per-epoch `val` predictions kept by the training run are written one file per epoch under the run's prediction directory, from epoch 1 to epoch 40.

`maxDets`	n	`prem_val` mean	`prem_test` mean	realization rate (ratio of means)	95 % CI
100	6	**+0.0809**	**+0.0551**	**68.2 %**	[26.9 %, 123.5 %]
300	6	+0.0809	+0.0551	68.2 %	[26.9 %, 123.5 %]
1000	6	+0.0809	+0.0551	68.2 %	[26.9 %, 123.5 %]

Per-run detail (seed 42 is the run reported in the previous version; it is the *first* of the six, not a separate experiment). Rates are `prem_test / prem_val` per run, so they vary far more than the aggregate:

run	`best` ep	`prem_val`	`prem_test`	per-run rate

original (seed 42)	30	+0.0932	+0.0641	68.8 %
seed 43	27	+0.0753	+0.0422	56.1 %
seed 44	33	+0.0670	+0.0147	22.0 %
seed 45	24	+0.1017	+0.1777	**174.8 %**
seed 46	28	+0.0315	+0.0009	2.8 %
seed 47	21	+0.1165	+0.0310	26.6 %

Table S34b — the G3 clean re-run (early stopping disabled, four budgets, one stack)

Table S34b. *3.2 (G3, clean re-run)* — the four-budget experiment re-run with the protocol's early stopping **disabled** (injected at the call site; the training script is byte-identical, md5 `316e074050dde23cde33e4a223ec46a8`). Same corpus, recipe, two arms and five seeds as G3; only the `patience` setting differs. The whole four-budget series was run **on one stack**: in the earlier clean batch one of its runs was silently degraded under memory pressure, which is why the series was repeated rather than patched. **40/40 runs completed**, each to its configured budget. The E = 50/100/200 rows coincide with the published budget series because no run early-stopped at those budgets; only E = 400 changes. Source: `g3cleanb_20260925/` (per-run `results.csv`; recomputation `work/clean_series_b_20260925.py`, output `work/_clean_series_b_20260925.txt`).

`E`	arm	n	premium mean (pp)	seed sd	bootstrap 95 % CI	`ê/E`
---	---	---	---	---	---	---
50	base	5	**+0.724**	0.094	[+0.663, +0.806]	0.760
50	lr005	5	**+0.163**	0.176	[+0.030, +0.298]	0.876
100	base	5	**+0.897**	0.101	[+0.820, +0.974]	0.788
100	lr005	5	**+0.468**	0.220	[+0.290, +0.633]	0.878
200	base	5	**+1.187**	0.084	[+1.119, +1.250]	0.813
200	lr005	5	**+0.756**	0.163	[+0.628, +0.900]	0.909
400	base	5	**+1.295**	0.083	[+1.222, +1.349]	0.760
400	lr005	5	**+1.248**	0.085	[+1.180, +1.317]	0.814

> **Verdict under the pre-declared criteria, on the clean re-run series.** The arm-equal means are

> **+0.4439 / +0.6824 / +0.9712 / +1.2717**, i.e. **monotone non-decreasing ⇒ C1 is *met*** (per-run
> regression on ln E: slope **+0.3999**, se **0.0503**, t = **7.950**, p = **1.325×10^{-9}**, n = 40),
> where the with-early-stopping reading scored **not met**. **C2 is out of band**:
> premium(400)/premium(100) = **1.8636** against the predicted 1.141, outside the pre-declared
> [0.86, 1.43] ⇒ the σ√(2 ln E) scaling **stays withdrawn** and the level reads as shape-driven.
> **C3 — the shape control — passes on the pre-declared *mean* reading** (ê/E **0.833 → 0.787**,
> |Δ| = **0.046** ≤ 0.05) and **fails per arm** (lr005 |Δ| = **0.065**; base |Δ| = **0.028**); the
> pre-registered criterion names the mean, so both readings are reported side by side and neither is
> dropped. **The registered criteria still forbid reporting the rise as an `E` effect**: the peak's
> relative position moves **in one arm and not the other** across budgets (base 0.760 → 0.760, lr005
> 0.876 → 0.814), which is the movement C3 exists to catch.
> **Superseded reading, kept on the record (first clean batch).** That batch carried only
> E = 50 and E = 400 and was split across two hosts; its per-arm values were

`E`	arm	n	premium mean (pp)	seed sd	bootstrap 95 % CI	`ê/E`
50	base	5	**+0.672**	0.180	[+0.526, +0.820]	0.744
50	lr005	5	+0.094	0.110	[+0.013, +0.185]	0.928
400	base	5	**+1.298**	0.082	[+1.223, +1.343]	0.836
400	lr005	5	**+1.211**	0.284	[+1.058, +1.469]	0.725

> and its verdict read: *using the published E = 100 and 200 means (0.682, 0.971), the four-budget
> series is **monotone** (0.383 / 0.682 / 0.971 / **1.255**) and the ln E regression gives slope

> **+0.419** (se 0.060), **t = 7.03, p ≈ 2×10⁻¹²** ⇒ the monotone-rise criterion **met**, where the

> with-early-stopping readings scored **not met**. The **shape control fails**: `max_arm |ê/E(400) −

> ê/E(50)| = 0.203` against the pre-declared 0.05, and the movement is **opposite in the two arms**

> (lr005 falls 0.928 → 0.725, base rises 0.744 → 0.836). *Those values are replaced** by the

> single-machine table above; they are retained so that the earlier reading can be checked against its

> replacement. Two things about that batch are the reason for the re-run, and both are on the record:

> one of its runs was **silently degraded under memory pressure** (the training path changed without a

> marker, recorded in this revision's run audit), and its comparison points were mismatched. **Its being

> split over two hosts is *not* among the reasons** — the A and B machines calibrated as equivalent. Its shape control

> also compared **E = 400 with E = 50**, because that batch had no E = 100 point — a mismatch of

> comparison points against the pre-declared `ê/E(400) − ê/E(100)`, and the reason its "failure" and

> the single-machine "pass on the mean" are not in conflict.

> **Result.** The `val`-selected premium is **partly realized** outside the YOLO lineage

> (`prem_test` = **+0.0551 pp** against `prem_val` = **+0.0809 pp**, **68.2 %**, 95 % CI

> **[26.9 %, 123.5 %]**), in the **same direction** as the five arm-cells of §8.6 — and in the same

> direction in **all six runs** (`prem_val > 0` and `prem_test > 0` in 6/6; no run has a negative rate).

> All three `maxDets` conventions give **identical** numbers, confirming (for this corpus) that the

> parameter is not binding here.

>

> **Pre-declared criteria for this escalation** (fixed before the five added seeds were run):

> ① same direction as the original single run; ② a 95 % interval reported; ③ *if that interval contains

> zero*, the wording stays "an anecdote". ① met, ② met, ③ not triggered** — the interval excludes

> zero — so this table is upgraded from an anecdote to an estimate **with** an interval.

>

> **Limits. One detector, one corpus**, six seeds — so the result establishes that the

> mechanism is **not confined to the YOLO lineage**, and **not** that non-YOLO pipelines realize ~68 %

> generally. The aggregate's agreement with the single-run value (68.8 %) is a **coincidence of a wide

> per-run spread**: individual rates run from** 2.8 % to 174.8 %**, one of them above 100 %. The

> realization rate also has a **small denominator** (mean `prem_val` = +0.0809 pp; the original run's was

> +0.0932 pp), so the ratio is far more

> sensitive to single-run variation than the directional statement `prem_test > 0`. The absolute

> premium is an order of magnitude below the YOLO-side cells of §8.6 (+0.865 … +1.469 pp) and

> **must not be compared with them**: the detector, corpus, dataset scale and absolute mAP level all

> differ. The evaluation-side self-test (`ground truth as predictions` = **0.990099** at

> `maxDets = 1000`, 0.962854 **at 100) is reported in the project log for that pass (2026-09-16).** Why it is not

> exactly 1.0.** Feeding the ground truth back in is not capped by the interpolation grid: measured locally

> (`work/c23_eval_ceiling_20261001.py`, `pycocotools`, gt = predictions, score 1.0), a **one-box self-test reads
> exactly 1.000000 **at both `maxDets` settings. The 101-point grid turns a** recall cap** `r` into
> `round(101 r)/101` -- a self-test with 100 detections against 150 ground-truth boxes reads **67/101 =
> 0.663366 **(measured) -- so the logged** 0.990099 = 100/101** says this pass's maximum recall was
> **100/101 ~ 0.9901**: one ground-truth box went unmatched. It is a **shortfall to investigate, not the
> evaluator's ceiling**; the lower reading at
> `maxDets = 100` is the evaluator's own cap truncating a crowded image's boxes. Neither figure is
> evidence about the detector — it is a check on the evaluation stack, and it is **1.0 up to that
> ceiling**.

Figure and table inventory (moved here from the article, B3, 2026-09-26). The article carries its argument, its four headline tables and its disclosure; this is the map of which tables and figures are printed there and which live here. It sits at the end of §S8 so that it directly follows the moved tables it indexes.

Where the evidence lives. The article keeps its claims, argument, **four headline tables** and its disclosure; the other tables are in the Supplementary **unchanged**, and **four of the six figures are printed in the article** — §6.1 (Fig. 1), §8.6 (Figs. 2 and 3) and §8.7 (Fig. 4) — with the remaining two in §S9, **and the printed numbers map to the deposited files as `Fig. 1` = `fig3_four_units.png`, `Fig. 2` = `fig2_realization_rate.png`, `Fig. 3` = `fig6_sigma_vs_rate.png`, `Fig. 4` = `fig5_epoch_curve.png`, `Fig. S1` = `fig1_dgap_decomposition.png`, `Fig. S2` = `fig4_public_logs.png`**, so a reader can pair every caption with its file, every number backed by an artifact pointer. Four tables are printed **in the article** too, each carrying a headline number to check:

In this article	§S8 counterpart	What it settles
Table 1	Table S4	**Tier 1** — premium per arm and the paired difference

Table 2	Table S6	**Tier 2** — Δgap by corpus, with its reading
Table 3	Table S11	the four counting units over the 19 audited benchmarks
Table 4	Table S20	the premium selected on `val`, and what it realizes on `test`

Everything else is in §S8–§S9, keyed by section, including **S24–S27** (the registration, now the companion's), plus **three tables this revision adds in §S12 rather than moves from the article** — **Table S35** (the estimand and scope map) and **Table S36** (the analysis-status ledger), and **Table S37** (the article's former Table 5: the six pre-registered experiments, their criteria and their verdicts, moved here in this revision) — **S35 and S36 are not rows of the §S8 index** (which lists the tables moved from the article); **S37 is**, having been moved from the article in this revision. The six figures are:

Figure	Main-text section
Fig. 1	**this article**, §6.1
Fig. 2	**this article**, §8.6
Fig. 3	**this article**, §8.6
Fig. 4	**this article**, §8.7
Fig. S1	Supplementary §S9, §5.6
Fig. S2	Supplementary §S9, §5.5

Not in the index above. Seven tables in this section were not moved from the article and are therefore not rows of the index: **S34** (cross-lineage realization test, non-YOLO detector), **S34b** (the G3 clean re-run, early stopping disabled, one machine, four budgets), **S34c** (the cross-lineage test on a second corpus, SFCHD, and the carve that gives it a disjoint selection set), **S34e** (the cross-lineage test on a

second detector family, RetinaNet, run on both corpora, added 2026-10-06), **S38** (the selection term, cell by cell, for all thirteen scanned corpora), **S39** and **S40** (the second and third intervention axes, `weight_decay` and `mosaic`), **S40b** (the three intervention axes side by side, added 2026-10-01) and **S41** (the `p_vistod15` clean re-run, and what it does to the memory-pressure bound). Two further tables of the same kind live in §S12: **S35** (the estimand and scope map) and **S36** (the analysis-status ledger).

Numbering. Two numbers are not used anywhere in this supplement: section **§S10** and **Table S30**. The numbering is not contiguous, and nothing is missing from the file under either number.

§10's defect table is **kept in the article**, deliberately: disclosure belongs in the article, as do first-report attribution and methodological premises. The Supplementary's §S6 records the **shared process protocols** — the same set of protocols used by this paper and its companion — with their file hashes.

Table S34c — cross-lineage realization test, second corpus (SFCHD, non-YOLO detector)

Table S34c. *6.5* — the same question as S34, asked of a **second corpus**, so that the cross-lineage reading rests on direction agreement across two corpora rather than on one. Detector, recipe and evaluation are held fixed: torchvision `fasterrcnn_resnet50_fpn`, COCO-initialised, 640-px input, batch 8, SGD lr 0.005 with a 1,000-iteration warmup, cosine schedule, 40 epochs, **one run per GPU**, `best` by the `val` curve's argmax, `test` inferred **once** for `best` and once for `last`. The probe lineage is our own `run_3way_v2.py` (a seed-carrying variant of the S34 script); the two differ in warmup and in the final-epoch endpoint, so the two tables are **direction-comparable, not magnitude-comparable** — consistent with the S34 column being no basis for a cross-detector comparison either.

Corpus: **SFCHD** (two classes, `hat` / `person`). SFCHD is one of the six corpora whose released configuration gives `val` and `test` the **same directory** (**S1**, row 13) — that is, one of the corpora this article counts as lacking an independent held-out split. Rather than inherit that, we **carved a verified-disjoint selection set** from its 20 % training pool (2,413 images) by a **fixed permutation seed** (`SEED=20261002`, written into the carve script; first 400 images), leaving train 2,013 and keeping the readout split (6,033) **read-only**; the three sets are pairwise disjoint (asserted in the script). The corpus was carved **twice, independently, on the two hosts** with that same script and seed (they calibrated as equivalent, so this is a **positive control**, not a confound), and the two carve manifests are **identical line by line** (md5 `8be82f1f1bfb41a0cd5b4f96cbaa356c`), so the 12 runs are one corpus, not two. Positive control: with ground truth as predictions, every checkpoint reaches **0.990099** (ideal 1/1.01) at `maxDets=1000` — the caliper used in S34 — and no reading is written if it fails. Per-image detections do not reach the `maxDets` cap on this corpus, which is why all three tiers agree.

`maxDets`	corpus	n	`prem_val` mean (SD)	`prem_test` mean (SD)	sign, `val` / `test`

| --- | --- | --- | --- | --- | --- |
100	SFCHD	12	+0.5216 (0.1544)	+0.1233 (0.1144)	12/0 · 11/1
300	**SFCHD**	**12**	**+0.5216** (0.1544)	**+0.1282** (0.1150)	**12/0 · 11/1**
1000	SFCHD	12	+0.5216 (0.1544)	+0.1233 (0.1144)	12/0 · 11/1
300	VisDrone (S34, for direction only)	6	+0.0807 (0.0390)	+0.0383 (0.0868)	6/0 · 5/1

Pre-declared reading (fixed before the runs). **C3 — direction agreement — is met**: both corpora have a positive `prem_val` mean and a positive `prem_test` mean, and each corpus is same-signed in at least five of its six runs (VisDrone 6/6 and 5/6; SFCHD 12/12 and 11/12). **Wording cap**: this licenses "two-corpus cross-lineage probe" and nothing stronger; the detector is still one.

Limits. Direction only: the corpora differ by an order of magnitude in absolute mAP50-95 (≈8 versus ≈35), so their premiums are not comparable. The VisDrone `prem_test` interval **includes zero** (**[−0.022, +0.103]**, bootstrap), so the directional verdict rests on the **sign counts** and on SFCHD; the per-run spread is wide (individual `prem_test` **−0.0853 … +0.3097 pp**), no host effect is present (`prem_val` correlation **+0.040** across the two hosts, n = 6 shared seeds — consistent with the A/B equivalence finding), and removing any single run leaves every corpus/arm mean same-signed. Two runs differ between S34 and S34c that are worth naming: S34's six runs are the original run plus five seeds aggregated at the time, whereas S34c's twelve runs were all run under `run_3way_v2.py`; the two VisDrone `prem_val` means (+0.0809 and +0.0807) agree to **0.0002 pp**, which we report as consistency, not as replication.

Table S34e — cross-lineage realization test, second detector family (RetinaNet, two corpora)

Table S34e. *6.5* — S34 and S34c hold the detector fixed at `fasterrcnn_resnet50_fpn` and vary the corpus; this table does the converse, holding **both corpora** and the three-way protocol fixed while changing the detector to a **different family**: torchvision `retinanet_resnet50_fpn_v2` (anchor-based one-stage with FPN, COCO-initialised). Recipe held to the S34c probe: 640-px input, batch 8, SGD lr 0.005, 1,000-iteration warmup, cosine schedule, 40 epochs, **one run per GPU**, `best` by the `val` curve argmax, `test` inferred once for `best` and once for `last`; `maxDets = 300`, the script default

used by S34c. Twelve seeds per corpus (42–53). The pre-flight caliper is applied unchanged: ground truth as predictions must reach **0.990099** (ideal 1/1.01), and no reading is written if it fails. Source: our `work/run_3way_family.py`, results `family_out/*.json`, aggregated by `work/collect_optionC.py`; `prem = V_max − V_final` / `T_best − T_final` in pp, and the rate is a **ratio of means** with a 2,000-draw bootstrap interval on that ratio (the S34 convention).

corpus	n	`prem_val` mean (SD)	`prem_test` mean (SD)	realization rate	95 % CI	sign, `val` / `test`
VisDrone	12	+0.0630 (0.0510)	+0.0710 (0.0786)	**112.6 %**	[48.8 %, 211.9 %]	10/2 · 8/4
SFCHD	12	+0.2887 (0.2761)	+0.0566 (0.1190)	**19.6 %**	[−3.7 %, +41.6 %]	11/1 · 7/5

Reading, and the wording cap. The sign pattern of S34/S34c reappears on **SFCHD**: a positive `prem_val` mean with a **smaller** `prem_test` mean, i.e. the selection gain does not fully carry to the disjoint readout. **On VisDrone it does not** — there `prem_test` (mean **+0.0710**) exceeds `prem_val` (**+0.0630**). **The two corpora do not agree on strength**: on VisDrone the realized premium is *not* discounted (rate **112.6 %**, interval excluding zero), whereas on SFCHD the rate is **19.6 %** and its interval

includes zero — so on this family the discount is visible on one corpus and not on the other. That is a weaker statement than S34c's, and it is the statement the data support: **the cross-family reading here is direction, not magnitude, and not a replication.** As in S34c, the corpora differ by an order of magnitude in absolute mAP50-95 (≈6 versus ≈34), so their premiums are not comparable to each other or to S34/S34c's.

> **Placement in the S34 series.** S34 = one detector, first corpus; S34c = same detector, second corpus;

> **S34e = second detector family, both corpora**. Read together the three tables say: the direction (a

> positive `val` premium that a disjoint readout realizes only in part) survives a corpus change and a

> **detector-family change**, while the **strength** of the discount does not — it is corpus-dependent in

> S34/S34c and corpus-dependent again here.

Per-run detail (`prem` in pp; two runs per corpus peak at the final epoch, so their `prem_val` is exactly 0 and their per-run rate is undefined — they are counted in the means and excluded only from the per-run rate column):

corpus	seed	`best` ep	`prem_val`	`prem_test`	per-run rate
VisDrone	42	31	+0.0723	+0.0609	+84.2 %
VisDrone	43	34	+0.1412	-0.0071	-5.0 %
VisDrone	44	38	+0.1115	+0.0183	+16.4 %
VisDrone	45	28	+0.1351	+0.2283	+169.0 %
VisDrone	46	30	+0.0650	+0.1206	+185.5 %
VisDrone	47	40	+0.0000	+0.0000	— (peaks at final epoch)
VisDrone	48	23	+0.0147	+0.1870	+1272.1 %
VisDrone	49	25	+0.0848	+0.1057	+124.6 %
VisDrone	50	26	+0.0314	+0.0917	+292.0 %
VisDrone	51	40	+0.0000	+0.0000	— (peaks at final epoch)
VisDrone	52	38	+0.0132	-0.0082	-62.1 %
VisDrone	53	29	+0.0871	+0.0543	+62.3 %
SFCHD	42	35	+0.2961	-0.0733	-24.8 %
SFCHD	43	33	+0.0070	-0.1474	-2105.7 %
SFCHD	44	35	+0.6023	+0.1639	+27.2 %
SFCHD	45	34	+0.1870	+0.1006	+53.8 %
SFCHD	46	34	+0.3541	+0.2131	+60.2 %
SFCHD	47	37	+0.0750	-0.0651	-86.8 %
SFCHD	48	40	+0.0000	+0.0000	— (peaks at final epoch)
SFCHD	49	30	+0.6915	+0.1913	+27.7 %
SFCHD	50	39	+0.4042	+0.1024	+25.3 %
SFCHD	51	34	+0.7597	+0.0466	+6.1 %
SFCHD	52	36	+0.0392	-0.0272	-69.4 %
SFCHD	53	36	+0.0486	+0.1747	+359.5 %

> **What this table does, and what it does not do.** It **adds a detector family, not a claim level**: the two

> corpora, the protocol, the caliper and the aggregation convention are S34c's. It does **not** support any

> cross-detector magnitude comparison — the two families differ in warmup, in the final-epoch endpoint and in

> backbone, exactly the reason S34c already gives for not reading its column as a cross-detector comparison. It

> does **not** change the article's §6.5 sentence, which rests on S34 and S34c.

S9 · Figures moved from the main text

> Same reason as §S8. Captions are unchanged apart from the `S` in the figure number.

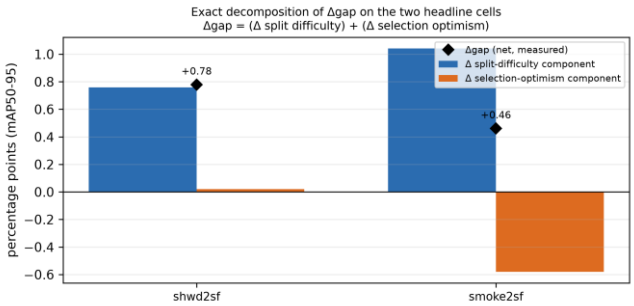


Fig. S1. Exact decomposition of Δgap into a split-difficulty component and a selection-optimism component on the two headline cells. The split-difficulty component dominates (+0.759 of +0.779 for shwd2sf; +1.041 against −0.580 for smoke2sf, where the two components partly cancel). This is an identity, not a fit: the three components sum to Δgap for all 13 cells. Source: `gap_mechanism_20260916.txt`, `xeval_analysis_20260916.txt`.

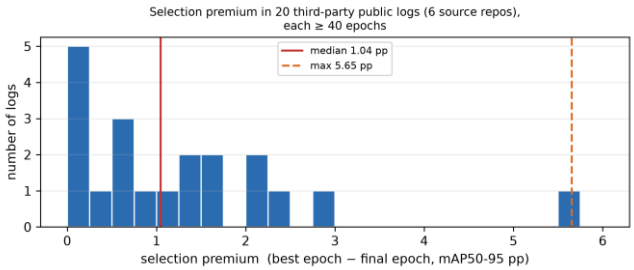


Fig. S2. Selection premium (best − final epoch, mAP50-95 pp) in the 20 complete third-party logs (≥ 40 epochs, 6 source repositories). Median +1.04 pp, maximum +5.65 pp, 18/20 positive, minimum +0.00. The same component measured on our own runs (main text §5.1) is present in logs we did not produce. Source: main text §5.5; `figures/FIGURES.md`.

S11 · Formalisation: propositions and proofs (companion to §3.1 of the article)

Notation as in §3.1 of the article: $V(e) = M_{S_val}(W(e))$, $T(e) = M_{S_test}(W(e))$, $\hat{e} = \text{argmax}_e V(e)$, e_f = the final checkpoint, $\text{premium_val} = V(\hat{e}) - V(e_f)$, $\text{premium_test} = T(\hat{e}) - T(e_f)$.

Proposition 1 (non-identifiability of the raw gap)

Statement. $V(\hat{e}) - T(\hat{e}) = [V(e_f) - T(e_f)] + [(V(\hat{e}) - V(e_f)) - (T(\hat{e}) - T(e_f))]$; hence for any candidate difficulty term δ' the residual $o' = G - \delta'$ reproduces the observation exactly, and no estimator using only the endpoint pair $(\hat{e}, e_f)$ identifies δ or o separately.

Proof. Substitute $V(\hat{e}) = V(e_f) + (V(\hat{e}) - V(e_f))$ and $T(\hat{e}) = T(e_f) + (T(\hat{e}) - T(e_f))$; the two expressions are equal identically. For the second claim, fix any $\delta' \in \mathbb{R}$ and set $o' = G - \delta'$; then $\delta' + o' = G$, which is the only equation the data impose. ■

Remark (which assumption buys identifiability). Differencing two arms with matched seeds gives $\Delta\text{gap} = E[G_base - G_strat] = E[(\delta_base - \delta_strat) + (o_base - o_strat)]$. The pairing assumption is $E[\delta_base - \delta_strat] = 0$, which holds when the two arms are trained and evaluated on one corpus and one split pair. Under an aliasing configuration both arms' δ are maxima of the *same* split's noise, so the assumption fails by construction: this is the formal content of §12.

What that does and does not say: it says the equal-difficulty assumption cannot be *assumed* under an aliasing configuration; it does **not** say that any realized contrast is non-zero, nor that aliasing makes a shift automatically detectable — those are statements about measurements, and they are made (or not) where the measurements are.

Proposition 2 (exact three-component identity)

Statement. $\Delta\text{gap} = \Delta\delta_f + \Delta\text{prem_val} - \Delta\text{prem_test}$, where $\Delta\delta_f = (V(e_f) - T(e_f))_base - (V(e_f) - T(e_f))_strat$, $\Delta\text{prem_val} = (\text{premium_val})_base - (\text{premium_val})_strat$, and $\Delta\text{prem_test}$ is the same for the realized premiums.

Proof. Apply Proposition 1 to each arm and subtract; the split-difficulty terms do not cancel but combine into $\Delta\delta_f$, and the bracket terms are $\Delta\text{prem_val} - \Delta\text{prem_test}$. ■

Verification. The identity holds to the printed digit on all 13 scanned cells (**Table S10 now prints all thirteen**; the seven it originally carried are typed

from the escalation analysis, the other six are derived from `work/gap_mechanism_20260916.txt` as its caption states) and on both escalated cells at n = 10: −1.226 = (−1.583) + (+0.090) + (+0.267) and −3.315 = (−7.072) + (−0.213) + (+3.971) (`work/x1_aux_20260918.txt`); the three components sum to **−3.314** at full

> precision, so the printed −3.315 differs from the component sum by **0.001 pp** — rounding, not a residual.

Remark 3 (the sign of Δgap does not identify a mechanism; not a separate result)

Statement. If the three components of Proposition 2 are of the same order, then for every target sign s ∈ {+1, −1} there exist admissible component values whose sum has sign s; hence the observed sign does not determine which component dominates.

Proof. Formalise "of the same order" as **K/2 ≤ |a|, |b|, |c| ≤ K**. Take (a, b, c) = (K, K/2, −K/2), whose sum is +K > 0, and (a, b, c) = (−K, −K/2, K/2), whose sum is −K < 0. In both choices every component lies in [−K, K] and has magnitude in [K/2, K], so both are admissible under the hypothesis and both are of the same order. Since the target sign s was arbitrary, either sign is reachable. ■

Empirical support. Measured medians of the three components are **1.07 / 0.54 / 0.93 pp** (same order), the 13-cell scan contains **both signs** (ten positive, three negative; −2.46 to +13.66 pp), and candidate mechanisms M1/M2/M4/M5 are each refuted by sign mismatches (Table S9). **Two corrections of record.** ① An earlier version of the companion script tried to demonstrate reachability by resampling *cells*; that inherits the cells' own sign split (98.4 % negative on the four cells used) and tests nothing. ② A stronger claim — that the sign is *fragile* — is **not supported**: perturbing one arm-difference term by ±10 % flips the sign in **0 %** of cells (±50 %: 23 %), with a median cancellation ratio of **0.485**. The proposition is therefore stated as "the dominant term varies by corpus and the sign is a sum of same-order terms", which is what the data support (`work/theory_p3a_correct_20260919.txt`).

Proposition 4 (realization rate = 1 − winner's-curse share)

Statement. Fix a **candidate set E** of checkpoints **before looking at any curve**, and let `e_f` be a **pre-specified member of E** (the final epoch). Assume V(e) =

v(e) + ε_e and T(e) = v(e) + τ + η_e with E[ε] = E[η] = 0, ε, η independent of v, and η independent of the selection event {ê = e}. Then E[premium_test] = E[premium_val] − WC with WC := E[ε_ê − ε_{e_f}] ≥ 0, so realization rate := E[premium_test] / E[premium_val] = 1 − WC / E[premium_val] ≤ 1.

Proof. (i) E[premium_test] = E[T(ê) − T(e_f)] = E[v(ê) − v(e_f)] + E[η_ê − η_{e_f}]; the second term is 0 by the stated independence and mean-zero assumption. (ii) E[premium_val] = E[V(ê) − V(e_f)] = E[v(ê) − v(e_f)] + E[ε_ê − ε_{e_f}]. (iii) Subtracting (i) from (ii) yields E[premium_test] = E[premium_val] − E[ε_ê − ε_{e_f}]. (iv) Because ê maximises V = v + ε **over E**, and `e_f` is a member of the same pre-specified set, V(ê) ≥ V(e_f) at the realised values, i.e. v(ê) + ε_ê ≥ v(e_f) + ε_{e_f}, hence ε_ê − ε_{e_f} ≥ v(e_f) − v(ê). The comparison is therefore between the selected index and the pre-specified one **within the same fixed set**; that is what the candidate-set clause buys, and it is the only place the argument uses it. Taking expectations, E[ε_ê − ε_{e_f}] ≥ 0 whenever the selection does not put the pre-specified index at a systematic **signal** disadvantage; selection on the signal alone is the equality case. ■

> **Why the candidate-set clause is not decoration.** Without it the step fails: if `e_f` were allowed to

> be chosen *after* the curve is seen, ε_{e_f} need not be the noise of a fixed element, and

> E[ε_ê − ε_{e_f}] ≥ 0 no longer follows — the final epoch could itself be a high-noise draw. The bound is

> therefore a statement about **selecting the argmax of V relative to a fixed reference member of the same

> candidate set**, which is exactly the `best`-vs-`last` convention this paper audits (`last` = the final

> epoch of a fixed budget, fixed before training).

Which assumption carries which half. The inequality (≤) needs **two** things: that ê maximises V

over a candidate set fixed in advance, and that the reference checkpoint `e_f` is a **pre-specified member of that same set** (its position not chosen after seeing the curve). Both hold under the convention this paper audits — `last` is the final epoch of a budget fixed before training — so the

direction is robust; but the inequality is a claim about **that** reference, not about "the final epoch" however it was arrived at. The closed form additionally needs η to stay out of the selection; under aliasing (`val` and `test` the same split) the two noises coincide and the closed form does not apply — which is why §12 treats the aliased cells' realization figure as a magnitude statement about bias, not as mechanism evidence.

Tests on the archive (`work/theory_p4_test_20260919.py`, `work/theory_meta_and_tests_20260919.py`):

test	result
inequality per run (WC > 0)	**+0.463 pp**, t = 8.34, **p = 1.8×10^{-12}**; realized premium smaller in **68 / 81** runs (sign test p = 3.8×10^{-10})
rate vs `val` noise (terciles)	**65.6 % → 49.8 % → 13.8 %**; Spearman **ρ = −0.466, p = 1.3×10^{-5}, n = 80**
premium level vs noise (σ√(2 ln E))	**withdrawn**: slope 0.026 ± 0.222, p = 0.82, n = 101 — the premium's level is set by the curve's shape, not its jitter

Proposition 5 (family dependence of Benjamini–Hochberg)

Statement. For a fixed multiset of p-values and level q, BH's rejection set is determined by the step-up condition on i/m·q; consequently the rejection count is not monotone under adding hypotheses: adding sufficiently small p-values raises the thresholds behind them and can turn a previously non-rejected hypothesis into a rejected one.

Proof. BH rejects H_(1..k) for the largest k with p_(k) ≤ (k/m)q. Since (k/m)q is increasing in k, inserting new small p-values shifts existing ones to higher ranks k, which relaxes their thresholds; a hypothesis that failed at rank j can therefore pass at rank j + t. ■

Instance in this paper. Adding the two escalated cells' n = 10 p-values (6.5×10^{-9}, 1.3×10^{-8}) moved `shwd2sf` from "does not survive" to "survives" (0.0053 against a line of 0.01154), and the 13-cell count went from **0 to 3** (`work/recompute_bh_20260917.py`). Declaring the family is therefore a precondition for the count to be meaningful, not a presentational choice.

Corollary to Proposition 4 (shape v. noise) — derivation and measurements

Derivation (exact). For one arm and one budget E let `V_r(e)` be run r's `val` metric at epoch e, `premium_r = max_e V_r(e) − V_r(E)`, and `m(e) =`

`mean_r V_r(e)`. Put `shape(E) = max_e m(e) − m(E)` and `v_r = premium_r − shape(E)`. Then `premium_r = shape(E) + v_r` **by construction**: no assumption about the noise distribution is used. Two immediate consequences: `shape` is a functional of the mean curve alone and is therefore **not** an independent check on σ — its uncertainty has to be propagated by resampling runs; and `E[v_r] ≥ 0`, because replacing each run's curve by the mean curve can only lower the maximum — this is the `val`-side member of the winner's-curse family of Proposition 4.

Measurement 1 — the premium's level is shape, and the noise term is flat in E (same corpus, pretraining, two arms, five seeds; only the epoch budget changes; run as the **single-machine clean batch** with early stopping disabled — `work/prop6_clean_b_20260925.py`, the same pipeline as `work/prop6_shape_noise_20260920.py` and the generator of `release_shape-noise-by-budget.csv`):

E	premium (mean of `shape`/premium arms)	`shape`/premium
50	+0.444	71.2 %
100	+0.682	90.5 %
200	+0.971	87.7 %
400	+1.272	87.6 %

Two columns are printed rather than three: `shape` is `premium` x (`shape`/premium), so a `shape` column would only duplicate a product of the two columns beside it (an earlier printing carried it empty, together with two trailing blank columns, which is what the review round flagged).

Selection rule behind these four rows, and the `n` of each cell. The series is the

clean batch: early stopping **disabled** (`G3CLEAN_NOSTOP=1`), and each budget keeps only the runs that **completed epochs 1..E**; two arms (`base`, `lr005`) × seeds 42–46, so **n = 5 per arm** (8 arm-cells, 40 runs), the merged row weighting the two arms equally after averaging within an arm. `shape` is the arm's **mean-curve peak minus its final value**, exactly as in the derivation above. Recomputation and release file: `work/prop6_clean_b_20260925.py` → `release_shape-noise-by-budget.csv` (the same pipeline as `work/prop6_shape_noise_20260920.py`, the entry point the reproduction map names, which computed the earlier reading).

The with-early-stopping reading is not deleted: `+0.907` over all ten E = 400 runs — four of them stopped early, at effective budgets 388/318/362/375 — remains the reading the article's §3.1 corollary carries, where it holds the pre-declared verdict, and Table S34b keeps the re-run's per-arm record. Only this table's premium and share columns move to the clean reading, which is why E = 400 now reads

+1.272 and **87.6 %** and agrees with that corollary's primary reading (**+1.2717**).

Four-budget outcome (pre-registered criteria C1–C3). On the **with-early-stopping** runs the verdict is the pre-declared one and stands: C1 — monotone rise of the premium with E — is **not met**, the series peaking at E = 200 and falling at E = 400 (four of the ten E = 400 runs stopped early under the protocol's own `patience`, at effective budgets 388/318/362/375; on the six full runs the E = 400 mean is +1.278 pp). C2 — the σ√(2 ln E) form — is **not rejected**: premium(400)/premium(100) = 1.329 against the predicted 1.141, inside the pre-declared band [0.86, 1.43]. C3 — the shape control — **passes** (ê/E 0.833 → 0.815). **The single-machine clean re-run (early stopping disabled, all four budgets present) gives a different verdict, and it is the one now reported**: arm-equal means **+0.4439 / +0.6824 / +0.9712 / +1.2717**, so **C1 is met** (monotone non-decreasing; per-run regression on ln E: slope **+0.3999**, se **0.0503**, t = **7.950**, p = **1.325×10^{-9}**, n = 40); **C2 is out of band** (premium(400)/premium(100) = **1.8636** against the predicted 1.141, band [0.86, 1.43] ⇒ the scaling **stays withdrawn** and the shape reading holds); **C3 passes on the pre-declared mean reading** (ê/E **0.833 → 0.787**, |Δ| = **0.046** ≤ 0.05) while **failing per arm** (lr005 |Δ| = **0.065**; base **0.028**). Table S34b carries the per-arm values, the superseded earlier reading and the reasons the series was re-run (**not** the two hosts: they calibrated as equivalent).

E = 50 → 200: `shape` ×2.70, premium ×2.19, `sd(v)` ×0.92; the σ√(2 ln E) form would predict ×1.24. The pre-registered decisive test of that form is criterion C2 (band [0.86, 1.43] on premium(400)/premium(100)); its verdict is reported in Tables S31–S33 and in §3.2.

Measurement 2 — where the realized gain goes (40 runs with per-epoch `test` readouts on a 5-epoch grid, `val` from each run's own `results.csv`; selection on `val` only, readout on `test` only):

quantity value

premium(`val`)	**+0.981 pp**
premium(`test`), oracle peak on `test`	+0.704 pp (**71.7 %** of the `val` premium)
premium(`test`), using the `val` argmax	**+0.211 pp (21.5 %)**
runs whose `val` and `test` peaks agree	**17.5 %**
median |	ê_val − ê_test|	10 epochs (grid step 5)

So the low realization rate is a **peak-alignment** phenomenon, not an absence of gain: an oracle that could pick the `test` peak would realize 71.7 % of the `val` premium, and `val`-selection retains 30 % of that.

Measurement 3 — selection rules (same 40 runs; every rule chooses on `val`, readout on `test`):

rule	realized `test` premium	captured share	cost to the reported number	within one step of the `test` peak
argmax (current convention)	+0.211	**30.0 %**	0.000	45.0 %
mean of top-2 `val` epochs	+0.406	57.7 %	0.690	55.0 %
mean of top-3 `val` epochs	+0.380	54.0 %	0.596	72.5 %
mean of top-5 `val` epochs	+0.490	**69.6 %**	0.727	62.5 %
argmax of a 5-epoch moving average	+0.459	65.2 %	0.526	70.0 %
final epoch (no selection)	+0.000	0.0 %	0.981	2.5 %

What these measurements are not. They are a **post-hoc comparison on the same runs**, not a pre-registered experiment: the rules were evaluated after the runs existed. The grid is 5 epochs wide, so "within one step" means ±5 epochs;

all choices are made on `val` and every readout is on the other split; and the oracle row is an upper bound, not an attainable protocol.

A candidate mechanism for the plateau rule — stated as a candidate, not as a result. The natural explanation of Measurement 3 is that the argmax of a noisy curve is pulled towards epochs whose noise happens to be high, so its underlying value sits below the true peak (the classical peak-shift penalty, of order $\sigma^2/|v''|$), while a top-k average estimates the peak *region* with lower variance. That story makes a sharp prediction: the plateau advantage should **grow with the noise of the `val` curve**. We tested it, and it is **not established**:

* noise estimator: `σ̂_seed` = mean over epochs of the cross-seed standard deviation at that epoch (`V_r(e) - mean_r V_r(e)`), i.e. the run-to-run dispersion the ten-seed design is about — not the residual around a smoothing of one run's own curve (which mixes curvature into "noise" and, with that estimator, gives ρ = +0.04, p = 0.79);

*per run (n = 40): ρ(`σ̂_seed`, plateau − argmax) = +0.309, p = 0.052**;

*per cell (n = 4 cells, the honest counting unit): ρ = +0.800, slope +0.406 ± 0.101 per pp of `σ̂_seed`, r = +0.943, but p = 0.057**;

* the four cells read: `σ̂_seed` 0.629 → +0.194 pp, 0.639 → +0.154 pp, 1.125 → +0.318 pp, 1.201 → +0.447 pp — i.e. both noisy cells show about twice the advantage of both quiet cells, with no exceptions.

What this is and is not. The *advantage* of the plateau rule is established (Measurement 3, and the paired test +0.278 pp, t = 3.678, p = 7.1×10^{-4}, positive in 26 of 40 runs). The *mechanism* is

consistent but not significant on four cells, so we report it as a candidate and not as a result. Settling it needs more corpus cells with per-epoch `test` readouts — the direction of the effect is already pinned down by the four we have, but four points cannot carry a significance claim. The `test` side of the remaining corpora has since been closed (Table S38), so the extension can be run on the release; we do not read it here rather than report it as settled.

13.2 §3.2: the per-comparison reasoning, moved from the article

Where the bias does move, it moves with what the selection sees: the split-size comparison held the test side fixed and varied only the size (both arms moving the same way), and the epoch-budget comparison left the peak's relative position essentially unchanged, so neither is a shape artefact. The third-corpus

replication evaluated its test side separately and points the same way as the 81 runs of §8.6 and the 40 of §5.1.

Sources: `work/g1p_verdict_20260919.py`, `work/g2_and_g1p_verdict_20260919.txt`, `work/g1pp_g3_g4_verdicts_20260919.txt`, `work/theory_p4_test_20260919.py`.

8.7b The per-epoch sign detail, moved from the article

`shwd2sf`'s Δgap is positive at **all 19 checkpoints**; `smoke2sf`'s is **negative at the first three — epochs 5, 10 and 15 — and positive at all 16 checkpoints from epoch 20 onward**; the four-epoch grid began at epoch 20, so it could not have seen this. At epochs 60 and 80 the val-side arm difference is −0.176 / −0.384 and the test-side one −1.051 / −1.445 — **reproducing the published values to the printed digit** — while Δgap stays positive: **the arm ordering flips on both splits at once, their difference does not.**

13.2b The four-budget detail, moved from the article

Four of the ten E = 400 runs stopped early under the protocol's own `patience` (effective budgets 388/318/362/375); on the six that ran to 400 the E = 400 mean is +1.278 pp — and that one restriction flips both criteria: on the six full runs the series is monotone (C1 met) and premium(400)/premium(100) = 1.873 falls *outside* the band (C2 rejected, the shape reading restored).

5.4 The two external critiques, in full (moved from the article)

The correction the critique forced: the paired-difference form cancels the difficulty term under the pairing assumption stated in §3.1, so the quantity we report is identifiable where the raw `M_val - M_test` is not.

13.3 The peak-agreement predictor (companion to §3.2 of the article)

Setup. The released per-run file carries, for each of the 40 paired runs, the validation peak (`e_val_peak`), the test peak (`e_test_peak`) and the two premiums. Both peaks are taken on the

same 20-point checkpoint grid (`0, 5, 10, …, 95`); that is the convention under which `same_epoch` is meaningful, and it is verified here: recomputing both peaks on that grid reproduces the released `e_val_peak` **40/40**, `e_test_peak` **40/40** and `same_epoch` **40/40**.

Per-run values for the group where the peaks agree (the other 33 runs' values are the released columns themselves):

run	`e_val_peak`	`e_test_peak`	Δgap
`r10_shwd2sf_base100_3way_s49n`	75	75	**+0.0561**
`r10_shwd2sf_lr005_100ep_3way_s45n`	75	75	**+0.4099**
`r10_smoke2sf_base100_3way_s42n`	80	80	**−0.2144**
`r10_smoke2sf_base100_3way_s46n`	85	85	**−0.5838**
`r10_smoke2sf_base100_3way_s47n`	65	65	**−0.2019**
`r10_smoke2sf_base100_3way_s49n`	75	75	**−0.2174**
`r10_smoke2sf_lr005_100ep_3way_s43n`	85	85	**+1.6649**

Group summaries. Peaks agree (n = 7): mean Δgap **+0.1305 pp**, median **−0.2019 pp**, four of seven non-positive. Peaks differ (n = 33): mean **+0.9056 pp**, median **+0.9537 pp**, four of thirty-three non-positive. Across all 40 runs Δgap spans **−0.7724 to +2.0454 pp**.

Permutation test. Reassigning the agreement label at random across the 40 runs 20 000 times, the absolute difference of the two group means is at least the observed **|0.7751| pp** in **0.0091** of draws (seed 42, `random.shuffle`); the agreement indicator's AUC for Δgap > 0 is **0.2969**, with a two-sided permutation p of **0.0196**.

The same relabelling on the median, which is the estimator these groups actually support. The agreement groups are separated far more clearly by their medians than by their means. Observed

|median(agree) − median(differ)| = |**−0.2019** − **+0.9537**| = **1.1556 pp**; 20 000 relabellings of the 40 Δgap values (seed 42, same convention) reach that or more in **0.0011** of draws (seeds 7 and 2026 give 0.0009 and 0.0014; the lower-median definition gives 0.0006). **The group means are the weaker summary, and the mean of the agreement group is carried by one run**: the agreement group's mean is **+0.1305 pp** on a group sum of **+0.9134 pp**, of which the single run `r10_smoke2sf_lr005_100ep_3way_s43n` contributes **+1.6649 pp**; dropping it moves the group mean to

−0.1252 pp while leaving the median at **−0.2019 pp**, and the median-difference permutation p becomes **0.0007**. **The reading this licences is a lower median Δgap under agreement — not "the premium is realized almost in full"**: four

of the seven agreement runs are non-positive, and the summary that survives the single-run sensitivity is the median. Script (read-only, no retraining): `work/r98_median_perm_20260927.py`, over `release_selection-transfer_perseed.csv` and `xeval_perepoch_20260918/matrix_perepoch.csv`.

What this table is not. The agreement group is **not** a group in which the premium is realized: its **median Δgap is negative** and four of its seven runs are non-positive, so agreement is a condition under which the *mean* deviation is small, not a guarantee that the selected checkpoint carries over. The result is also a **partial** predictor: it conditions on one observable and leaves the sign's variation inside each group unexplained, it rests on **7 runs** in the agreement group, and it is defined **only on the 5-epoch grid used here**, which was not refined.

Sources: `work/partial_predictor_final_20260922.py` (grouping, permutation test, AUC), `work/probe_evalpeak_convention_20260922.py` (the grid convention and its 40/40 reproduction); the grid and premium columns are the released `matrix_perepoch.csv` and `release_selection-transfer_perseed.csv`.

S12 · Method detail moved from the article

This appendix holds the measurement, protocol and record detail whose **conclusions stay in the article**. It is moved rather than cut: a reader who wants to check a claim reads the same measurements here, while the article keeps one sentence at each site. Identifiers that describe one machine's internal bookkeeping have been replaced by descriptors; the measurements themselves are unchanged.

3.1 Formalisation — the statements displaced from the article

§3.1 of the article now keeps each statement in compressed form, the adoption declaration, the two testable consequences of the adopted identity, and the corollary; the full statements with their derivations and proofs are in §S11. What the compression displaced is reproduced here **verbatim**, so that no number and no qualification leaves the paper — including the attribution sentence that separates what the generalization-gap literature establishes from what this paper

states as its own formalisation, and the disclosure that the sign of Δgap is not claimed to be *fragile*.

Definition (the raw gap is not identifiable from the pair alone). V(ê) − T(ê) = [V(e_f) − T(e_f)] + [(V(ê) − V(e_f)) − (T(ê) − T(e_f))], i.e. **G = δ + o** with δ a split-difficulty term and o a selection-optimism term. **The additive decomposition, and the qualitative reading that a single-model endpoint pair cannot separate its two terms, are established in the generalization-gap literature [17]**; its **strong form** — that *any* δ′ fits the observation equally well, so that no endpoint-pair estimator separates them [19] — is **stated here as a formalisation rather than attributed to that literature**. **The pairing assumption is what makes Δgap meaningful**: with matched seeds and one split pair, E[δ_base − δ_strat] = 0, and only then does the arm difference identify the difference in selection optimism. Under an aliasing configuration that assumption

fails by construction — the two arms' δ are two maxima on the same split — which is why §9 cannot be read as a clean-protocol result.

Adopted identity, not a claim of this paper (the realization rate is one minus the winner's-curse share). With V(e) = v(e) + ε_e and T(e) = v(e) + τ + η_e, **E[premium_test] = E[premium_val] − WC** where **WC := E[ε_ê − ε_e_f] ≥ 0**, so the realization rate is **1 − WC / E[premium_val] ≤ 1**, with equality only if selection is driven by signal alone [13] — Smith & Winkler's optimizer's curse written in this paper's notation; **we adopt it as the frame for §8.6–§8.7 and claim no part of it**. **What the per-run check can and cannot settle** (moved from the article): the check reads the **observed** `prem_val - prem_test`, so it corroborates that *consequence* of the inequality; it does **not** settle the latent `E[ε_ê - ε_e_f] ≥ 0`, which no archive decides without the identifying assumption. The inequality needs that ê maximises `val` **over a candidate set fixed in advance** and that the final reference checkpoint is a **pre-specified member of that same set** (both hold for the budget-fixed `last` this paper audits); the identity additionally needs the `test` noise not to enter the selection. **Two consequences, both tested on the archive** — the per-run inequality and the monotonicity in the `val` noise; per-run values, terciles and rank correlation: **Supplementary S11**. This is why §8.6's realization rate and §8.7's per-cell variation are the same statement at different noise levels, not a contradiction. A

third form — that the premium scales as σ√(2 ln E) — was **tested and withdrawn**; the regression and its diagnostic are in **Supplementary S11**. Contemporaneous work measures ≈**5**× on the same σ√(2 ln K) bound [18].

Proposition 2 (exact three-component identity). Differencing the definition across arms gives

Δgap = Δδ_f + Δprem_val − Δprem_test — an identity, not a fit, holding value by value on all 13 scanned cells and both escalated cells at n = 10 (§5.6).

Remark (the sign of Δgap does not identify a mechanism; not a separate result). Three measured facts carry this: the components are of the same order (medians **1.07 / 0.54 / 0.93 pp**), so which dominates differs by corpus; **both signs occur** in the scanned cells (per-cell table: **Table S6**); every candidate mechanism we could state is refuted by a

sign mismatch (M1/M2/M4/M5, Table S9). **What we do not claim** is that the sign is *fragile*: a perturbation analysis (numbers in **Supplementary S11**) leaves it stable under a ±10 % change of a single arm-difference term, so the honest statement is "the dominant term varies by corpus and the sign is a sum of same-order terms", not "the sign is noise".

Proposition 5 (Benjamini–Hochberg depends on the declared family). The step-up thresholds are i/m·q, so adding hypotheses with smaller p-values **raises** the thresholds behind them and the rejection count is not monotone in the family (proof: **Supplementary S11**). We are ourselves an instance: escalating two cells to n = 10 moved the 13-cell count from **0 to 3** surviving cells (§5.1) [20] — so declaring the counting unit and the family is a condition for the number to mean anything.

§3.2 as worded in the previous revision — the per-experiment criteria and their values, replaced in the article by the verdict list plus the Table S31–S33 pointer. Kept here verbatim for the same reason:

The propositions above are testable, so we ran them on rented GPUs with the **criterion fixed before the runs**. Six experiments were run; **five of them had their pre-declared criteria judged** — G1 is **void** (its design was diagnosed as confounded), so no verdict is entered for it. **Two of the five judged outcomes are negative** (G1′ and G2) and one design had to be redone — reported as they came out.

The six experiments, their **pre-declared criteria** and how each came out are tabulated in

Tables S31–S33: G1 **void** (the two splits differ by 14.7 pp in difficulty on the same checkpoints, so its −0.582 pp mixes the effect with a level effect); G1″ **met**; G1′ **not met** (WC = **−0.069 pp**, t = −0.23, p = 0.83, n = 10 — no inflation detectable); G2 **direction met, magnitude not met**; G3 **met** (and, in the clean re-run without early stopping, **C1 flips to met while the shape control fails** — S34); G4 **met** (WC = **+5.21 pp**, CI [+4.82, +5.66], 20/20 runs positive). Where the outcome is negative it is reported as negative.

The criteria that were met and those that were not carry one message: **the selection bias moves with the absolute amount of selection noise and with the number of selection opportunities, but not with whether the two splits are the same file.** Two consequences do not follow from the numbers alone: the configuration-level fact of §4 stands — the reported number *is* the selected number — while the stronger reading, that aliasing *adds* a measurable amount on top of it, does not. That identity's WC is best read as the winner's-curse share of the selected gain, a quantity set by the noise. **That reading yields a partial predictor of Δgap's sign, which we test here.** On the shared 5-epoch checkpoint grid the `val` and `test` peaks agree in only **17.5 %** of runs, and conditioning on that single observable splits the 40 runs cleanly: where the peaks agree the `val`-selected premium is realized almost in full (mean **+0.13 pp**, n = 7), and where they differ it is **systematically not** (mean **+0.91 pp**, n = 33; permutation p = **0.009**). **This is a partial predictor, not an explanation of the sign, and its limits are part of it:** the agree-group contains four of seven runs whose gap is non-positive, so agreement is not a guarantee of realization; n is small; and the split is defined on the 5-epoch grid, which we did not refine. The per-run table and the permutation test are in **Supplementary S11**. The per-comparison reasoning and the sources are in **Tables S31–S33**.

5.3 The dota15 cell: now reproducible

The local version repaired it for **two reasons**: a stale root snapshot (trees only up to 2026-09-14; it adds two `*_results_20260915` archives), and **the matcher could not see unseeded runs** — `prefix + '_s'` matched nothing while `dota15_base100` and `dota15_lr005_100ep` (**no seed suffix**) exist, so it reported `no runs found` forever. The seed suffix is now optional.

It agrees **value by value** with the published readings: three paired best-endpoint differences **+1.12 / +0.95 / +0.93**, mean **+0.999 ≈ +1.00** (last endpoints +1.77 / +1.46 / +1.54 — **again larger for the baseline arm**, as in §5.1 Tier 1).

> **A wording correction, retracted (2026-09-21).** An earlier version of this note said that only

> two of the three runs carried a seed. **That is wrong.** The third run's seed is **s42** — the

> campaign's fixed `--shuffle-seed 42`, recoverable from its recorded configuration; what it lacks is

> the `_s42n` **suffix in its directory name**, not its seed identity. The correct statement is

> **three runs on seeds s42/s43/s44, the third's directory name omitting the suffix** — a naming

> convention, not a missing seed. Each of the three readings **+1.12 / +0.95 / +0.93** is now owned

> (the suffix-less pair contributes the **+1.12** best-epoch reading). The companion made the same

> correction and kept its own anchor count at 567/567.

> **Status after the §5.1 escalation (2026-09-19).** The cell now exists as **ten seeded runs**

> (seeds 42–51: `dota15_base100_s*n` / `dota15_lr005_100ep_s*n`). All twenty `args.yaml` files —

> including the two pre-existing seeds — are identical in every recipe field (same pretrain, same

> YAML, 100 epochs, batch 32, imgsz 640, SGD, lr0 0.001 vs 0.005), so the old and the new runs are

> poolable; and the seed now appears in every directory name. **The wording defect above stays on

> the record as a defect of the old three-run cell, not of the row now printed.**

9.3 The seed convention: how the ten seeds are realized

What this implies for the intervals in the article's Tables 1–2 (added 2026-10-10, at review): those intervals take the **replicate** as the sampling unit, so what they bound is the **file-order** component of the reported mean — **not** the variance between independent training seeds. The **initialization** component

measured in §8.4 is of the **same order** (**0.146 → 0.439 pp**) and is **not** folded into them; the two tables carry a pointer here for that reason, and their `n` of eleven paired units is ten runs carrying a `--shuffle-seed` suffix plus one whose directory records none. Read the intervals with that axis named.

Per-run randomness is installed as a **deterministic permutation of the training file order** (`--shuffle-seed`, set to the number in the run's own name), not through the framework's `seed` argument: ultralytics 8.4.120's data pipeline does not draw from `args.seed`, and on two of these corpora different `--seed` values produced **identical** epoch-by-epoch replays.

The same convention and the same check hold for the cells escalated in §5.1: all twenty `args.yaml` of the `dota15` cell record `seed: 42` while its ten per-seed premiums are ten distinct values, and the forty runs of the two escalated Tier-2 cells likewise give ten distinct values in each arm-cell. Script: `work/check_seed_convention_20260918.py`.

5.4 Two external critiques, checked (one refuted, one judged non-identifiable)

Full report: `the two-critique verification record`; script `analysis/work/verify_two_critiques_20260916.py` → `eval_validity/verify_two_critiques_20260916.txt`.

(I) "The premium difference may be noise-level" — refuted. The critique compared it against **0.515 pp**, the **single-run** data-order component of §8.4, and concluded "underpowered, not significant". The quantity under test is a **difference of arm means**, with uncertainty the SE of that difference (**0.058–0.120 pp**); the t and p values of §5.1 Tier 1, and their four CIs excluding zero, settle it.

The wrong denominator was used; the data are sufficient.

(II) "You should report `M_val(ê) − M_test(ê)` directly" — judged non-identifiable. That quantity mixes the two splits' difficulty with selection optimism and is not identifiable here; we report the **paired difference Δgap** instead and **state honestly that its sign is not uniform** (§5.1 Tier 2). **A hard fact that came with it:** in shwd2sf the arm ordering on `val` is the reverse of that on `test` — see §5.1.

What these two checks changed. We used to write "the problem **understates** rather than manufactures a difference between arms"; (II) corrects this to "**it**

systematically shifts arm-to-arm comparison, and the direction varies by corpus" — the abstract, §2.1, §2.2 and §5.1 rewritten accordingly.

5.5 The same component in third-party public logs

Sections 5.1–5.4 use **our own** 569 runs; this asks whether the component **exists in other people's published logs**.

Sample. 24 per-epoch training logs; the **20 complete logs (≥ 40 epochs), from 6 source repositories**, are analysed. The definition is identical to ours: `premium = best-epoch mAP50-95 − final-epoch mAP50-95`. Ultralytics-style logs report the

best-validation checkpoint, so this difference is **an upper bound for** the component contributed by selecting on the reported split.

> **Table S7** → Supplementary §S8.

What matters most. The largest premium (**+5.65 pp**) occurs in an **ordinary published log** (a 47-epoch segmentation run; final epoch 0.6521, best epoch 42 at 0.7086). Beside this paper's own magnitudes —

+1.434 pp (the companion paper's reading, §8.1) and a published controlled ablation of **0.040 mAP50-95** (§2.1) — **the premium's long tail is larger than both**, its median of the same order as the former. (A magnitude comparison, **not** a same-unit subtraction: a within-run selection difference versus a between-arm effect.)

What it can and cannot support. Can: the component exists in **other people's** published logs, with a per-log magnitude (all 20 → Supplementary S5).

Cannot: a **field-level** distribution — 20 logs from 6 sources, most not stating dataset or protocol.

The sample is a convenience sample, not a frame: logs came from public dataset hosts and one code repository, filtered only by the search terms stated above; no platform-side query is named because none was systematic. **Any quotation of these magnitudes** (§2.2, §5.5) carries the four limitations listed below.

Verifiability (upgraded to "byte-checkable *and* traceable"). Each log's **SHA-256 and byte count** are in `published_logs/SHA256SUMS.txt` (**author-side directory, not part of the released package**); at source level, **24/24 files** matched to a source path by **content hash or unique byte count** —

8 Kaggle datasets plus 1 GitHub repository pinned at commit `2417aff0457c`, with **0 untraceable entries** and **24/24 agreement** with `SHA256SUMS.txt`.

Four limitations:

1. **It cannot establish aliasing.** Public logs carry **no YAML**, so we cannot tell whether that run's `val` and `test` share a directory. **"The premium exists" and "the benchmark aliases" are two independent pieces of evidence and cannot substitute for one another** — this section supports only the former (§4, §6 the latter).

2. **The exact downloaded version is unprovable.** What is provable is "**the bytes we analysed equal these hashes**", not "these bytes equal the Kaggle version at the time" (the source table records `currentVersionNumber` *at probe time*: `ikram0703` 2, the rest 1).

3. **The endpoint reading is an upper bound**: repositories rarely record which epoch was reported, so we assume it was the best.

4. **Licensing** (of **these third-party logs** — a different object from the seven **open** corpora of §4–§8, all of which we hold no local copies of only because the images are not in the archive): **5 sources do not state a licence** (Kaggle `Unknown`), and the GitHub source's `LICENSE` was not retained. We **do not redistribute** these CSVs, only hashes and source pointers; derived statistics should be cited with that caveat.

A defect we must record (one silent exclusion). The first script version identified metric columns by the substring `'mAP50-95' in column`, recognising **only the new Ultralytics spelling**, so the YOLOv5/v3 spelling `metrics/mAP_0.5:0.95` used by `ayyappakorla-yolov3-results` made `read_log` return `None` and the file **disappeared from the sample with no diagnostic anywhere**. It is a

200-epoch complete log (final epoch 0.9011, best epoch 198 at 0.9041, premium **+0.31 pp**). After the fix the sample went **19 → 20 logs**, **5 → 6 sources**, the median **+1.18 → +1.04 pp**, and positives

17/19 → 18/20.

Three conventions adopted from that failure (project-wide): ① annotate per row **which column spelling** the log uses; ② a "**logs that could not be parsed must be listed**" section — **a guard may exclude, but it may not exclude**

silently; ③ read the best epoch from each log's **own epoch column**, not the row index (most start at 1, the `ayyappakorla` one at 0).

The four logs not analysed run for 30, 20, 5 and 1 epoch; the 5-epoch one's "best − mean of last five" gives a meaningless +12.94 pp — **which is why a minimum length must be imposed**, and we report only `max - final`, for 20 logs.

> **Relation to §5.1.** This section supplies the existence and magnitude of the same component in **other

> people's **data. It** does not enter** §5.1's two-tier conclusion (whose evidence is our own paired

> design), and the two bodies of evidence do **not** corroborate one another.

6.1 Which reported number the COCO row judges, and this table's first-version correction

> **Table S11** → Supplementary §S8.

Which number the `reported` unit judges. For one row — COCO — it depends which reported number is meant: community tables report the **server-scored `test-dev`** value, which no local run can be selected on (we count it independent), while a paper's own ablations report its **`val`** value, which is also the selection split. First reading: `reported` **12/19**, clean **3/19**; second: **13/19** and **2/19**. No other row is affected; the abstract prints the first.

Correction to the first version of this table. It **put 2/19 under `yolo_dist`**, contradicting the recomputation: `yolo_dist` is **4/19** (PASCAL VOC and SFCHD's **literal two-key** alias, SHWD's `train == val`, Mendeley's generator alias), **2/19 being only the subset** the audit reported. The table makes **a unit a whole column**, ordered as §3's four definitions.

6.2 "Same directory" has four levels, and the audit reported only the narrowest

> **Table S12** → Supplementary §S8.

L1 is what the audit enumerated; L3 carries most of the *reported* layer's non-independence.

6.3 We do not reproduce the original audit's two numbers

> **Table S13** → Supplementary §S8.

6.5 Across frameworks: is the convention specific to YOLO?

Is "the reported metric comes from a non-independent split" a YOLO writing problem, or the default of the field? §6.1–§6.3 rest on the **generic YOLO distribution package**.

We retrieved, at **pinned commits**, the official configurations and training scripts of **five pinned repositories — 25 files**, covering six frameworks in total, since the Ultralytics distribution package is the `yolo_dist` unit of §6.1 and is supplied separately (SHA-256 and byte counts per file in `_xframe/SHA256SUMS.txt`; pins in `PINNED_COMMITS.txt`; script `work/analyze_xframe_20260916.py` → `xframe_evidence_20260916.txt`), then read mechanically which split each framework **reports** on and whether that split is **also** used for checkpoint selection.

> **Table S14** → Supplementary §S8.

Two conclusions: one widens the problem, one narrows our own claim.

(C-1, widening) Reporting on `val` is the shared default of all six frameworks, because of annotation availability. COCO's test annotations are not public — Detectron2 writes the test split's annotation file as `image_info_test2017.json` (**image information only**), PaddleDetection reads the

same val annotation file through a ground-truth-free `ImageFolder`. This is why the reported-layer count is what it is: where annotations are not public, the default path is to report on whichever split can be scored locally, and that split is usually also the selection split. **COCO sits on the boundary** — its `test-dev` is server-scored and cannot be selected on, so we count its community-reported number as independent, while a paper's own ablation numbers are not; §6.1 states both readings and both counts. **On benchmarks like COCO the coincidence is not an author's oversight — annotation availability pushes everyone onto the same split**, the default path itself.

(C-2, narrowing) But "select the checkpoint on that split" is not the shared default — only the YOLO lineage does it. Ultralytics (`best.pt`) and YOLOX (`best_ap`) select by the validation metric; Detectron2 (`EVAL_PERIOD = 0`), MMDetection (no `save_best`), DETR (no best tracking) and PaddleDetection

(none found) **report the final epoch**. §5's selection-premium claim therefore reads:

"within the pipeline family that selects by the validation metric by default (the YOLO lineage), the convention systematically shifts arm-to-arm comparison" — **not** "the default of the field". The abstract and the §2.1 table claim about YOLO distributions only, consistent with this.

Four limitations of this section (so it is not read as a census): ① we read **default configurations**, which users may change; ② only **5 pinned repositories / 18 official files printed with line-by-line evidence (S3)** — **fixed-point sampling, not a census**; ③ the pins are **default-branch heads** and will move, hence the SHA-256 and commit per file; ④ as in §4 and §6, a **configuration-level audit — no dataset was downloaded**.

8.1 The protocol

Split the pool three ways — a training subset, a **carved validation** subset for selection, a **test** subset for reporting; carved and test are disjoint, so the reported number is no longer the maximum a selection rule saw. Three requirements make it trustworthy:

carve out of the pool (the test split was never in training), **verify all three intersections are zero** (by filename), and **freeze the endpoint definition in writing** — which checkpoint counts as "the" reported number, whose omission is defect 3 in §10 and the source of the conflict in §7.3.

8.3 A direct measurement of selection inflation

Extending the same cell from n = 6 to n = 9, the mean falls **systematically**:

> **Table S18** → Supplementary §S8.

The magnitude is 9 %–29 %, taken against each cell's own n = 6 mean rather than the pooled headline effect. The estimand is the change in the cell mean when the seed count rises from 6 to 9 (two nested counts of the same runs; no new training), so this measures **seed inflation**; reading it as "how much was selection rather than signal" is an interpretation we do not test here.

8.4 The three variance components: the excluded one is the smaller in this sample

> **Table S19** → Supplementary §S8.

Reading (favourable to this paper, at three initializations): on the same unit, **initialization (0.146 pp) fluctuated less than data order (0.515 pp)** when this

was written — **the component we excluded was not the dominant one** — so "the reported power is an optimistic upper bound" tightened to "the excluded initialization component is smaller than the dominant seed component". **The five-initialization extension (2026-09-30) removes that reading**: the gain SD becomes **0.439 pp** with a bootstrap interval

[0.058, 0.580] that overlaps the data-order component, so the ranking is retained only as a **sample statement at n = 3**, not as an established ordering.

Two limitations must be stated with it: ① **three** initializations stood behind the original reading and **five** behind the 2026-09-30 extension, so the SD estimate stays uncertain (the 95 % interval for σ is roughly [0.5σ̂, 6σ̂] at 2 degrees of freedom and [0.6σ̂, 2.9σ̂] at 4); the extension's interval overlaps the data-order component, so we write "consistent with", **not** "established" — and the same now applies to the **ranking** itself. ② That experiment used A-side third-party YAMLs and the n = 10 cells used the same protocol; **the two hosts are not a confound — repeated calibration of the A and B machines found them equivalent, so no cross-host term is carried anywhere in this document**.

8.5 Why "same-pipeline pairing" is a necessity, not a habit

Three initializations move the **absolute endpoint** of one configuration by 43.250–43.630 (range 0.38 pp, SD 0.209 pp), comparable to seed noise; **five** (the 2026-09-30 extension) move it by

43.070–43.630 (range **0.56 pp**, SD **0.253 pp**). **Absolute endpoints are not comparable across source endpoints; only paired differences within one source endpoint are stable.**

5.6 Why Δgap's sign varies by corpus — measurement detail

§5.1 Tier 2 reports the **fact**; with the archived quantities

the mechanism is not identifiable, though three candidates are **refuted** and two unsupported.

Data: 13 three-way corpus cells × 2 arms (paired seeds n = 3 or 10). Script `work/gap_mechanism_20260916.py` → `gap_mechanism_20260916.txt`; the data path matches §5.4's and every cell's Δgap agrees **value by value** with the §5.1 Tier 2 table.

> **Tables S8–S10** → Supplementary §S8.

First, an identity: Δgap = (arm difference on val) − (arm difference on test), hence

Δgap > 0 ⇔ the relative arm difference on `val` is smaller than on `test`. It compares the two splits only and **points to no generating cause**.

An exact decomposition (an identity explaining the unstable sign — **not** a mechanism). Because `M_val(ê) = V_final + premium_val`, Δgap expands into three components, **all computable from the archive**:

> Δgap = (arm difference at the **final** epoch on val) + (**val-side selection gain** difference) − (test-side arm difference)

This is an **identity, not a fit**: for all 13 cells the "check" column (the sum of the three components)

equals Δgap. The three components' median magnitudes are **1.07 / 0.54 / 0.93 pp — the same order**, so

any one of them can flip Δgap's sign, consistent with §5.1's instability.

The most informative cell is `mask20`: the baseline leads the strategy by **+15.78 pp** at the **final epoch on val**, but the strategy's **val-side selection gain** is **13.10 pp** larger; the two nearly cancel, leaving Δgap **+1.75** (p = 0.66, not significant). **This cell shows that selection alone can absorb an arm difference of order 13 pp** — it supports "the convention changes arm-to-arm comparison" (§5.1 Tier 1) while showing that Δgap's sign **cannot** be used to infer any single cause.

What this section claims. Δgap's sign **varies by corpus** is firm (§5.1). Its generating **mechanism is identifiable in this archive only in part**: separating "the two splits' difficulty" from "selection optimism (one part on val, one on test)" requires **test metrics at non-selected epochs**, and the archive stores only the test value of the **selected** checkpoint — supplying it means re-running the test evaluation (see §8.6 and §10, defect 6, for what that closed).

What remains is bounded and stated, not left as "mechanism to be explained".

7.2 The two clean-protocol cells clear significance by n = 6–8 — measurement detail

The clean-protocol seed extension (seeds 42–51) reaches **n = 10**:

> **Table S15** → Supplementary §S8.

> **First-report attribution (a convention we hold to).** The two Δ values here (**+1.434 / +0.507 pp**),

> the three-way remediation protocol itself, and the pre-registered multi-target replication (§9) belong

> to the companion paper and are **cited, not claimed**: both modules were transferred to it by agreement

> (2026-09-21) and are its first report there. What this paper reports first is **the selection premium

> (§5), the 2 × 2 realization of that premium on a disjoint split (§8.6–§8.7), and the formalisation of

> the realization rate (§3.1)**. The companion manuscript is named, and its relation to this paper stated, in the cover

> letter; it carries no identifier while under review, and is available to the editor and the

> reviewers on request. A formal citation follows once both papers are accepted.

Seed extension is therefore not optional: it upgrades the conclusion from "significant under the parametric test only" to "significant under both".

7.1 The smallest attainable permutation p at each n — moved from the article

Why the floor is $2/2^n$, and what each n licenses. For a sign-flip permutation test on n paired differences the smallest attainable two-sided p is **$2/2^n$**, because the observed sign pattern is the most extreme of the 2^n equally likely relabellings; at n = 3 that is **0.25**, so such a cell has **no resolution at all** and may be read only as "significant under the paired t-test; the non-parametric test is unavailable at this n". The floors at the n used in this paper are: n = 3 **0.25**; n = 6 **0.031**; n = 7 **0.0156**; n = 10 **0.0020** — the last being the same quantity as **$2/2^{10}$ = 0.001953**, which is how the tables print it, so the two spellings denote one number and not two. **After §5.1's escalations no headline cell (Tables 1–2) is n = 3.** The article keeps the general form, the n = 3 case and the fact that no headline cell is n = 3, and points here for the derivation and the remaining floors.

The power note for the n = 3 cells, with its derivation (moved from the article). Recovering each Tier-2 cell's sd from its own printed 95 % CI as `sd = half-width ÷ (t_{9,0.975}/√10)` gives **0.20–0.67 pp** across the four escalated cells; a paired comparison at n = 3 then needs `(t_{2,0.975} +`

`t_{2,0.80})·sd/√3 = 3.10·sd`, and the **exact 80 %-power MDE from the non-central t (df = 2, two-sided α = 0.05) is `3.26·sd`** — i.e. **0.65–2.19 pp** for 80 % power at a two-sided α = 0.05, **at or above every Tier-1 effect the article reports** (+0.27 to +0.53 pp); at n = 10 the same derivation gives an MDE of `0.99·sd`, i.e. **0.2–0.7 pp**, the scale at which the four escalated cells are judged. The nine cells left at n = 3 cannot carry an inference at this design and are reported descriptively only.

7.3 A cross-archive endpoint conflict on the same run (investigated — and itself a disclosure) — measurement detail

One run, `t2_mask2mende_base100_s42n`, is recorded with **two different checkpoints**: one row reports the **final** epoch (67.12), the other the **best** (70.06). > **Table S16** → Supplementary §S8.

We follow B and report both. Not an error in either archive: evidence that **the project had no written convention for which endpoint constitutes "the" reported number** — §10's defect 3, the reason §3 insists that a count declare its unit.

5.1 Direction: the two tiers — measurement detail (moved from the article)

The pairing arithmetic, in full. The tested quantity is the paired difference d(seed) = premium_baseline(seed) − premium_strategy(seed), within cell and seed. The right scale is the SE of the difference (**0.058–0.120 pp**), and the four paired differences are

+0.416 / +0.526 / +0.272 / +0.449 pp, with p from 7.6×10^{-6} to 1.1×10^{-3} and all four 95 % CIs excluding zero (CI bounds **[+0.255, +0.578]**, **[+0.386, +0.665]**, **[+0.141, +0.403]**,

[+0.309, +0.589]; the unpaired Welch t is **5.68**).

Metric sensitivity, in full. Re-running the estimator on `metrics/mAP50(B)` keeps the sign in all four cells but moves one cell's paired difference into the noise: smoke2sf falls from

+0.416 pp (p = 1.9×10^{-4}) to **+0.063 pp (p = 0.54)**, while shwd2sf **+0.847**, a2d15 **+0.406** (p = **0.0019**) stay significant. Script `work/t9_ap50_sensitivity_20260924.py`.

The `dota15` escalation. It entered at n = 3 with a parametric p = **0.039**, below the resolution of any sign-flip test there; the pre-declared consequence was that

a ten-seed paired difference failing to be positive with ≥ 8/10 same-sign and a CI excluding zero would remove the cell. Ten seeds (42–51) gave **+0.449 pp**, t = **7.23**, p = **4.9×10^{-5}**, 95 % CI **[+0.309, +0.589]**, **10+/0−** — **met as stated**. Its mean is **24 %** below the n = 3 value (**+0.589 → +0.449**), a fourth instance of §8.4's seed inflation inside the published 9 %–29 % band.

The two overstating cells. The plan fixed before the runs (40 runs, seeds 42–51 × 2 arms × 2 cells; paired *and* sign-flip tests; `{best, last} × {val, test}`; a prespecified correction) named the strong branch of its outcome rule: *"both remaining negative and significant with permutation p ≤ 0.002"*. At n = 10: `p_aitovis` **−1.226 pp** (95 % CI **[−1.370, −1.081]**, t = **−19.17**, p = **1.3×10^{-8}**, **0+/10−**, permutation p = **0.001953**) and `p_vistod15` **−3.315 pp** (**[−3.676, −2.953]**, t = **−20.76**, p = **6.5×10^{-9}**, **0+/10−**, permutation p = **0.001953** — $2/2^{10}$, the attainable floor at n = 10). Both conditions are met, including **Holm across the two targets**.

The raw gap and the marginal cells. The raw quantity `M_val(ê) − M_test(ê)` ranges from **−33.0 to +21.2 pp** and cannot be separated into its difficulty and selection parts without per-epoch test evaluation — which has since been run for **all thirteen** corpora, so the separation is now available cell by cell (**Table S38**); it does not change the p-values, the confirmatory family or anything reported in this subsection. Four of 13 cells cross the uncorrected line; the three marginal ones are `aitod20` **+1.481** (p = **0.054**), `fire` **+13.658** (p = **0.061**) and `mende20` **+2.472** (p = **0.099**). The **declared descriptive** family is this 13-cell scan, and BH at q = 0.05 is applied to it (with **Holm across the two targets** of the escalation plan); the **confirmatory** family is the four escalated cells of §5.1, declared before their runs. The two are **not the same test** (the article's §5.1 states this); **Table S8 carries the raw p-values only** — the BH-adjusted step is stated here, not tabulated. The **0 → 3 move after escalation is descriptive**. The three cells BH leaves significant are `p_vistod15` (**6.5×10^{-9}**), `p_aitovis` (**1.3×10^{-8}**) and `shwd2sf` (**0.0053** against a line of **0.0115**), because the two very small p-values at the front of the ordering lift the lines behind them; `smoke2sf` (**0.0263** against **0.0154**) does not survive. Before the escalation the smallest p (**0.0053**) lay above the first line (**0.00385**) and nothing survived.

The 13-cell family, moved out of the article (effective base: four cells). The article's Tier 2 no longer carries the scan itself: it states the effective base (**four** cells at n = 10, not thirteen) and points here. The sentence displaced from §5.1, verbatim: *"Three marginal cells do not cross the line and we make no claim about them; the 13-cell scan is descriptive, and no formal power analysis was run."* The aggregate row that the article's Table 2 carried for the remaining cells is likewise here rather than there: `| other 9 cells | 3–10 | −2.46 to +13.66 | ≥ 0.054 | all include 0 | **not significant** |` — nine cells that do not cross the uncorrected line, with 95 % intervals including zero.

Why the base is four. Nine of the thirteen scanned cells sit at n = 3, and §7.1 of the article shows that a sign-flip permutation test has **no resolution** there (the smallest attainable two-sided p is $2/2^3 = 0.25$). The test is used here as a check at each cell's attainable floor, and it carries the usual assumption: under the null the paired differences are symmetric about zero, i.e. their signs are exchangeable, so those cells are descriptive and carry no inference; the four cells at n = 10 — the four rows the article's Table 2 retains — are the cells the direction statement rests on. The cell-by-cell values and the raw and BH-adjusted p-values side by side are above in this subsection and in Table S8; the 0 → 3 move after the escalation is descriptive.

The selection term, cell by cell. Table 2's caption names the selection contribution proper as the difference `Δprem_val − Δprem_test`; per arm-cell that term is `prem_val − prem_test`, the winner's-curse quantity WC of §3.1 read on the archive. **Verified against the archive, and complete as of 2026-09-30**: the scan record originally carried `prem_test` for **three of the thirteen** corpora, because separating the two sides needs per-epoch test evaluation. That evaluation has now been run — every archived 5-epoch checkpoint of the ten remaining corpora and of the `aitod20` strategy arm, **1,568 checkpoints over 140 runs** (1,848 evaluations including each run's `best`/`last`), on one machine, with the pipeline's registered call and the data YAML read from each run's own `args.yaml` — and the new `test` side **reproduces the archive's own readings**: over the 21 arm-cells carrying both, the val maximum, `prem_val` and the archived `T_best` agree to **max |Δ| = 0.005 pp**, which is the printed rounding; the coverage audit reports 140/140 runs,

1,848 rows, 0 anomalies. `prem_test`, and hence WC, is therefore formable for **all thirteen** corpora and both arms (**Table S38**). Nothing is imputed.

What is recorded for **all thirteen** scanned cells is the val-side selection gain `prem_val = V_max − V_final`, per arm, at the cell's scanned n — baseline → strategy, pp: `shwd2sf` (n = 10) **+1.47 → +0.87**; `smoke2sf` (n = 10) **+0.69 → +0.61**; `p_aitovis` (n = 3) +0.45 → +0.26; `p_vistod15` (n = 3) +1.11 → +0.88; `aitod20` (n = 3) +0.70 → +0.26; `d15d15` (n = 3) +2.13 → +0.67; `dota` (n = 3) +1.20 → +0.67; `fire` (n = 3) +10.95 → +8.87; `mask20` (n = 3) +16.07 → +29.17; `mende20` (n = 3) +4.83 → +5.68; `p_d15toai` (n = 3) +0.49 → +0.39; `p_masktomende` (n = 3) +2.33 → +1.34; `vis` (n = 3) +0.44 → +0.22. Source: `gap_mechanism_20260916.txt`, the same record §5.6 and Table S8 use. The two cells escalated in §5.1 (`p_aitovis`, `p_vistod15`) carry n = 10 after the escalation, which is the four-cell base the article's Table 2 prints.

The selection term `prem_val − prem_test` where the archive carries it is the five realization arm-cells of §8.6, read off Table S20's printed values (**the printed `+0.736` / `+0.509` supersede the verbatim S4 dump's third decimal, `0.735` / `0.507`**): `shwd2sf` baseline **+1.469 − 0.736 = +0.733** and strategy **+0.865 − 0.152 = +0.713**; `smoke2sf` baseline **+0.691 − 0.509 = +0.182** and strategy **+0.605 − (−0.157) = +0.762**; `aitod20` baseline (n = 10) **+0.766 − (−0.218) = +0.984**. The other ten scanned corpora and the `aitod20` strategy arm were closed on 2026-09-30 as described immediately above; their cell-by-cell `prem_val`, `prem_test` and WC are in **Table S38**, and read descriptively they carry the same asymmetry — WC positive in 25 of the 26 arm-cells, arm-cell median realization rate 17.2 %, and the realized premium negative in 8 of the 13 strategy arms. These are cell-level readings at each cell's recorded *n*, not an upgrade of any claim. **Ten of those cells, which stood at n = 3, were later escalated to n = 10 (Table S38b); with the escalation folded in the same three quantities read WC > 0 in 25 of 26 (unchanged), median rate 21.6 % (mean 11.6 %), and the realized premium still negative in 8 of the 13 strategy arms. In arm-difference form** the term is recorded for exactly two cells, from the clean-protocol batch: `shwd2sf` **+0.020** and `smoke2sf` **−0.580**, against Δgap of **+0.779** and **+0.461**. In `smoke2sf` the selection term is therefore **opposite in sign to the Δgap** the article's Tier 2 prints — which is why the direction sentence rests on the four-cell base and **not** on Δgap's sign.

The sharpest single fact. In the shwd2sf cell the arms differ on **val** by 60.669 − 60.397 = **+0.272 pp** (baseline better) and on **test** by 43.559 − 44.066 = **−0.507 pp** (baseline worse): **the sign reverses**, so reading only `val` inverts the arm ordering.

The endpoint convention, in full. On `val` the per-epoch maximum; on `test` the checkpoint the pipeline's own fitness rule kept (`best.pt`), read on the disjoint split. Handing a "selected checkpoint" to a split therefore always resolves to one of these two endpoints, and no other convention is used anywhere in this paper. The four paired differences and their intervals are in Table 1; the Δgap values and readings for the **four printed cells** are in Table 2; the other nine cells of the 13-cell scan are in §5.6 and **Table S10** (all thirteen rows) with the remaining components in `work/gap_mechanism_20260916.txt`.

Sign-flip permutation tests on the four Tier-1 paired differences. Table 1 of the article reports a paired *t* test per cell. Its non-parametric counterpart needs no distributional assumption and is exact at these *n*: enumerating all 2^n sign flips of the per-seed paired differences d(seed) = premium_baseline(seed) − premium_strategy(seed), where premium = best-epoch − final-epoch mAP50-95 on the run's own monitored split, gives

Cell	n	$\bar{d}$	sign split	paired *t* p	exact permutation p	attainable floor $2/2^n$
smoke2sf	11	+0.416	11+/0−	1.9×10^{-4}	**0.00098**	0.00098
shwd2sf	11	+0.526	11+/0−	7.6×10^{-6}	**0.00098**	0.00098
a2d15	10	+0.272	9+/1−	1.1×10^{-3}	**0.00391**	0.00195
dota15	10	+0.449	10+/0−	4.9×10^{-5}	**0.00195**	0.00195

The permutation test agrees with the paired *t* in every cell. Three of the four sit exactly at the smallest two-sided *p* attainable at their *n*; `a2d15` carries one negative pair and therefore reaches $4/2^{10} = 0.00391$, one step above its own floor. Wilcoxon signed-rank returns the same *p* to four decimals (0.00098 / 0.00098 / 0.00391 / 0.00195). Per-seed values, so that each *p* can be recomputed — the two

n = 11 cells carry ten seeded runs plus one unseeded run, listed last:

Cell	d(seed)
smoke2sf	+0.354, +0.247, +0.873, +0.217, +0.738, +0.394, +0.217, +0.088,

								+0.357, +0.482, +0.611
shwd2sf								+0.305, +0.506, +0.919, +0.466, +0.702, +0.280, +0.515, +0.756, +0.641, +0.354, +0.339
a2d15								+0.241, +0.211, +0.467, +0.560, +0.324, +0.415, −0.073, +0.216, +0.261, +0.096
dota15								+0.627, +0.821, +0.524, +0.190, +0.257, +0.369, +0.494, +0.545, +0.226, +0.437

`a2d15`'s negative pair is its seventh listed run (seed 48); the other nine are positive. The attainable two-sided floor for a sign-flip test on n pairs is $2/2^n$, so the $n = 11$ cells can in principle reach 0.00098 where the $n = 10$ cells cannot go below 0.00195. The `dota15` per-seed values are the ones tabulated in Table S28; the other three cells' per-seed values are recomputed from the archived per-epoch curves.

Metric sensitivity: the full mAP50 columns for Tables 1 and 4. The article reports the headline metric mAP50-95 throughout and states one metric-sensitivity datum in the text. Both tables recomputed on `metrics/mAP50(B)`, from the same run sets and the same code path, follow, so that the two metrics can be compared cell by cell rather than by assertion.

Table 1 on mAP50 (same cells and runs as the printed table):

Cell	n	premium baseline	premium strategy	paired difference	t	p	95 % CI	per-seed signs
smoke2sf	11	+0.972	+0.910	**+0.063**	0.63	0.545	[−0.160, +0.285]	**6+/5−**
shwd2sf	11	+1.678	+0.831	**+0.847**	7.00	3.7×10^{-5}	[+0.577, +1.116]	11+/0−
a2d15	10	+0.803	+0.396	**+0.406**	4.33	0.0019	[+0.194, +0.619]	9+/1−
dota15	10	+1.318	+0.625	**+0.693**	5.67	3.0×10^{-4}	[+0.417, +0.970]	9+/1−

Table 4 on mAP50 (the same five arm-cells and runs as the printed table):

Cell	Arm	n	`prem_val`	`prem_test`	realization rate
shwd2sf	baseline	10	+1.387	+1.082	78.0 %
shwd2sf	strategy	10	+0.866	+0.283	32.6 %
smoke2sf	baseline	10	+0.995	+0.839	84.3 %
smoke2sf	strategy	10	+0.840	+0.001	0.1 %
aitod20	baseline	10	+1.744	−0.369	−21.2 %

Three things the columns show, stated rather than left to the reader. (i) **The sign of the Tier-1 paired difference is positive in all four cells on both metrics**, and all three cells other than `smoke2sf` keep significance on mAP50. `smoke2sf` is the single cell that does not: its paired difference is +0.063 pp with a 95 % interval [−0.160, +0.285] that includes zero, and its per-seed signs are **6+/5−** rather than unanimous, so on this metric the weakest cell loses both significance and sign unanimity — which is why the article attributes metric-independence to the *sign* claim and not to any per-seed claim. (ii) On Table 4's realization rate, mAP50 gives **31.5 %** with a run-level 95 % interval [15.0 %, 48.9 %] — the three weightings again coincide, every arm-cell being at n = 10 — against 23.3 % on mAP50-95, and the number of arm-cells with a negative realized premium is **1 of 5** rather than 2 of 5. Partial realization is therefore directionally metric-robust, while its magnitude and that count are metric-specific. (iii) Both tables' positive controls are the printed mAP50-95 values, recomputed on the same code path: the four Tier-1 rows reproduce Table 1 exactly at the printed precision, and the five realization cells reproduce Table 4 to within 0.001 pp, the aggregate returning 23.26 % against the printed 23.3 %.

mAP75 is now reported on both sides: the archive still carries no per-epoch mAP75 *reading*, but the per-epoch checkpoints are in register, so the `val` half is re-measured rather than missing — on a five-epoch grid, for four of the five arm-cells.

A same-estimand ratio exists elsewhere: the genomics relative-bias literature. The article's §2.2 scopes its "no ratio of gains" statement to a ratio of *difference* premiums on disjoint splits in detection; a ratio of the same estimand — how much of a selected effect survives on an independent split — is however long established in quantitative genetics, where it is written as a **relative bias** and reported as a number. Würschum & Kraft,

Cross-validation in association mapping and its relevance for the estimation of QTL parameters of complex traits,

Heredity 112(4):463–468 (2014), doi 10.1038/hdy.2013.126, define it as `1 - (pG-TS / pG-ES)` — the complement of a realized-to-selected ratio on a disjoint estimation/test split — and report a mean overestimation of **38 %** for the proportion of genotypic variance explained by QTL, **30 %** averaged over four further traits, and a literature range of **10–60 %**; Utz, Melchinger & Schön, *Genetics* 154(4):1839–1849 (2000), had already reported that in independent validation `p(TS.ES)` "was on average only **50 %** of `p(ES)`". What is not found in that literature, and what this paper reports, is the ratio of two **difference** premiums — `(T_best - T_final)/(V_max - V_final)` — on disjoint splits, in detection, on our own five arm-cells, with measured `val` noise as moderator; and the sign flip of §8.6 (in two of the five arm-cells the realized premium is negative) has no counterpart there either.

No per-epoch mAP75 reading exists in the archive: the per-epoch training logs and every evaluation file in this release record `mAP50` and `mAP50-95` only — a scan of the 55 metric files behind the two tables above finds no mAP75 column. That is why the **`val` half** of a mAP75 sensitivity column cannot be *read*; it does **not** block *measuring* it, and it does **not** block the **test half**. Both halves are analyses of what exists rather than new experiments: `best.pt` and `last.pt` are the released endpoints for every cell in the two tables, so re-scoring those endpoints at IoU 0.75 needs no further training; and the per-epoch checkpoints are in register for the four `*2sf` arm-cells on a **five-epoch grid**, so re-evaluating those checkpoints on `val` at that threshold needs none either. Both passes have been run and are reported with their scopes below: the test half (82 evaluations, 41 runs, exact positive control against the archived matrix) and the `val` half (880 evaluations, 40 runs, positive control against the training logs at a median absolute difference of

0.0161 pp, which is what a re-measurement of a recorded training run should look like). We report the mAP50 columns (recomputable from the released artifacts) **and** both mAP75 columns — the `val` one on the five-epoch grid, labelled as such.

8.6 The realization rate — measurement detail (moved from the article)

Scope of the headline number. The **pooled 55 %** covers the four arm-cells §5.1's escalation adds (**82 % / 104 %** on `p_aitovis`, **56 % / 29 %** on `p_vistod15`); the two batches are **not pooled** (different runs, different

protocol), so the headline is **not** re-averaged — it is defined over §8.6's five arm-cells and Table 4 stands. The **104 %** is within noise on ~0.27 pp. Source: `work/x1_aux_20260918.py`.

The theory-driven check, in full. §3.1 derives the realization rate as **1 − WC / premium**, WC being the winner's-curse part of the selected gain. Two consequences are testable: ① the proposition itself establishes only an expectation, `E[premium_test] = E[premium_val] − WC ≤ E[premium_val]`, and the ratio is defined only where the denominator is positive (`premium_val` > 0); its **per-run shadow is testable rather than proved** — over the **81** runs with both sides measured, WC averages **+0.463 pp** (t = **8.34**, p = **1.8×10^{-12}**) and the realized premium is smaller in **68 of 81** runs; ② the rate should **fall as the `val` curve gets noisier** — sorting those runs by their own curve's noise scale $\hat{\sigma}$ gives terciles of **65.6 % / 49.8 % / 13.8 %**, Spearman **ρ = −0.466** (p = **1.3×10^{-5}**, n = **80**). **This section's 23.3 % and §8.7's per-cell variation are therefore the same statement at different noise levels.** What does **not** hold is the stronger form we first wrote down — the premium scaling with the noise as $\sigma\sqrt{(2 \ln E)}$: slope **0.026 ± 0.222** (p = **0.82**, n = **101**) — because the premium's *level* is set by the `val` curve's shape; §12 records the withdrawal. Scripts: `work/theory_p4_test_20260919.py`, `work/theory_meta_and_tests_20260919.py`. **Re-read with the arm-cell as the resampling unit, as the κ fit of Table S22 already is (81 runs in 9 arm-cells: the 41 three-way runs of §8.6 plus the 40 paired runs of §5.1): WC = +0.463 pp, cluster-robust 95 % CI [+0.204, +0.721], p = 0.0033 (CR1 sandwich on 9 clusters; a cluster bootstrap gives [+0.253, +0.661]); the noise-rate Spearman ρ = −0.466, cluster-bootstrap 95 % CI [−0.651, −0.131], p = 0.0115 — and at the 9-cluster level ρ = −0.433, p = 0.244 (work/r102_cluster_20260927.py).**

The three weightings. The three figures are ratios of means computed three ways — runs weighted equally, unweighted arm-cell means, and arm-cells with n ≥ 5 — and because every cell carries n = 10 they coincide at **23.3 %**. On the previous **41**-run sample they did not, under the same declared `prem_val = V_max − V_final` endpoint as Table 4: the arm-cell-equal reading was **19.6 %**, the run-weighted one **32.4 %**, and the arm-cells-with-n ≥ 5 one **34.5 %** — a ~15 pp spread coming from a single cell holding **1** run. Each arm-cell is at

n = 10 here, so the estimator's own interval is reported instead: 95 % CI **[8.4 %, 37.4 %]**, run-level bootstrap, P(rate ≤ 0) = **0.002**.

The two intervals, and the primary one. The run-level bootstrap on the same 50 runs gives 95 % CI **[8.4 %, 37.4 %]** with P(rate ≤ 0) = **0.002**; every `P(rate ≤ 0)` in this subsection is a **bootstrap tail proportion** (B = 20,000 resamples, seed 20260927), not a frequentist p-value, and the only multiplicity correction this paper applies is the **Benjamini–Hochberg step over the declared 13-cell family** (Holm, where it appears, belongs to the **two escalation targets**); p-values printed elsewhere, including the paired t values, are **raw**, the **arm-cell-equal (five-cluster) bootstrap** gives **[−17.3 %, 53.2 %]** with P(rate ≤ 0) = **0.118**, and it is the interval the claim is read at, because the claim is about five arm-cells. The five-cell rates are 50 %, 18 %, 74 %, −26 % and −28 % (Table S20), i.e. a median of **17.6 %**, a span of **−28.5 % to +73.7 %** (**102.2** pp from the printed endpoints, 102.1 pp from the unrounded cells) and a leave-one-out range of **9.77 % to 34.16 %** — dropping the `shwd2sf` baseline arm-cell gives 9.77 %, dropping the `aitod20` baseline arm-cell gives 34.16 %. Script `work/item2_final_v2_20260925.py`; the two-cell negative set is Table 4's **2/5**.

§5.1's escalation in this section's units. The four arm-cells it adds are **82 %** / **104 %** (`p_aitovis`) and **56 %** / **29 %** (`p_vistod15`), pooled **55 %** against **23.3 %**.

Metric and endpoint detail. `best.pt × val` versus the per-epoch maximum in `results.csv` differs by up to **0.083 pp**: the selection action and the readout action can differ; the earlier attribution to a `best.pt` selection rule other than mAP50-95 is **withdrawn** — in the pinned v8.4.120 `fitness ≡ mAP50-95` on the monitored split — and the residual is recorded as **unidentified**. `last.pt`'s val recomputation differs from the final `results.csv` row by **~0.02 pp**.

The closest concurrent work, moved out of the article. The article's §8.6 no longer carries this comparison; it is recorded here. **[15]** splits selection leakage into diversity and noise over **2,047** tabular classification datasets (the noise share is reported, the non-tabular case is undone). **[12]** measures checkpoint selection raising a reported score by **6.24 pp** on EEG when the candidate set grows from **5 to 80** — and, in the same table, **−1.24 pp** (95 % interval **−2.67 to 0.26**) on its other pool, so **its own sign does not reproduce across its two**

pools. This paper instead uses **our own five arm-cells**, reports a **ratio**, on **detection**, with **measured `val` noise** as moderator (the third-party logs of §5.5 corroborate the same component, but they carry no `test` side and so cannot form this ratio). **No number left the paper when this paragraph moved**: every value above is the article's former wording.

8.6b The eleven arm-cells with both sides measured: a batch-stratified re-aggregation, no new training

> **What this is, and what it is not.** The article's realization rate rests on **five** arm-cells, and a

> reviewer asked for every arm-cell whose **two sides** are both in the archive, aggregated with the

> **batch as the stratum**. This section is that aggregation, and it is a **re-reading of measurements this

> submission already reports** — the per-epoch `val` curves and the `best`/`last` × `test` readouts of runs

> that already exist. **It contains no new training, no new seed, no new evaluation and no new data.**

> Nothing in the article changes because of it: the five-cell rate of §8.6, its interval and its scope

> stand, and this is a wider reading of the same archive, not a replacement headline.

The eleven, by batch. A cluster is one `corpus` × `arm` pair measured on **both** sides, paired inside the run: `prem_val = max_e V(e) − V(e_f)` from the run's own `results.csv`, and `prem_test = T_best − T_final` from its `test` readouts; a cluster's rate is `prem_test / prem_val`, and a pooled rate is the ratio of the summed means.

Batch 1 — the article's headline batch (Table S20 / Table 4; three corpus pairs, n = 10 per cluster, 50 runs; 5 clusters). Cluster rates: `shwd2sf` baseline 50.1 %, `shwd2sf` strategy 17.6 %, `smoke2sf` baseline 73.7 %, `smoke2sf` strategy −26.0 %, `aitod20` baseline −28.4 %; median **17.6 %**, span **−28.4 % to +73.7 %** (**102.1** pp from these cells; **102.2** pp from the printed endpoints) — the article prints the lower end as −28.5 %, from Table 4's 3-decimal cells, the full-precision value being **−28.39 %** — **2/5** negative,

run-weighted **23.26 %**, cell-level 95 % CI **[−17.3 %, 53.2 %]** (P(rate ≤ 0) = **0.118**). This batch
is §8.6.

*Batch 2 — both arms of the two Tier-2 cells §5.1 escalated** (§5.1; n = 10 per cluster, 40 runs;
4 clusters). `p_aitovis` baseline **81.7 %**, `p_aitovis` strategy **103.7 %**, `p_vistod15` baseline **55.7 %**, `p_vistod15` strategy **28.8 %** — the four values the article prints as
82 % / 104 % and **56 % / 29 %**, pooled **55 %**; median **68.7 %**, span **+28.8 % to +103.7 %**,
0/4 negative, run-weighted **55.0 %**.

*Batch 3 — the third-corpus replication** (§3.2, G4; `mende20`, two arms × ten seeds, 20 runs;
2 clusters). Baseline **41.5 %**, strategy **33.8 %**; median **37.6 %**, span **+33.8 % to +41.5 %**, **0/2** negative, run-weighted **37.5 %**. Its winner's-curse term is the **+5.212 pp** that Table S33 already prints.

*All eleven, batch as the stratum. Median 41.5 %, mean 39.3 %, span −28.4 % to +103.7 % (132.1 pp), 2/11 negative; run-weighted pooled 36.9 % over 110 runs. A cluster bootstrap that resamples arm-cells within** their batch (B = 20,000; seed 20260927) gives 95 % CI
[19.3 %, 60.4 %] for the arm-cell-equal statistic, **[28.9 %, 44.4 %]** for the run-weighted one, and
[26.2 %, 60.8 %] for the median — each with P(rate ≤ 0) = **0.000**.

*Difference scale, the reviewer's other request (`prem_test − prem_val`, pp, clusters equally weighted): −1.367 pp, 95 % CI [−1.59, −1.15]. It is reported with a warning rather than as a co-primary, because a pp difference is not scale-free: the mean is carried by the batch with the largest `prem_val` — batch 3 is −5.212 pp, against −0.675 pp (batch 1) and −0.310 pp** (batch 2).

*The other axes. By arm: six baseline-style (`base100`) clusters, median 52.9 %, one negative, against five learning-rate-style (`lr005_100ep`) clusters, median 28.8 %**, one negative. By
budget: ten clusters ran **E = 100** and one (`aitod20` baseline) ran **E = 30**, too lopsided to be read as a budget effect. By **corpus family**: four clusters transfer into **SFCHD** (`shwd2sf`, `smoke2sf`), four come from the `p_aitovis` /

p_vistod15 pair, one is **AI-TOD** on its 20 % subset, and two are mende20. By **metric**: the five-cluster mAP50 column is the one printed in the **5.1** measurement detail of this supplement (**31.5 %**, **1/5** negative, against 23.3 % and 2/5 on mAP50-95); **this section does not extend that column to the other two batches**, and no eleven-cluster mAP50 rate is stated.

The five arm-cells' denominators, moved from the article. The article says the difference-scale reading rests on a denominator that spans **+0.605 to +1.469 pp** across the five arm-cells; the per-cell values are in §S4's verbatim dump above (smoke2sf strategy **+0.605**, shwd2sf baseline **+1.469**, and the three further cells in Table S20).

What the wider reading does not license. At eleven clusters the arm-cell-equal interval no longer includes zero, where at five it did; that is the outcome and it is stated rather than smoothed. It does

not upgrade the article's claim. ① The six added clusters **were not run for this question**: the two Tier-2 cells were escalated *because* their val overstatement had already been flagged, and the third corpus replicates the same protocol, so the eleven are not an exchangeable sample of corpora. ② The strata differ in **level** — batch 1 spans **−28.4 % to +73.7 %**, batch 2 **+28.8 % to +103.7 %** — which is why the batch is the stratum and no single pooled rate is printed as *the* rate. ③ It is still a **ratio of small means**: prem_val across the eleven clusters runs from **+0.268 pp** to **+8.723 pp**, so a few hundredths of a point in a numerator move a cluster's rate by tens of points. ④ The reviewer's own power note applies: at this between-cluster spread, clearing zero at cluster level would take of the order of

25 clusters, and the interval above excludes zero **for these eleven archived clusters**, which is all it says.

What is paired, what is single-sided, and what has no denominator.

*Paired on both sides, on the same run: all eleven clusters**, and within a batch the two arms share their seeds. No cluster in the eleven is assembled from different runs on its two sides.

Single-sided or off-metric, and therefore outside the eleven. ① The epoch-budget series (§3.2, G3; Tables S32 and S34b) carries `prem_val` only: its readout side is the disjoint carve C3 (100 images), not the published `test` split, and the archive holds no per-epoch `test` curve for it, so no `prem_test` — and

hence no rate in this section's sense — can be formed; its own readout is recorded where it belongs (WC = `prem_val` − `prem_C3` is negative in all 16 readouts, mean −0.616 pp, Table S32). ② The IoU-0.75 pass now carries both sides for the four `2sf arm-cells: the val half is re-measured from the in-register checkpoints (880 evaluations, five-epoch grid), so the one cluster it touches (smoke2sf strategy) has a denominator at mAP75 — **+1.2862 pp** against **−0.3201 pp**, a rate of **−24.9 %** — while the aitod20` baseline cell stays single-sided at that threshold because its nine remaining runs' weights are not in this archive. ③ The test-carve sub-sample pass (25 % / 50 % / 100 %) re-scores both endpoints but reports premium means over pairs, not a per-cluster rate.

*Paired and double-sided, but outside the eleven for its sample size. The selection-split-size batch (§3.2, G2; Table S31) is measured on both sides at n = 5 per cluster — four clusters, the small and large selection splits × two arms — with rates 23.5 % / 5.1 % / 27.0 % / 15.6 %, median 19.5 %**,

0/4 negative and run-weighted **20.5 %**; it is outside the reviewer's set because that set is the

n = 10 clusters, and adding it would make **fifteen**.

*No denominator, recorded as such rather than imputed. ① The `aitod20` strategy arm: batch 1 could have been six clusters and is five, because that arm's `prem_test` was not recorded when this ledger was written — it had no per-epoch `test` curve. That curve has since been evaluated (Table S38: n = 10, `prem_val` +1.111 → `prem_test` −0.543, WC +1.654), so the arm now carries a rate; we leave batch 1's registered composition at five clusters and record the addition here rather than re-cutting the set after seeing it. ② Ten of the thirteen scanned corpora** of §5.1: for those ten neither `prem_test` nor a rate could be formed when this ledger was written. They are now formed for

all thirteen corpora and both arms (Table S38), at each cell's recorded *n* — five of them at n = 3, which is why the new rates are reported there as cell-level descriptives and not as a significance claim. ③ The **mAP75 `val` half**: the per-epoch `val` curve at IoU 0.75 is still **not recorded** as a reading (the 55 metric files carry `mAP50` and `mAP50-95` only), but it has now been **re-measured** from the in-register checkpoints for the four `*2sf` arm-cells (880 evaluations, five-

epoch grid), so a mAP75 realization rate exists for those cells (arm-cell median **+26.8 %**, run-weighted **+22.3 %**, **1 of 4** negative); what remains without a denominator is the **`aitod20` baseline cell**, its other nine runs' weights not being in this archive. ④ **Initialization** is not the axis these replicates vary, so the across-initialization realization rate was **not recorded** in this archive; **it is now recorded** (2026-09-30, five source-side initializations, 10 arm-runs, `work/G3_init_rc_20260930.py`): **18.5 % run-weighted** (Σ`prem_test`/Σ`prem_val` = 2.269/12.250), **median 19.2 %**, positive in **8/10** arm-runs — the same order as the five-arm-cell headline of §8.6, with two negative cases (`src1` strategy 44.200 → 44.258, `src4` strategy 44.030 → 44.116).

> **An independent replication of that arm, pre-registered (2026-10-02).** A fresh batch of **ten** runs of the

> same arm was registered before it ran (same corpus, same recipe, same `--shuffle-seed`

> 42-51), and the batch's own reading is `prem_val` **+1.4013 pp**, `prem_test` **-1.0169 pp**, realization rate

> **-72.6 %** (n = 10, cell-level bootstrap 95 % CI [-133.15, -42.70] %; the ten per-run values ship in

> `02_release_data/release_G1_replication_20261002.csv`, the criterion frozen before the runs staying

> author-side). The sign reproduces the archive's: the strategy arm's

> selection action **costs** rather than failing to pay off. The *magnitudes* do not agree run for run: against

> the archive arm, paired by `--shuffle-seed`, the differences average **+0.2904 pp** (`prem_val`) and

> **-0.4736 pp** (`prem_test`), larger than the same-configuration **replicate** band this submission measured

> (fresh replicates on one configuration: mean |delta| 0.362-0.442 pp, maximum 0.810-1.110 pp; §12, Table S39) —

> **a band set by seeds and data order, not by hosts, since the A and B machines calibrated as equivalent and no

> cross-host term is carried anywhere **— so the difference is recorded as** unidentified **and is** not** read

> as evidence about the phenomenon. **Batch 1's registered
> composition stays at five clusters**: this replication is disclosed here and does not enter Table 4, whose
> headline is unchanged.
> **One released reading corrected (2026-10-02), with no printed number affected.** In the released registered-
> replication table the saturated control's `test_best_epoch` row had been computed with one run's **`last`**
> reading in place of its `best` reading (`t2_mask2mende_base100_s42n`: 67.1200, against 67.1183 for that run's
> `last` and 70.0603 for its `best`; the row's `source_test` also came from a **different evaluation batch** from the
> rest of its family). The row is corrected in `release_T2_saturated-control_summary.csv` and in the long table:
> `-0.9020` becomes **`-1.4901`** pp (sd 0.8002, paired t -4.164, p = 0.0141, 0+/5-), which is what the row reads
> once every run contributes the quantity the column names. **Every number printed in this article and this
> supplement is unchanged**: the figure quoted for this control above is the `last`-epoch row, which was already
> correct, and the **`best`**-epoch reading of the same cell is **−1.490 pp, p = 0.0141** — i.e. under the
> article's frozen `best` convention this control is **significant**, not non-significant; the article's §7.3 and
> Table S16 now say exactly that, and Table S25 is endpoint-independent (all three of its questions answer the
> same way under either reading). What is *unchanged* is every number the article prints from the **archive-A**
> row. A systematic check of all released per-run `test` values against fresh re-evaluations found this to be
> the **only** row that mixed the two quantities; seven further rows (`t1d_dotatod15_*`) differ from the fresh
> re-evaluation by 0.012-0.164 pp in the direction of `best`, i.e. within the same-configuration replicate band this

> submission measured, and are not corrected.

Provenance, and the positive controls that bind this section to the printed record. The four batch-2 cluster values are recomputed from `x1_tier2_20260918/runs/` and `teval.csv`; the two batch-3 values from `g4_mende3way_runs/` and `g4_mende3way_20260919_g4_test.csv`; the five batch-1 values from the code path behind Table S20 (`work/item2_final_v2_20260925.py`). Three checks make the re-aggregation auditable: the five batch-1 clusters reproduce Table S20 to **0.0019 pp**, with the aggregate returning **23.26 %** against the printed **23.3 %**; the four batch-2 clusters return **81.7 / 103.7 / 55.7 / 28.8 %** against the printed **82 / 104 / 56 / 29 %**, and their Δgap selection terms **+0.090** and **−0.213** pp reproduce the printed decomposition of §5.1's two cells; and the two batch-3 clusters reproduce Table S33's printed per-run `prem_val` / `prem_test` pairs to **0.0003 pp**.

8.7 The per-epoch view — measurement detail (moved from the article)

κ accounting, in full, and the relation between the two estimates. Two estimates of the same quantity appear in this submission, at two grids, and they must be read together. **0.734** is the earlier fit over the **five-epoch grid** (R^2 **0.902**, n = **161** points); **0.708** is the corrected value over the **complete grid** (R^2 **0.937**, n = **765** (seed, epoch) points over 41 runs), and the completed sweep moves the estimate by **−0.026** with overlapping intervals. **The interval printed in the article is the run-clustered (sandwich) one, taken on runs as the clusters: 95 % CI [0.671, 0.745]** (sandwich SE **0.018**); a run-level **bootstrap** cross-check gives

[0.672, 0.744], i.e. the same interval to the printed precision, and the within-run estimator, which removes run-level differences entirely, gives **0.698**. The five-epoch grid carries its own run-clustered interval, **[0.683, 0.784]** (sandwich SE 0.025, bootstrap [0.685, 0.784]), and that interval belongs to the **0.734** fit, not to the value the article prints. Scripts: `work/x2_kappa_clustered_20260918.py` (which reproduces the 0.734 / 0.902 / n = 161 fit exactly before adding the interval) and `work/x3b_perepoch_corrected_20260918.py` (the complete-grid refit; its printed lines give the sandwich interval, the bootstrap cross-check and the within-run estimator).

Per-arm κ and the five-epoch comparison. baseline κ = 0.600 (95 % CI [0.581, 0.619], R^2
0.946) against **strategy κ = 0.767** ([0.735, 0.800], R^2 **0.951**); the per-arm fits and the resampling are in **§S8, Table S22** (and the fit itself in §S11).

The per-checkpoint sign detail. On the complete curve `smoke2sf`'s Δgap is negative at epochs
5/10/15 and positive from epoch **20** on; the four-epoch sample had suggested uniformity. The per-checkpoint values are in **Supplementary S11** and Tables S21–S23.

The three-layer cross-check. The same checkpoints were evaluated twice independently and agree on all **202** overlapping (run, epoch) points to **0.0000 pp**; the full sweep covers **806** rows (**820** minus the 14 the single-run `aitod20` cell does not have, its 30 epochs contributing **6** checkpoints where the others contribute **20**).

10 Our own defects — item-by-item detail (moved from the article)

Defect 1. The clean-protocol results document cites `_r10_final_stats.txt`, and the file later identified as the "real log" is `_r10_stats_0403.txt`; **neither** contains any of that document's headline values (6 probes each, **0/6 hits**), so those numbers have **no surviving stdout ledger**. We do **not** point at a replacement file — renaming would only manufacture a new false citation — and the p-values remain recomputable from the local `pvalue_table_local.py`.

Defect 2. The dota15 cell of `selection_premium.py` was not reproducible because the root directory was a stale snapshot and the matcher used `prefix + '_s'` while the actual run names carry no seed suffix; after the fix it reproduces M3's **+1.12 / +0.95 / +0.93** value by value. The "3 seeds" wording note it once triggered was **retracted on 2026-09-21**: the third run's seed is **s42** (the campaign's fixed shuffle seed, recoverable from its configuration), so what is missing is the directory-name suffix, not the seed.

Defect 4. The two headline scripts originally depended on a rented pod (`import paramiko` plus a plaintext password); `selection_premium_local_20260916.txt` and `work/pvalue_table_local.py` now read the local archive, and the pod versions are listed only as cross-checks.

Defect 5. For a long time no single script computed the σ decomposition: the three components came from three unrelated analyses. **Two of the three are now recomputed by one script** (`work/sigma_decomposition_20260926.py`) — the data-order component and the initialization component are recomputed from the archived per-run values and each is asserted against the number printed in Table S19.

The augmentation component is the exception and is named as such: the archive records its SD without its per-run values, so the script checks only that the record carries it, and its output says `cited, not recomputed` rather than printing a constant as though it had been computed. Still open, and still disclosed: there is **no frozen effect-size rule file**, and **no script generates `benchmark_split_audit.md`** (hand-written, **28** judgments as literals; its **counts** are in fact recomputed, by `work/recompute_split_units_20260916.py`).

Defect 6. The local archive holds no copy of the three-way carve directories and no copy of 6 of the 7 corpora. The 2026-09-16 GPU run repaired part of this: the per-epoch curves of the 41 three-way runs, `sio_b_results.csv` and the **2 × 2 evaluation matrix** (**164** evaluations) are now local with a SHA-256 list in `xeval_20260916/`, so §8's readings and §5.6's decomposition are locally recomputable;

what remains non-recomputable is retraining, which needs the corpora themselves.

Defect 9. Our first per-epoch report claimed curve-wide stability from a four-point sample, and reading the complete curve overturned it for one of the two cells (§8.7): `smoke2sf`'s Δgap is negative at epochs 5/10/15 and positive from epoch 20 on, so the claim is **narrowed** rather than deleted, with the retracted sentence quoted and the sampled grid's apparent uniformity named as partly an artefact of where it sampled.

Defect 10. *Moved here from §10, item 10, verbatim, so that the article can state the disclosure and the bound without carrying the event counts.*

> **Memory pressure silently changes the training path**: under GPU memory pressure Ultralytics switches implementation without changing the recorded configuration — (a) `CUDA out of memory with batch=32. Reducing to batch=16` (a **configuration-level** change), or (b) `OutOfMemoryError in`

`TaskAlignedAssigner, using CPU` (same batch, but label assignment re-runs on CPU, where a boundary argmax can differ and the trajectory forks). Case (a) is **invisible in the standard artifacts** — `args.yaml` still records `batch: 32`, and the only fingerprint is the log's per-iteration denominator (32 → `0/9` / `0/41`; 16 → `0/18` / `0/81`).

>

> **Disposition. Disclosed, not retracted; we do not call the affected batches "clean".** Configuration-level hits: **2** places — §3.1's E = 50 clean batch, **1** run (Δpremium **+0.308 pp** against the same-machine re-run), and §5.1 Table 2's `p_vistod15` cell, **3** runs — so the whole clean four-budget series was re-run on one machine. Assigner fallback occurs in **every** batch, **1–4** times per run (≈**1** per 1000 iterations). **A reproducibility residual, not an event size.** The two fresh replicates of a *same* configuration are bit-identical (**0.000 pp**, four pairs, one machine, one day): that is **observation-level reproducibility**, and it is **not** a size for the event. The quantity a bound would be about — archived run against clean re-run — is **0.000 / 0.020 / 0.150 / 0.000 pp** (Table S41): on the clean E = 50 batch **7 of 10** same-config pairs are bit-identical, so **≤0.05 pp** applies *there* and is **not extrapolated**, and it is not a bound on those four pairs, one of which is **0.150 pp**. In the `p_vistod15` and `aitod20` cells the treated runs' A→B differences (Δpremium **≤0.35 pp**) fall **inside the spread of the zero-event controls** (up to **+0.704 pp**), so at this scale the event effect is **unidentified** rather than bounded, and we say so; the one clear case is the configuration-level run (ΔVmax **+0.769 pp**). This sits against the paper's own realization-rate 95 % CI **[8.4 %, 37.4 %]**.

>

> **The clean re-run, and what it can and cannot test (2026-09-30).** That cell was re-run twice over — four `(arm, seed)` pairs, each **twice** — on one machine with ample VRAM. Reading the **test** endpoint: the two fresh replicates of the *same* configuration are **bit-identical in all four pairs** (`|Δ| = 0.000 pp`; the recorded `args.yaml` carries `deterministic: true`), which is the strongest reading the `≤0.05 pp` bound above can ask for; the **archive** differs from the clean run by up to **0.150 pp** (`0.000 / 0.020 / 0.150 / 0.000`, mean 0.043), and `args.yaml` is **identical field for field** (only `name` and `save_dir` differ), so that difference is not a recorded configuration change and,

with a deterministic pipeline, not run-to-run noise either. **A same-configuration control that had no such event is itself not reproducible across dates**: re-running the archived `iv_src1/base100` configuration today gives **43.190 against the archived 43.630 — a 0.440 pp drift** (its per-epoch `val` curves move by up to 3.198 pp), i.e. **larger than the whole difference on the cell defect 10 flags** (up to 0.150 pp). ⇒ **the bound stays `≤0.05 pp`, read as an archive-internal statement about same-configuration pairs; cross-date comparisons cannot test it**, because the drift alone is several times the effect the bound is about. Table S41; scripts `work/G2_jitter_20260930.py`, `work/ctrldrift_check_20260930.py`.

The five realization arm-cells: what each one pairs, and whether the corpora are public

All five are public corpora, and the corpus pair is in the cell name. A cell name reads `source→target`: `shwd2sf` and `smoke2sf` transfer from one public corpus to **SFCHD**, and `aitod20` is the **AI-TOD** corpus on its own 20 %-budget subset. None of the five is an internal corpus, so no cell's result rests on data a reader cannot obtain; what is *ours* is the reproduction and the three-way split built on top of them, and the per-cell values are in Table S20.

Cell	Source → target	Corpus status
`shwd2sf` (baseline, strategy)	SHWD → SFCHD	both **public**
`smoke2sf` (baseline, strategy)	public smoke corpus → SFCHD	both **public**
`aitod20` (baseline)	AI-TOD → AI-TOD (20 % budget)	**public**; the subset is ours

What this does not settle. That the corpora are public does not make these cells a sample of the field: they are the cells this design measured, and §8.6's rate is scoped to them.

What the 19 benchmarks are, and what is recorded about them

Every row carries an evidence class and a confidence level; the inclusion rule is not recorded, and we do not reconstruct one after the fact. The marking table was **established 2026-09-16**, and each row carries its **evidence class**, its **confidence level** and the **document version** the split configuration was read from — so a reader can trace any row back to the release archive or the dataset's own paper it came from, and the four subtotal equalities are recomputed mechanically rather than hand-summed.

What is *not* in the archive is a stated rule for why these 19 and not others. Two things can still be said without inventing one. First, the rows are the benchmarks for which this archive holds the split configuration read from **primary documentation** rather than from a secondary description — that is a property of the rows, and it is visible in them. Second, **no sampling frame of papers was built**, which is why the article says it does not sample papers per benchmark and makes no prevalence claim: the counts describe **these** benchmarks, and a stated-frame sample remains future work.

Read as: a census of the configurations this archive could read, not a sample of the field.

Two later measurement batches: what they add, and the bounds they carry

Both are evaluation-side; neither changes a headline number, and both are stated with the scope they have. They answer two questions a reader is entitled to ask about §8.6: is the paired premium an artefact of *which* sub-sample of the held-out split was scored, and is it an artefact of the metric?

The paired premium does not move with the size of the test sub-sample. For the four Table 4 families that are not the `aitod20` baseline, the same two endpoints were re-scored on random sub-samples of the held-out split at 25 %, 50 % and 100 % of its images, ten independent draws per size (21 carve points, **1,974 completed evaluations**; the driver is `work/b51_eval.py`, the summary `work/b51_summary.py`). The paired `best - last` premium keeps its sign and its order of magnitude at every size: **+0.2096 pp** at 25 % (470 pairs), **+0.2538 pp** at 50 % (470 pairs), **+0.2325 pp** at the full split (47 pairs), with the run-to-run spread (SD 0.49–0.59 pp) an order of magnitude larger than the shift between sizes. **What this does not licence**: it shows the premium is not an artefact of one particular sub-sample, not that these cells are a sample of the field; and the 100 % row rests on fewer pairs than the other two because **126 of the 2,100 scheduled evaluations could not be run** — the weights of three runs (`r10_aitod20_base30_3way_s43n`, `…s44n`, `…s45n`) are not held in this archive, so their 42 carve points each are missing rather than reported.

The sign instability is not an artefact of the metric. The same `best`/`last` endpoints were re-scored at **IoU 0.75** (82 evaluations, 41 runs, no failures; drivers `work/b55_m75.py` and `work/b55_val.py`, controls `work/b55_m75_control.py` and `work/b55v_control.py`, pairing

work/b55v_pair.py, same-sample metric comparison work/b55v_metric_compare_20260928.py). The positive control is exact: the mAP50-95 values from this pass reproduce the archived matrix to **0.0000 pp on all 82 run-checkpoint pairs**. The paired premium is **+0.2933 pp** at mAP50-95, **+0.2982 pp** at mAP75 and **+0.5201 pp** at mAP50, and the arm-cell that is negative at mAP50-95 (smoke2sf, strategy arm, **−0.1571 pp**) is **still negative at mAP75** (**−0.3201 pp**, positive in 2 of its 10 runs). **Both halves are now measured on the same runs, and the rate is metric-specific.** The val half was obtained by re-evaluating the in-register checkpoints of the same four arm-cells on their own val split at IoU 0.75 — **880 evaluations** (20 five-epoch-grid checkpoints, last.pt and best.pt for each of 40 runs), no failures — with a positive control against the training logs: the re-measured mAP50-95 at those checkpoints reproduces results.csv at a **median absolute difference of 0.0161 pp** (**800/800** within 0.2 pp, none above 1 pp; that the training log's epoch column is 1-based against the checkpoint filenames is itself part of the record, and comparing the two at the same index instead produces a spurious median of 0.6462 pp). Per arm-cell the mAP75 pair (prem_val, prem_test) and its rate read **+1.8828** / **+1.0072** = **+53.5 %** (shwd2sf baseline), **+1.8043** / **+0.0017** = **+0.1 %** (shwd2sf strategy), **+0.7205** / **+0.5792** = **+80.4 %** (smoke2sf baseline) and **+1.2862** / **−0.3201** = **−24.9 %** (smoke2sf strategy), against **+50.1 %** / **+17.6 %** / **+73.7 %** / **−26.0 %** on mAP50-95 over the same 40 runs: arm-cell median **+26.8 %** against **+33.8 %**, run-weighted **+22.3 %** against **+34.2 %**, and **1 of 4** cells negative in both. Three cells move by at most a few points; the one that moves materially is the shwd2sf strategy arm (**+17.6 %** → **+0.1 %**), where the realized premium essentially vanishes at the stricter threshold.

What this does not licence: the val side is a **five-epoch grid** whose peak is not the per-epoch argmax — the grid peak falls short of the per-epoch peak by **0.13 to 0.33 pp** on mAP50-95 for these cells — so these prem_val values are grid values and are labelled as such; the aitod20 baseline cell is not covered on the val side, its other nine runs' weights not being in this archive; and a per-run rate stays unreadable, the median per-run ratio being **+0.016** against a mean of **+0.023**. Both batches cover the runs whose weights this archive holds, and the

per-run files are held with this revision's measurement batch rather than inside the archived reproduction package.

Read as: two robustness checks on §8.6's readings, scoped to the cells and the sub-samples they cover — one on the carve, one on the metric — not as further evidence about the field.

Independent re-coding of the 19-row marking table (current instrument)

The 19 x 4 = 76 judgements printed in S1 were re-coded independently by **five current-generation language models from distinct vendors**, each in a fresh session, **blind to the printed marks and to one another**. Each coder received (i) a **decision procedure** -- a per-unit decision tree plus a single value-mapping table -- and (ii) per row, an **evidence card** containing only verbatim evidence with its source, line number and pinned commit. Coders were forbidden to use outside knowledge of the benchmarks, and a cell the tree cannot decide must be declared `**blocked**` rather than guessed.

Two rounds were run, differing **only in the value scale**; both are reported in full below.

Earlier rounds of this table (three coders, then eight, then a ten-coder round under a different instrument) are superseded by these two and are not reported here.

Two rounds on this archive, and what they change (2026-10-09). Earlier rounds of this table measured how well the marks survive re-coding *under the published rules*; these two rounds measure something the earlier ones could not separate -- whether the marks are reproducible **from the evidence the archive records for each row**, with that evidence placed in front of the coder. Both rounds used **five current-generation language models from distinct vendors**, in fresh sessions, **blind to the printed marks and to one another**; the task sheet, the answer tables, the returned codings, the raw matrices and the agreement scripts are held with this revision's measurement batch. **The earlier rounds are superseded by these two and are not reported here.**

The instrument. Each round hands the coder a **decision procedure** (a per-unit decision tree plus a single value-mapping table) and, per row, an **evidence card** carrying only verbatim evidence with its source, line number and pinned commit -- no marks, no confidences, and no access to this supplement. A cell the tree cannot decide is declared `**blocked**` (the tree reaches a step whose

evidence the card lacks) rather than being guessed, and the two rounds differ **only in the value scale**:

*Round 7 (coarse, six values per unit). The coders agree with each other** at **Fleiss' kappa 0.836** -- all five give the identical value on **38 of the 51** cells where no coder declared `blocked` -- yet they reproduce only **31.4 %** of the printed marks (majority of five, after mapping both scales onto a common six-value equivalence). Their agreement is highest on `release` (**9/10** cells) and `yolo_dist` (14/18), and lowest on `reported` (5/9).

*Round 8 (isomorphic, nine values for `release` and seven for `protocol`). The scale is changed to the exact vocabulary the printed table uses (`no_test`, `no_val`, `no_split`, `no_yaml`, `n_a`, `test_gated`, `contradictory`, ...), and the `release` tree asks first whether the layer supplies a usable independent test artifact. Agreement among coders falls a little (kappa 0.811) and 26 cells move to `blocked`, but agreement with the printed marks rises to 42.2 %, and the `protocol` column is recovered exactly: the majority of the five returns the same number of non-independent rows as the table prints (13 of the 17 decidable cells, against the printed 13 of 19**).

Round 9 (isomorphic scale + first-party evidence; the round this revision reports). Every row was re-verified against first-party artifacts -- official release listings, READMEs and papers at pinned commits -- and that evidence, together with an explicit statement of the counting convention, was placed in front of five fresh coders. Agreement among coders is kappa 0.824, and 26 cells are declared `blocked`. The decisive number, however, is not the agreement but the recovery: the re-verification changed 30 cells, and the five coders -- who never see the new values -- independently reach the corrected value in 19 of those 30 (63.3 %), eleven of them unanimously. The residual splits three ways: three cells are differences of convention (which of two test variants counts as the *test, and whether a layer whose test images cannot be downloaded at all is `test_gated` or has no usable test artifact), three are rule thresholds that this round's instrument states too loosely, and five remain genuine evidence gaps. The corrections are therefore evidence-driven rather than a matter of coder taste**; what stays open is convention and rule wording, and the correction record above states both.

What the two rounds settle, and what they do not. The coders agree with each other at a level the earlier rounds never reached, so the residual is **not rule wording**: what the five independently cannot recover is the printed **mark**, and the gap widens or narrows with the

granularity of the value scale rather than with the amount of rule clarification. The `release` column is the clearest case: with the fine scale the coders assign `n_a` much more often than the table does, because "no usable independent test artifact" and the table's several other reasons for a non-independent mark are **not separable from the recorded evidence**.

Twenty-five of the 76 cells (Round 7) and 26 (Round 8) could not be decided from the public record at all -- for `DIOR`, `D-Fire`, `MAFA`, `UAVDT`, `SFCHD`, `SHWD`, `Mendeley face-mask` and `NWPU VHR-10`, whose released artifacts, protocol texts or file listings are behind login walls, paywalls or dead domains. That is a **property of the public record**, not of coder diligence, and it is reported here as such.

Corrections to the 19-row marking table from first-party re-verification (2026-10-09)

The marking table in S1 was coded by one rater, partly from third-party mirrors and low-confidence sources. As part of this revision every row was re-verified against **first-party artifacts** (official release listings, official READMEs and papers, at pinned commits). The re-verification

confirms some cells and corrects others; both are listed, because a correction record that reported only the corrections would misrepresent the audit's reliability.

Counting convention used for `release` in this record (the *reported-test* convention). A row's `release` value describes whether the layer supplies **public annotation for the test split that the literature actually reports on**. A public annotated `val` split, on its own, does **not** make `release = independent_test`; it is recorded as `no_val` when no test artifact exists, and as `test_gated` when a test split exists but its annotation is withheld or server-bound. Under the alternative convention (any public annotated held-out split, `val` included) five rows -- COCO, DOTA v1.0, DOTA v2.0, WIDER FACE and CrowdHuman -- would all read `independent_test` instead, and the `release` subtotal would change with them. The two conventions are each self-

consistent; **this revision uses the reported-test convention throughout**, and the alternative is stated here so that the subtotal is not recomputed under a mixed reading.

Corrections (value changed).

Row	Unit	Was	Now	Ground
4 Open Images v7	`yolo_dist`	`no_test_key`	`absent`	The distribution config's `test:` key is **present but empty** (`open-images-v7.yaml`; docs: "The `test:` key in the configuration is left empty"), so the distribution supplies no test artifact
4 Open Images v7	`reported`	`alias`	`independent`	Official facts page: a validation set and a **125,436-image test set**, with "**exhaustive box annotation** for all object instances"; `HEAD test-annotations-bbox.csv` returns 200 (77,484,237 B)
5 DOTA v1.0	`release`	`independent_test`	`test_gated`	First-party Drive listing: train and val carry `labelTxt-v1.0`/`-v1.5`; the **test tier carries `images/` only**
5 DOTA v1.0	`protocol`	`independent_test`	`test_gated`	"For evaluation, you must registrate and submit on the Evaluation Server" (Task 1 and Task 2)
6 DOTA v2.0	`release`	`independent_test`	`test_gated`	"We released the images **but not the ground truths**" (test-dev); test-challenge "available only during the challenging"
6 DOTA v2.0	`yolo_dist`	`clean`	`absent`	The 52 distribution configs contain `DOTAv1.yaml` and `DOTAv1.5.yaml` only; the official devkit ships v1.0/v1.5 evaluators and **no v2**
7 VisDrone-DET	`release`	`independent_test`	`test_gated`	Official README: "testset-challenge is used for VisDrone2020 Challenge and **the annotations is unavailable**"
7 VisDron	`protocol`	`independent_test`	`test_gated`	Same source; the distribution config is

e-DET				annotated "# Download (ignores test-challenge split)"
8 AI-TOD	`release`	`contradictory`	`independent_test`	The official Drive listing **enumerates the test annotation file**; measured test 14,018 images / 347,617 annotations, and train+val+test = 28,036 with 700,621 annotations, which reconciles exactly
8 AI-TOD	`protocol`	`contradictory`	`independent_test`	As above (the paper's "images without annotations" sentence conflicts with the shipped annotation file; the measured artifact governs)
8 AI-TOD	`yolo_dist`	`unknown`	`absent`	Twelve search surfaces (full trees of three Ultralytics repositories, the unpacked 8.4.174 wheel, the documentation index, and four other frameworks) return **zero** hits; the same surfaces do hit the other rows, so the search is sensitive
8 AI-TOD	`reported`	`unknown`	`independent`	As above; the test annotations are public and the counts reconcile
10 xView	`release`	`no_test`	`no_val`	The official listing carries three items only (`train_images`, `train_labels`, `val_images`) and the tutorial states "**Labels are not available for the validation set**"
10 xView	`protocol`	`no_test`	`test_gated`	Official site: "**Neither images nor labels from the holdout set are available for download**"; scoring requires container submission (the download endpoint returns 401 anonymously)
11 DIOR	`yolo_dist`	`no_test_key`	`no_yaml`	Zero `dior` hits across the full tree and the 51 shipped configs -- there is **no config at all**, so `no_test_key` (which presupposes one) is wrong

13 SHWD	`release`	`no_yaml`	`no_split`	The official repository's 26 files contain no annotation, no split and no config; the released artifact is a VOC-format archive (a second-party conversion adds `ImageSets/Main/{train,val,trainval,test}` and is **not** the official listing)
13 SHWD	`yolo_dist`	`train_val_alias`	`absent`	No distribution config exists for this dataset in any of the six packages swept
13 SHWD	`reported`	`alias`	`not_recorded`	The README prints benchmark scores (map 88.5/86.3/75.0) but **never states which split produced them**, and never gives split sizes; the only split statement is in code (`splits=[(2028,'trainval')]` / `[(2028,'test')]`)
14 SFCHD	`release`	`alias`	`contradictory`	The release ships **three mutually inconsistent splits**: `new_split_yolo` (9,897 / 2,475 / 6, with `test` a subset of `val`), `yolo.zip` (9,898 / 2,475 / **no test**), and `annotations/` (9,898 / 2,475 / **`test.json` is 0 bytes**); the A-versus-B/C train intersection is only **7,942** and the val intersection only **520**
14 SFCHD	`protocol`	`no_val`	`contradictory`	As above; the four YOLO cache files embed **two different developer root paths** and contain **zero** `.yaml`/`.yml` references
14 SFCHD	`yolo_dist`	`alias`	`absent`	Zero hits for this dataset across the full trees of the Ultralytics packages and four other frameworks; the only helmet-class config found has a different class list, size and download source
15 MAFA	`yolo_dist`	`unknown`	`absent`	Six packages swept with zero hits; the only MAFA-labelled YOLO directory carries no config file

15 MAFA	`reported`	`alias`	`independent`	The official SS5.1 states "we only report the performances ... on the **testing set**" and Table 1 is titled "Average Precision (%) on the **Testing Set** of MAFA"; the follow-up literature reports on the testing set, and no work re-cuts a validation split
17 WIDER FACE	`release`	`independent_test`	`test_gated`	The official split archive contains train and val bounding-box ground truth and **no test ground truth**; "we do not release bounding box ground truth for the test images"
17 WIDER FACE	`yolo_dist`	`no_test_key`	`absent`	No config or documentation entry for this dataset across the three distribution directories swept
18 CrowdHuman	`protocol`	`alias`	`test_gated`	The official listing carries `annotation_train.odgt` and `annotation_val.odgt` only; the first author's own mirror shows `CrowdHuman_test.zip` holding **5,018 `.jpg` entries and zero annotations**, and `annotation_test.odgt` returns 404
18 CrowdHuman	`yolo_dist`	`no_test_key`	`absent`	As WIDER FACE (the ground is upgraded from a third-party mirror to an exhaustive sweep of the distribution directories)
19 D-Fire	`release`	`independent_test`	`contradictory`	The full repository tree is six entries (**no config, no split list, no evaluator**; releases and tags empty) while the README claims "pre-split training, validation, and test sets"; the linked OneDrive folders are unreachable (301 then 000), and a 2026 record of the official archive states it "defines only a training list of 17,221 images and a test

				list of 4306; **it ships no validation list**"
19	D-Fire	`yolo_dist`	`clean`	`absent`	Zero hits for fire/smoke across the full Ultralytics tree (1,185 entries) and the shipped configs
19	D-Fire	`reported`	`unknown`	`alias`	The authors' own thesis states the database was "divided **arbitrarily** ... 80% for training (17,221) and 20% for test (4,306)", matching the mirror's split and its interleaved file names

One direction change inside the table's own history. The pair {WIDER FACE, CrowdHuman} was recorded as a disagreement between coders. The two rows are indeed structurally identical, and the re-verification resolves them in the direction of **lowering WIDER FACE to `test_gated`**, not of raising CrowdHuman to `independent_test`.

Confirmations under first-party evidence (value unchanged).

Row	Unit	Value	What was upgraded
11	DIOR	`release`, `protocol`	`independent_test`	From a third-party mirror (confidence low) to the official package: `Annotations.zip` (32,064,775 B, downloaded and unpacked) carries annotation for **all 11,738 test images**, and `ImageSets.zip` carries `Main/test.txt`; the paper's own sentence gives the 11,725/11,738 trainval/test division and there is no `trainval.txt`
9	UAVDT	`release`,	`no_val`	The official split

	`protocol`		is "30 training sequences (24,143 frames) and 70 test sequences (53,676 frames), **with no official validation split**"; the official package `UAVDT-Benchmark-M` carries frames and annotations
15 MAFA	`release`, `protocol`	`no_val`	The official site is reachable (its TLS certificate has expired, so retrieval requires `-k`) and lists the release artifacts; the **author-signed** readmes inside the label packages give 25,876 training and 4,935 testing images, and the CVPR paper's SS5.1 gives the same two-subset division with no validation subset
19 D-Fire	`protocol`	`independent_test`	No evaluator, leaderboard or release exists, so the test list is used verbatim locally (a 2026 record: "Official test list (4306) used verbatim")
13 SHWD	`protocol`	`no_val`	The repository's 26 files contain no split; the split statement exists only in code, and gives trainval/test with no independent

10 xView	`yolo_dist`, `reported`	`alias`	val The distribution config's `val` is an autosplit of 10% of the training images ("10% of 847 train images"), not the official 282-image val

Mis-citations corrected in the article and in this supplement.

*Row 16's source was cited as a Mendeley Data record. A title sweep of DataCite and Mendeley Data (2026-10-09) returns no record with that title; the 853-image, 3-class, PASCAL-VOC form is the Kaggle** record `andrewmvd/face-mask-detection` (CC0). The reference in the article has been changed accordingly.

*Row 16's `train == val` evidence was cited at `data.yaml:18`. The `data.yaml` is four lines (110 bytes); the literal evidence is in the generator** `convert_voc_to_yolo.py:17-18`, which writes the same image directory under both keys. The citation now names the generator.

* Row 12's TorchGeo source is now at `torchgeo/torchgeo` (the old `microsoft/torchgeo` path redirects); the `positive`/`negative` split in the dataset module is target presence, and the train/val/test split is created at runtime by `random_split(seed 0, 80/10/10)`.

Still not decidable from the public record. Eight rows keep at least one cell that first-party retrieval could not settle: Objects365 (`release`, `protocol` -- the official download page serves a single-page application with no listing and the official repository's README is 53 bytes, so the honest statement is "the official side does not publish a listing", **not** "the official side declares there is none"); COCO, PASCAL VOC, DOTA v2.0, WIDER FACE and CrowdHuman (the `reported` unit, which needs a literature sweep rather than a first-party artifact); CrowdHuman (no written prohibition sentence equivalent to WIDER FACE's, and the evaluation platform is unreachable); and Mendeley face-mask (`reported`: neither the Kaggle card nor any first-party statement gives the evaluation ratio).

A further trap worth naming. The frequently cited Chetoui & Akhloufi *Fire* (2024) baseline is

not this corpus: it reports on 11,667 images, not 21,527. Likewise, a Kaggle face-mask dataset of **853** images is cited as MAFA by at least one paper ("MAFA[17] contains 6k data"); it is not MAFA (25,876 + 4,935 images). Both were excluded from the audit.

The four subtotals after the 2026-10-09 re-verification (added 2026-10-10)

The correction record above changes **30 cells**. The article's Table 3 printed the four subtotals before that record and §13 repeated them; **neither had ever been recomputed**, which is the gap this section closes. They are recomputed here from the S1 marking table plus this record and nothing else, by the **same per-column rule that reproduces the printed subtotals exactly** (`work/recompute_subtotals_corrected_20261010.py`; the script **asserts the printed values before it reports the corrected ones**, so a wrong value-to-count mapping fails loudly instead of producing a quiet number). **The article's printed counts are not modified: they stay, and the recomputed values are printed beside them.** Holding the rule fixed is what makes every movement below a movement of **evidence** rather than of convention.

The rule, per column — which values enter the non-independent numerator.

Column	counts as non-independent	does **not** count	undecidable
`release`	`no_test`, `no_val`, `no_split`, `no_yaml`, `test_gated`, `alias`, `contradictory`	`independent_test`	— (`n_a`: 2 rows, denominator only)
`protocol`	`no_test`, `no_val`, `no_split`, `test_gated`, `alias`, `contradictory`	`independent_test`	—
`yolo_dist`	`alias`, `train_val_alias`	`clean`, `no_test_key`, `no_yaml`, `absent`	`unknown`
`reported`	`alias`	`clean`, `independent`	`unknown`, `not_recorded`

Three of these four rows are the rule the printed table already used; the `release` row **is** the `n_a` rule stated in "The `n_a` rule the `release` subtotal rests on", above. The `yolo_dist` row is worth spelling out, because it is the one a reader is most likely to get wrong: this column asks a single question — **does the generic distribution package's own config bind `val` to `test`?** A package that ships **no config at all** for a dataset (`no_test_key`, `no_yaml`, `absent`) cannot bind the two, so it does not enter the numerator; but it is not `clean` either, because nothing positively shows a clean independent split. Under that rule the 2026-10-09 record moved **four** of the marks that had counted or read `clean`: `alias` → `absent` (SFCHD), `train_val_alias` → `absent` (SHWD), `clean` → `absent` (DOTA v2.0, D-Fire). That is why this unit **drops** rather than rises.

The recomputed subtotals.

Unit	printed (single rater)	after first-party re-verification	of decidable rows
`release`	10/19 = 52.6 %	**14/19 = 73.7 %**	10/19 → **14/19**
`protocol`	13/19 = 68.4 %	**14/19 = 73.7 %**	13/19 → **14/19**
`yolo_dist`	4/19 = 21.1 %	**2/19 = 10.5 %**	4/15 = 26.7 % → **2/17 = 11.8 %**
`reported`	12–14 of the 19	**10–11 of the 19**	12/15 = 80 % → **10/16–11/17 = 62.5–64.7 %**

The two **layer** units rise and the two **package/report** units fall. The article's qualitative headline — that a **majority** of the nineteen benchmarks report a number which is not a sample from an independent split — survives every cell of the table: the weakest recomputed majority is `reported` at 10/19 = 52.6 %.

Two boundaries a reader should carry. ① COCO `release` is `n_a`. The convention paragraph of the record above implies that under its own *reported-test* convention COCO reads `test_gated` (its `test-dev` annotation is private), which would take `release` to **15/19**; the record does **not** list that cell as changed, so the recomputation leaves it as printed. The **14-versus-15** difference is therefore a **convention boundary, not a disagreement about evidence**, and it is where two independent recomputations of this table can legitimately part company. ② **`not_recorded`** (SHWD `reported`, a value this record introduces) is deliberately **not assigned** above, and it is the reason the

`reported` unit is given as a range: **10/19** if a split that is nowhere recorded is treated as evidence-insufficient (its `unknown` count rises 2 → 3), **11/19** if a split nowhere recorded is simply not an independent one (its `unknown` count falls 2 → 1). As with the printed `12–14`, **the range spans two readings, not an interval.**

Two `14/19`s, and they are not the same number. This supplement prints `14/19` for two different cells of two different instruments. **Above**, in the independent re-coding record, the `**reported**` column reaches `14/19 = 74 %` under the second, model-assisted pass's reading of two rows the printed table leaves `unknown`. **Here**, the `**release**` column reaches `14/19 = 73.7 %` after the 2026-10-09 first-party re-verification. They are different units, different evidence and different instruments; they must not be added, averaged or interchanged, and **neither is the audit's original `14/19`**, which §6.3 shows this paper does not reproduce.

What this recomputation is not. It is not the second, model-assisted non-blind pass, whose instrument differs and which moves **eight** cells: that pass is reliability evidence about the

instrument, whereas this is the **printed marking table** re-tallied after the first-party record. Nor does it re-adjudicate any cell — the thirty values are the correction record's, and this section only tallies them. In the article, Table 3's re-verified column prints **counts only**, no percentages, for the reason §6.1 gives for the whole table: the count is the datum and the percentage is a reading aid.

Clauses moved out of the article's §6.1 (added 2026-10-10)

Five clauses move here from the article's §6.1 (the self-check's column and subtotal equalities to **S1**): the COCO row's judged number, the `yolo_dist` first-version correction (**4/19**), the literal-two-key self-check (**2/19**), Mendeley face-mask's **generator** provenance, and the audit's unused "63–79 %" bound. The clause is **relocated, not dropped**: it is kept here so that the article's §6.1 stays inside the journal's page limit.

Table S45 — §2.1 relocated from the article (detailed related-work comparison)

> Moved here at revision because the journal counts text inside the 20–35 page limit. The article keeps

> the one-paragraph positioning statement in §2 and the two delimiting sections (§2.2, §2.3).

2.1 The closest work, and where this paper sits

> **Table S1** → Supplementary §S8 — verified against both 2026 studies' **full texts**, with the PDFs and extraction script archived.

None of these works is wrong — **they audit data (image identity, or the splitting unit), we audit configurations** (which path each key points to). Configuration-level aliasing shifts the

comparison between arms, and its direction is **not single-valued** — a **weaker** statement than [2], whose rankings reverse across budgets on **all** of its benchmarks. Here **2 of 13 cells are underestimated on `val`, 2 overestimated and** 9 are not testable at n = 3**, **so no more than that is claimed. They instantiate the taxonomy of [3], of which duplication is one mismatch [4]; binding `val:` and `test:` to one directory creates the same mismatch** without duplicating an image**, so de-duplication cannot find it.

Non-collision, checked by literal search: three exact queries return **0** hits and the channel is **metadata-only** — which establishes that these strings are unused, **not that the idea is new** (**Supplementary S12**).

Why this is a selection question, not only a leakage question. The bias we measure is the one the model-selection literature has described for decades: **choosing a model by its score on a validation set makes the reported score optimistic** [5], as does the **choice of validation criterion** — accuracy-based early stopping being the worst rule tested [6] — and the classical remedy is to *nest* the selection inside the evaluation, not to de-duplicate the data. In detection the selection step is implicit: the training script writes `best.pt` by fitness and the paper reports that checkpoint, often from a single run [7]. A reader trusting the split names therefore sees a held-out evaluation where the pipeline performed a **maximum** — the bias the benchmark-erosion literature [8] describes, reached here with no data reused or mis-split. We take [5] as the general statement and report the detection-specific instantiation: **the selection premium (best − last), and how much of it transfers to a disjoint split** (§5, §8).

The sharpest contrast is the two closest works: *Technologies* 2026 [9], which audits a detection pipeline's **splitting unit** (frame versus sequence), and *Drones*

2026 [10], whose six configurations differ between `val` and `test` by **between −0.41 and +0.49** mAP@0.5, "with no systematic direction". We measure a **different quantity**: on three-way corpora we **measure the paired difference's sign**. We therefore do **not** write "understated" unqualified, and state the difference from [10]: [10] **names the mechanism** — "the validation split is what early stopping and checkpoint selection saw" — but reports a raw val-minus-test gap, mixing two splits' difficulty, **not identifiable**, whereas our **paired difference** cancels difficulty and still shows a residual shift. Reference [11] is not competing: cited only as evidence that the genre has a venue.

That separation is the paper's own contribution. The literature above measures how much of the choice survives as a level [12]; it does not predict that in these curves the premium is **dominated by the peak's position, not by run noise** (§8.6). Its remedies are therefore not interchangeable with ours, and **a `val`-side difference stays uninterpretable as a generalisation difference even after selection is removed**.

Reproduction map: headline number → entry point

Every headline number that a script produces maps to one entry below — a producing script, or an explicit note that no script produces it — so a reader can either recompute a number or see at once that we cannot offer a script for it. Three headline counts come from the audit's own records rather than from a produced table — §4.1's 6 of 14, §5.5's 20-log sample and §8.3's 9 %–29 % bound — and are marked at their point of use rather than mapped here. Paths are inside the archived reproduction package (see the Data availability statement in the article); `02_release_data/` holds the derived tables the article's tables are generated from.

Headline number	Derived table (package)	Producing script
23.3 % realization rate and its three (coincident) weightings (§8.6, Table 4)	`release_selection-transfer_perseed.csv` (**four of the five arm-cells**; the `aitod20` cell's completion to n = 10 is **not** in this package, see the scope note below), `release_shape-noise-by-budget.csv`	`work/item2_final_v2_20260925.py`
κ and its clustered	`x3b_tableS22_20260918.csv`,	`work/x2_kappa_clustered_20260918.py`,

error (§8.7)	`x3b_tableS23_20260918.csv`	`work/x3b_perepoch_corrected_20260918.py`
shape share of the premium (§3.1 corollary)	`release_shape-noise-by-budget.csv`	`work/prop6_shape_noise_20260920.py`
transfer slope for the shape/noise decomposition	`release_selection-transfer_shape-share.csv`	`work/prop6_transfer_20260920.py`
Tier 1 / Tier 2 paired and permutation statistics (§5.1)	`release_X1_tier2_best-last_summary.csv`	`work/xeval_analyze_20260916.py`
registered replication and its saturated control (§9)	`release_T1_registered-replication_summary.csv`, `release_T2_saturated-control_summary.csv`	`work/preregister_E_ext_20260919.py`
G1″ / G1′ / G2 / G3 / G4 verdicts (§3.2)	`release_G1p_AB-half-level-match.csv`, `release_G1pp_thirds_levels.csv`, `release_G2_selection-split-size_summary.csv`, `release_G3_epoch-budget_summary.csv`, `release_G3ext_E50-400_perrun.csv`, `release_G4_mende20_3way_perseed.csv`	`work/g1p_verdict_20260919.py` and the verdict scripts named in Tables S31–S33
the 19 × 4 counting-unit table (§6.1, Table 3)	recomputed in-package	`work/recompute_split_units_20260916.py`
the framework-default check (§6.5)	line-by-line evidence in §S3	`work/x10_alias_audit_20260918.py`
the three variance componen	recorded in Table S19 only	`work/sigma_decomposition_20260926.py` — recomputes the data-order and initialization

ts (§8.4, Table S19)	components from the archived per-run values; **cites** the augmentation component, whose record has no per-run values (Defect 5)
the 289 in-package audit anchors (Declarations)	`work/audit_anchors_fupaper.py`

The three variance components of §8.4 are the one entry in this map whose script covers two of its three components rather than all three. The data-order and initialization components are recomputed from the archived per-run values; the augmentation component's record holds an SD and no per-run values, so the script cites it and labels that line `cited, not recomputed`. §10 defect 5 and the item-by-item Defect 5 below state the same thing, and we would rather name the gap than write a script that only re-prints a recorded constant.

The pre-declared criteria are files, not recollection. `work/preregister_E_ext_20260919.py` writes the G-batch criteria *before* the runs; `work/g1p_verdict_20260919.py` applies them *after*. Each verdict in §3.2 is therefore recomputable from a criterion that existed in advance of its data.

The audit's coding rules are stated, not implied. The 19-row marking table of §S1 is normative: each row records its evidence class and the reading it was coded under, and the four subtotal equalities are recomputed mechanically by `work/recompute_split_units_20260916.py` rather than hand-summed. What is recomputable here is the *mechanical* consistency of the table with its own markings, and that recomputation ships with the table (§S1, "Recomputation results"); the second-rater point is made once, below.

Row-level evidence pointers for the 19-row marking table of §S1. The marking table is coded by hand, by one coder, from the configuration audit: no script assigns a mark, and a second rater would be needed for a true inter-rater statistic, which this paper does not claim. **The pointer for each row is the audit's own row, printed in §S1 beside the mark together with that row's evidence class and confidence level**, so any mark can be traced to the excerpt it was read from without re-deriving it; the audit file, its version and the reading policy are named in §S1's header, and §S0's table gives that file's md5 and byte

count. What *is* mechanical is the step from the marks to the numbers — and that step, and only that step, is scripted.

Where the mechanical recomputation starts and stops. Every count and percentage in §S1's "Recomputation results", the "among decidable rows" denominators, the four levels of "same directory" and the cross-check against the audit's published counts are recomputed from the 19 rows' own marks by `work/recompute_split_units_20260916.py`, with no percentage hard-coded; the text of §S1's table and of its recomputation block is that script's own output, and the same script is the entry point this table is given in the reproduction map above. **It covers that step and nothing upstream of it**: it consumes the 19 rows as literals, and it does not choose a benchmark, open a release package, read a dataset paper, or check a document version. Two caveats follow. Running it is not a read-only reproduction — it writes its two artifacts to an absolute path and overwrites them in place, so a verifier should read the archived artifacts rather than re-run the generator. And its own upstream is a hand-written audit: the marks it recomputes *from* were not derived by any script either.

What each row's evidence pointer resolves to, and what is now on record. For every row, the recomputation artifact `split_units_recompute_20260916.txt` carries that row's own entry — its evidence class, its self-assessed confidence and its derivation or correction note — and `work/recompute_split_units_20260916.py` carries the same text as that row's literal. **The concrete source for all nineteen rows is now archived**: each row below names the upstream repository and commit the mark was read at, the file, the line the mark stands on, and the SHA-256 of the retrieved bytes. Read as a chain, one row is now walkable end to end — the English row, that row's entry in `split_units_recompute_20260916.txt`, the same row's literal in the script, the audit's mark, and the upstream file with its line anchor and hash.

What is still not on record, stated because it bounds the chain. The row-level source fields above are archived; the **audit's own line numbers** (`A:…`) remain unrecorded except for row 19, whose note names two of them, because the audit document's internal numbering is not regenerated by any shipped script. **Every one of the nineteen rows is pinned to a commit in this package**: the commit

was resolved with `git ls-remote` on the repository's default branch, the file was re-fetched at that commit, and its bytes were checked to be identical to the archived copy (`sha256`), so the pinned commit is the commit of the bytes read and not merely a later state of the same file. Where the currently published upstream file differs from the mark, the drift is recorded at the end of this list rather than silently reconciled. **No printed mark, count or percentage in §S1 is changed by this list.**

How to read the list below. Each line carries that row's evidence class and self-assessed confidence as §S1's table prints them, its four printed marks in column order (release / protocol / yolo_dist / reported), the archived file with its line anchor, and the SHA-256 of that file's first twelve hex digits. Every row's commit is resolved: re-fetching each file at its pinned commit reproduces the archived `sha256` byte for byte.

1. COCO — evidence class: Official counts + mirror yaml; confidence: high; marks: n_a/independent_test/clean/clean; concrete source: `ultralytics/ultralytics@ce1399439b93 ultralytics/cfg/datasets/coco.yaml:15` (sha256 `38ed1eb122fa`, archived as `ultralytics_ultralytics__ultralytics__cfg__datasets__coco.yaml`).

2. PASCAL VOC — evidence class: mirror VOC.yaml (both keys cited per row); confidence: high; marks: n_a/test_gated/alias/alias; concrete source: `ultralytics/ultralytics@ce1399439b93c490b648b55dedd0a6334f0199da ultralytics/cfg/datasets/VOC.yaml:19` (sha256 `6e2dae457723`, archived as `ultralytics_ultralytics__ultralytics__cfg__datasets__VOC.yaml`).

3. Objects365 — evidence class: mirror yaml + Ultralytics docs; confidence: high; marks: no_test/no_test/no_test_key/alias; concrete source: `ultralytics/ultralytics@ce1399439b93c490b648b55dedd0a6334f0199da ultralytics/cfg/datasets/Objects365.yaml:15` (sha256 `7ca822c2f9ed`, archived as `ultralytics_ultralytics__ultralytics__cfg__datasets__Objects365.yaml`).

4. Open Images v7 — evidence class: Official facts page + mirror yaml; confidence: high; marks: independent_test/independent_test/no_test_key/alias; concrete source: `ultralytics/ultralytics@ce1399439b93c490b648b55dedd0a6334f0199da ultralytics/cfg/datasets/open-images-v7.yaml:15` (sha256 `497468af9dc6`, archived as `ultralytics_ultralytics__ultralytics__cfg__datasets__open-images-v7.yaml`).

5. DOTA v1.0 — evidence class: Official download page + mirror DOTAv1.yaml; confidence: high; marks: independent_test/independent_test/clean/clean; concrete source: `ultralytics/ultralytics@ce1399439b93c490b648b55dedd0a6334f0199da ultralytics/cfg/datasets/DOTAv1.yaml:15` (sha256 `18e3ca2f9254`, archived as `ultralytics_ultralytics__ultralytics__cfg__datasets__DOTAv1.yaml`).

6. DOTA v2.0 — evidence class: Official/mirror documentation; confidence: high; marks: independent_test/test_gated/clean/alias; concrete source: `ultralytics/ultralytics@ce1399439b93c490b648b55dedd0a6334f0199da ultralytics/cfg/datasets/DOTAv1.yaml:5` (sha256 `18e3ca2f9254`, archived as `ultralytics_ultralytics__ultralytics__cfg__datasets__DOTAv1.yaml`).

7. VisDrone-DET — evidence class: mirror + converter; confidence: high; marks: independent_test/independent_test/clean/clean; concrete source: `ultralytics/ultralytics@ce1399439b93c490b648b55dedd0a6334f0199da ultralytics/cfg/datasets/VisDrone.yaml:15` (sha256 `627f60f7602d`, archived as `ultralytics_ultralytics__ultralytics__cfg__datasets__VisDrone.yaml`).

8. AI-TOD — evidence class: Official README (contains two mutually contradictory release statements); confidence: low; marks: contradictory/contradictory/unknown/unknown; concrete source: `jwwangchn/AI-TOD@7f56cb6b2aaa README.md:39` (sha256

0491ad0fc58a, archived as jwwangchn_AI-TOD__README.md); Chasel-Tsui/mmdet-aitod@e3e567111490 README.md:39 (sha256 f924c74b048b, archived as Chasel-Tsui_mmdet-aitod__README.md).

9. UAVDT — evidence class: Official paper + two mirrors (the audit calls these unverified beyond these two); confidence: medium; marks: no_val/no_val/unknown/alias; concrete source: CQNU-ZhangLab/SFFNet@1e6a3c9cab8e4129aba1b8196625026a5d482ccb ultralytics/cfg/datasets/UAVDT.yaml:4 (sha256 ff5b076b3bce, archived as CQNU-ZhangLab_SFFNet__ultralytics__cfg__datasets__UAVDT.yaml); forever208/yolov5_train_on_UAVDT@a14a46296fd7f370a615a6151742373e5e2435d7 data/UAVDT.yaml:4 (sha256 cb9616967927, archived as forever208_yolov5_train_on_UAVDT__data__UAVDT.yaml).

10. xView — evidence class: mirror xView.yaml; confidence: low; marks: no_test/no_test/no_test_key/alias; concrete source: ultralytics/ultralytics@ce1399439b93c490b648b55dedd0a6334f0199da ultralytics/cfg/datasets/xView.yaml:14 (sha256 9b6a565f0fd2, archived as ultralytics_ultralytics__ultralytics__cfg__datasets__xView.yaml).

11. DIOR — evidence class: Official description (no counting) + mirror DIOR-R yaml; confidence: low; marks: independent_test/independent_test/no_test_key/unknown; concrete source: Hamedlk80/DIOR-R-Rotated-Object-Detection-YOLOv8@6be5a006533e7b6a985f38d0004eaf58afcb8881 diorr_obb_analysis/data.yaml:2 (sha256 5fb9761fbb1c, archived as Hamedlk80_DIOR-R-Rotated-Object-Detection-YOLOv8__diorr_obb_analysis__data.yaml).

12. NWPU VHR-10 — evidence class: Official TorchGeo (the split means positive/negative); confidence: high; marks: no_split/no_split/unknown/alias; concrete source: torchgeo/torchgeo@ceb558ee1616d545431d6a59c41a2276b18025bd torchgeo/datasets/vhr10.py:120 (**repo renamed**: the old path microsoft/torchgeo redirects to torchgeo/torchgeo; HEAD at 2026-

10-09 is 379be63764354f078e626df239f4c1c7532959da, at which datasets/vhr10.py still carries split: Literal['positive','negative'] -- target presence, not train/test; the train/val/test split is created at runtime by datamodules/vhr10.py via random_split(seed 0, 80/10/10)) (sha256 32efdc6e5479, archived as microsoft_torchgeo__torchgeo__datasets__vhr10.py).

13. SHWD — evidence class: Official README + mirror Reflective_vests.yaml; confidence: medium; marks: no_yaml/no_val/train_val_alias/alias; concrete source: njvisionpower/Safety-Helmet-Wearing-Dataset@c952631dfc352e0defdd54b4785a9be6576451ad README.md:59 (sha256 245b47068ebf, archived as njvisionpower_Safety-Helmet-Wearing-Dataset__README.md); gengyanlei/fire-smoke-detect-yolov4@98b1fec0f82e09d67ef5fc657a80eaf0b1450360 yolov5/data/Reflective_vests.yaml:59 (sha256 c142a50b7a36, archived as gengyanlei_fire-smoke-detect-yolov4__yolov5__data__Reflective_vests.yaml).

14. SFCHD — evidence class: Official first-party release (machine comparison _ev_sfchd_check2.py); confidence: highest; marks: alias/no_val/alias/alias; concrete source: lijfrank/SFCHD-SCALE@7bfabb453361 dataset_SFCHD/new_split_yolo/train.txt:1 (sha256 e2ef62278eb4, archived as lijfrank_SFCHD-SCALE__dataset_SFCHD__new_split_yolo__train.txt); lijfrank/SFCHD-SCALE@7bfabb453361 dataset_SFCHD/new_split_yolo/val.txt:1 (sha256 4c2ad09163c3, archived as lijfrank_SFCHD-SCALE__dataset_SFCHD__new_split_yolo__val.txt); lijfrank/SFCHD-SCALE@7bfabb453361 dataset_SFCHD/new_split_yolo/test.txt:1 (sha256 80e54955c64b, archived as lijfrank_SFCHD-SCALE__dataset_SFCHD__new_split_yolo__test.txt).

15. MAFA — evidence class: Official CVPR PDF (extracted with pypdf); confidence: high; marks: no_val/no_val/unknown/alias; concrete source:

ElenaRyumina/AnnotationMAFA@5d0b451c0c69fbaeeed2ea798eb912fb9c6339fc README.md:5 (sha256 e0ab2118a8b6, archived as ElenaRyumina_AnnotationMAFA__README.md).

16. Mendeley face-mask — evidence class: **Kaggle data record** (andrewmvd/face-mask-detection, 853 images / 3 classes / PASCAL-VOC XML) + mirror generator; **no Mendeley Data record with this title exists** (DataCite title sweep 2026-10-09: total 29, none with this title; the nearest Mendeley item is Face Mask Dataset 2022, 10.17632/7bt2d592b9, 20,347 images / 2 classes / image-classification, with no split statement); confidence: lowest; marks: no_split/no_split/alias/unknown; concrete source: Prikshit7766/Face-Mask-Detection@ecd8be957c3a14e57d4f00cd9e62024d192479fd convert_voc_to_yolo.py:18 (sha256 75c5bf6055db, archived as Prikshit7766_Face-Mask-Detection__convert_voc_to_yolo.py); Prikshit7766/Face-Mask-Detection@ecd8be957c3a14e57d4f00cd9e62024d192479fd data.yaml (**4 lines / 110 bytes**, sha256 d75e524c03ba, archived as Prikshit7766_Face-Mask-Detection__data.yaml; contents: train: ../train/images, val: ../test/images). **Correction (2026-10-09)**: the train == val literal is in the **generator** convert_voc_to_yolo.py:17-18 (the same images_dir written under both keys), **not** at data.yaml:18; the earlier citation implied a longer config file than the 4-line file that exists.

17. WIDER FACE — evidence class: Official site (GT not released, verified word by word) + mirror; confidence: high; marks: independent_test/test_gated/no_test_key/alias; concrete source: spacewalk01/yolov5-face@84ef013381e3a155b7a746215974dca7894e0f3e data/widerface.yaml:1 (sha256 c39567becdf7, archived as spacewalk01_yolov5-face__data__widerface.yaml).

18. CrowdHuman — evidence class: Official paper + mirror; confidence: high; marks: test_gated/alias/no_test_key/alias; concrete source: yakhyo/yolov8-crowdhuman@18db0736cbdf4881cde3eb8c86a2d4232ba3ec52

`dataset.yaml:2` (sha256 `684bd619eb9b`, archived as `yakhyo_yolov8-crowdhuman__dataset.yaml`).

19. D-Fire (FireSmoke) — evidence class: Official README; confidence: medium; marks: independent_test/independent_test/clean/unknown; concrete source: `gaia-solutions-on-demand/DFireDataset@4bf9c31b18fadcd44d5f0b6d66f82bc56fa5e328 README.md:51` (sha256 `c5395ce455e0`, archived as `gaia-solutions-on-demand_DFireDataset__README.md`).

Two rows whose currently published upstream file differs from the printed mark (recorded, not reconciled). **Objects365**: the printed `yolo_dist` mark records that the distribution-package configuration carries no `test:` key, while the file archived here (Ultralytics `Objects365.yaml`, line 15) does carry one. **VisDrone-DET**: the printed `yolo_dist` mark is `clean`, while the archived `VisDrone.yaml` (line 15) points `test:` at `images/test`. Both are evidence that upstream configurations change; the marks record what the audit read on its own date, and this list records what was read on the date of this archive.

Scope of reproducibility: two things that are not the same

What	Status	Why
Evaluation-side recomputation — every premium, rate, κ, interval and verdict in the article	**reproducible from the package**	needs the per-epoch matrices and the released tables, all shipped
Retraining — re-running training to regenerate the checkpoints those numbers were computed on	**not reproducible from the package**	the archive holds no copy of the three-way carve directories and no copy of 6 of the 7 corpora (§10 defect 6)

This distinction is why the article claims **evaluation validity** rather than full reproducibility: every number we report is recomputable from what is shipped, while regenerating the checkpoints behind them would require downloading the corpora again — they are **open**, but the archive **holds no copies** of six of the seven, so the images are not on hand (`§10`, defect 6).

Why a second pass, and what it is not. The single-rater disclosure above is unchanged, and a second human rater is still what a true inter-rater statistic would need. What has been added is a **second coding pass, model-assisted and non-blind** (run 2026-09-27): it re-coded all 19 rows × 4 units of the §S1 table

under the published rules, after the rules were first separated from the printed conclusions into a standalone manual (**withdrawn from the released package**; the earlier coding rounds are not published). Because §S1 prints rules and marks in one place, the pass could not be blinded — the printed coding was visible throughout, and the statistics below are a consistency check against a visible target, not a blind inter-rater result. **The second pass is model-assisted, not a second human rater, and it does not replace the single-rater disclosure printed with the table.** Its per-cell marks are archived with the audit ground for each row (`work/the earlier-round coding table (**withdrawn from the released package**; not published)`), and every statistic below is recomputed mechanically by a script that reads the printed table and that file (`work/kappa_the earlier coding rounds (**withdrawn from the released package**)_20260927.py`; machine output `work/kappa_computed_20260927.txt`, readable summary `work/kappa_summary_20260927.md`). The pass coded against the archived configuration audit — the evidence base one step upstream of the printed table, whose English rows mirror a Chinese authoritative file.

Agreement with the printed coding. On the decision scale the printed counts use (non-independent / clean / no-test-key / unknown / not-applicable), the per-unit κ is

0.716 for release (95 % interval 0.422–1.000), **1.00** for protocol (all 19 cells agree; the interval degenerates at 1.00), **0.929** for yolo_dist (0.792–1.000) and **0.784** for reported (0.501–1.000), and the overall κ over the 76 judgments is **0.874** (95 % interval 0.778–0.971). Treating the four printed `unknown` cells as a third category or dropping the eight cells where either coding is unknown moves the overall κ between

0.874 and **0.922** (0.835–1.000); on the literal marks, exact agreement is 68/76. The intervals are wide at n = 19, and the two lower units sit in the "substantial" rather than "almost perfect" band — the disagreement concentrates exactly where the manual is under-specified, which is itself a finding.

The eight cells that differ (of 76), each with its ground.

- Rows 1–2 (COCO, PASCAL VOC) · release · `n_a` → `independent_test`: the printed `n_a` has no stated applicability rule in any published material, and

the audit's row facts yield a decidable mark — a manual gap, not an evidential conflict.

- Row 18 (CrowdHuman) · release · `test_gated` → `independent_test`, and protocol · `alias` → `test_gated`: the audit documents the same structure here as in row 17 (WIDER FACE), which the printed table itself marks `independent_test` / `test_gated`; under any fixed rule one of the two structurally identical rows must move. The release cell is decision-bearing; the protocol cell is not (both marks are non-independent).
- Row 9 (UAVDT) · yolo_dist · `unknown` → `no_test_key`: the audit's §1 verdict for the row is explicit ("F-B … empty `test:`"), so "the audit takes no position" — what `unknown` means — does not hold; the manual had to fix the precedence of a §1 verdict over a §4 generality caveat.
- Rows 11 (DIOR) and 16 (Mendeley face-mask) · reported · `unknown` → `alias`: the audit's §3b header and its 14/19 roll-up both enumerate these rows, so the audit does take a position; the manual had to fix the precedence of the enumerations over the rows' own caveats (missing primary counts; a naming ambiguity; a "treat as hypothesis" self-rating that the audit's count does not honor).
- Row 16 · yolo_dist · `alias` → `train_val_alias`: the verified generator writes `train == val` — the F-C shape, for which the mark vocabulary reserves `train_val_alias`; the printed `alias` follows the audit's own "F-A generated (`train == val`)" label, which conflates F-A with F-C. Both marks are non-independent, so no count moves.

The `n_a` rule the `release` subtotal rests on, and why that subtotal is not mechanically recomputable from the marking table. The printed `release` subtotal — **10/19 = 52.6 %**, with

unknown 0 — is computed under a rule the table itself does not state: **a row marked `n_a(A)` enters neither the non-independent numerator nor the undecidable column**. Rows 1–2 (COCO, PASCAL VOC) carry `n_a(A)` in the `release` column because that layer supplies **no independent test artifact at all** — `n_a` means *not applicable*, which is not the same thing as *undecidable* — so they stay in the 19-row denominator while counting nowhere else. Of the nineteen rows, **ten carry a non-independent mark, two carry `n_a`, and seven are independent**; read the other way, i.e. with `n_a` treated as non-

independent, the same column yields **12/19 = 63.2 %**. **The subtotal therefore cannot be recomputed from the marking table alone**: the step `12 - 2 = 10` *is* the rule, not a recomputation, and it is stated here so that a reader re-deriving the four subtotals can reproduce them. Under this one stated rule and no other, the four subtotals are mechanically recomputed (`work/recompute_split_units_20260916.py`); the `n_a` mark's lack of a stated applicability rule is also one of the eight cells the second coding moved, recorded above.

What this does to the printed counts. Two of the eight cells are count-bearing. Under the printed resolution of rows 11 and 16 the reported layer reads 12/19 = 63 % (12/15 = 80 % of decidable rows); under the second pass's reading of the audit's own enumeration, the same rules yield 14/19 = 74 % (14/17 = 82 %, two rows unknown). The printed coding is not modified here — the two readings bound the claim, and the qualitative conclusion (a majority of these benchmarks report a split that is not independent of checkpoint selection) is stable across the range. Where the reported-layer count is quoted, the strength of the claim should follow the bound rather than the point:

"12–14 of 19 benchmarks (63–74 %; 80–82 % of the decidable rows), where the range spans the printed coding's two undecidable rows and the second pass's reading of the audit's own enumeration; the printed coding is the conservative end."

What this does not settle. The marking is not double-rated: one human coder's marks remain the printed coding, and this pass is reliability evidence, not a replacement.

What the archive holds of the three-way carve, and what the two n = 11 cells pair

The carve manifest is archived; the other carve directories and 6 of the 7 corpora are not. The article's §10 defect 6 states the recomputability boundary, and this is the precise form of it. The archive holds the **manifest** of the SFCHD three-way carve: the dataset YAML (`sfchd20_3way.yaml`, naming `train/20_percent/images`, `carve_sfchd20/images` and `test/images`), the three filename lists (`train_20_percent_images.txt` **2413**, `carve_sfchd20_images.txt` **1000**, `test_images.txt`

6033), their **SHA-256 list** (`split_defs_20260916/SHA256SUMS.txt`) and the recomputed

zero pairwise intersections (`VERIFY_INTERSECTIONS.md`: all three intersections **0**, recomputed locally without a pod and without a GPU by `work/verify_split_defs_20260916.py`). The transfer set also holds the image and label files of that one carve directory (`rest/root__carve.tgz`, **1000** images with their labels). **What is not held**: any of the other carve directories, and 6 of the 7 corpora — so **retraining remains non-recomputable**, exactly as the article says. **What the manifest does buy** is the layer the protocol's premise rests on: the zero-intersection check by filename is now a local recomputation from archived lists rather than a pod-side report.

What the `n = 11` Tier-1 cells pair. Table 1's `smoke2sf` and `shwd2sf` rows pair **11** runs each, and the file that table cites lists the paired units explicitly (`verify_two_critiques_20260916.txt`): `[42, 43, 44, 45, 46, 47, 48, 49, 50, 51, unseeded]`, the same line for both cells. **The eleventh unit is therefore a run whose directory carries no seed suffix** — the archive records it as `unseeded` — while the two `n = 10` cells of the same file (`a2d15`) are fully seeded. This is stated rather than left implicit because "n = 11" can otherwise be read as eleven independent training seeds, which is not what the design varies: what varies across these replicates is the **training file order** (`--shuffle-seed`), and one of the eleven units records no seed suffix at all, so its data order is not identified by name. Runs without a seed suffix exist in both archives for these cells (the pod-side run lists carry entries such as `shwd2sf_base100` and `shwd2sf_lr005_100ep` beside the `_sNNn` ones), so this is a property of the run set, not of one machine's naming.

Method detail relocated from the article

> **Why these paragraphs are here.** The article is capped at 35 double-spaced pages with no margin left, and eight external reviews of the previous version said the same thing: the prose read as an *audit trail* rather than as an article, because round-by-round bookkeeping and repair history occupied the argument's place. These paragraphs are moved here **verbatim and unedited** — every value, every date, every file name and every disclosure sentence is unchanged. The article keeps a one-sentence bridge and a pointer at each former location, and keeps every disclosure that bears on a claim.

This paper and the budgeted fine-tuning study it re-measures — the companion paper — share **process protocols**: a multi-model review checklist and a local/cloud workflow discipline (scripted remote execution, all-or-nothing patching, audit anchors, a local artifact pointer behind every number). **Both are project-internal protocols, not publications, and not a methodological contribution of this paper.** This paper displays only their **outputs**: the ten self-reported defects in §10 and the artifact pointer behind every number. So does the **audit-anchor system** (a mechanically checked set of anchors that must all pass on every rebuild of this manuscript): shared tooling, no contribution claimed. What is asserted is only that the same protocols were used; **no originality is claimed** for them.

What is ours, and what is not. Of the statements that follow, the raw gap's definition [17], the impossibility result for endpoint-pair estimators [19] and the family-dependence procedure [20] are **machinery rather than results of ours**, and the identity is **adopted** from the selection-bias literature; **the corollary alone carries measurements of this paper**.

The same script on the **pod root** (347 runs, n = 10) gives a smoke2sf premium of **+0.786 / +0.390**; on the **local archive** (569 runs, n = 11) **+0.799 / +0.382** — agreeing in sign and magnitude, not value. **The main text uses the local version**; the pod version is recorded as a cross-check.

The refuted one compared the premium difference against a single-run component (0.515 pp) rather than the SE of an arm-mean difference; the other asked for the raw `M_val - M_test`, whose difficulty term cannot be separated from selection optimism.

The audit classified the FireSmoke mirror family row as "mixed" solely from a config with `train == val`, but **another line of the same document records that this config is based on SHWD**; all three FireSmoke/D-Fire copies have **three separated keys**, so D-Fire moves to the clean group.

A direct measurement of seed inflation — reproduced inside this setting: adding seeds lowers the mean by **9 %–29 %**, each figure taken **against that cell's own n = 6 mean** and not against the pooled headline effect, the **seed-budget discipline** measured rather than assumed (the setting in which [52] locates the first-order component, §8.4). **What these replicates are, and what the bound is a bound on.** The per-run randomness the ten runs install is a

deterministic permutation of the training file order (`--shuffle-seed`, the number in the run's own name), **not** the framework's `seed` argument: the Ultralytics data pipeline does not draw from `args.seed`, and on two of these corpora different `--seed` values produced **identical** epoch-by-epoch replays (**Supplementary S12**, section 9.3). "Seed inflation" here therefore means **data-order replication**: the 9 %–29 % it bounds is the **file-order** component of the reported mean, and is **not** an estimate of the variance between independent training seeds. That best-of-K selection inflates a reported score is **not new** — §8.6 positions the closest concurrent work, which measures the **dose–response** over 2,047 datasets — so **what is measured here is the magnitude under a protocol that selects cells for significance first and expands seeds afterwards**; design and the check against recomputed means: **Supplementary S12**, Table S18. **The title is deliberately the weaker, sample-scoped claim**: with three initializations the interval is wide, so the components are **consistent with, not established by**, these three initializations.

The three variance components, and the excluded one is the smaller in this sample: the single-run data-order component is **0.515 pp**, against initialization and seed components of the same order — the **same ranking** as the shuffle-order result of [52] and consistent with the run-to-run variance literature [23]; we therefore **reproduce the ordering inside this setting** rather than discover it. What was held fixed, and why this is a sample statement, is in **Supplementary S12** and Table S19.

Mixing two pipelines' `val`/`test` conventions would confound the split convention with the framework. The reasoning and the counterexample are in **Supplementary S12**. **Pairing is not neutrality, however**: the paired design used here covers the **budget and data** axes only, and **not** the optimiser's.

One run is recorded with two different endpoints in two archives — "final" (67.12) in one and "best" (70.06) in the other — i.e. **which checkpoint is recorded was never specified**; §10 discloses it and the two readings' consequences are worked out in **Supplementary S12** and Table S16. **Note (2026-10-04): the archived "best" row was computed with the run's `last` reading standing in for its `best` reading**; the row was corrected in `release_T2_saturated-control_summary.csv` and in the long table on **2026-10-02**, where the `test_best_epoch` entry goes from **−0.902 pp (p =**

0.2931) to **−1.4901 pp (p = 0.0141)**, i.e. **both endpoints now read ≈ −1.490 pp**, so the endpoint conflict does not change any verdict — see §S12 for the row-level detail:

Recomputation status. The readings were originally **not locally recomputable** (the archive lacks 6 of 7 corpora; **defect 6 in §10**); a GPU **2 × 2 evaluation** on **41** three-way runs of

YOLOv12n [54] × {`best`, `last`} × {`val`, `test`} makes the readings locally recomputable and

retraining is not. That batch reproduces the archived `best × test` values to within **0.0047 pp** (median **0.0021 pp**) — the positive control for this paper's endpoint rule (**Supplementary S4**, the verbatim dump that prints both values); the remaining record is in **Supplementary S12**.

Five limitations: ① the two cells escalated in §5.1 reach n = 10 with **local** runs and test-side evaluations, but the remaining 9 cells' runs are **not local**; ② `prem_test` uses only **best and last**, not a per-epoch curve; ③ `last.pt`'s val recomputation differs from the final `results.csv` row by ~0.02 pp; ④ this section does **not** touch §6's 19-benchmark census; ⑤ the aggregate is a **ratio of small means** — `prem_val` across the five arm-cells spans **+0.605 to +1.469 pp** (Table S20) — so a change of a few hundredths of a point in the numerator moves the rate by tens of points, and the aggregate summarizes a wide spread rather than a typical run's outcome.

The **41 original** runs behind §8.6 **retain intermediate checkpoints** (`save_period=5`: `epoch0,5,…,95`, **20 files**; **3,854** `epoch*.pt`), evaluated twice independently with exact agreement (**Table S21**; **§S12**, section 8.7) — a free reproducibility check of the endpoint rule.

Three limitations: ① only the two headline cells reach n = 10; `aitod20` has a single run (30 epochs, 6 checkpoints where the others contribute **20** — hence 806, not 820); ② **closed: every retained checkpoint is now evaluated**, and the *stability* reading is narrower than the sampled grid suggested; ③ κ is relative to each run's own final epoch and is **not** causal. The 806-row evaluation is archived as `xeval_perepoch_20260918/matrix_perepoch.csv` with a SHA-256 list.

- Four disclosed uncertainties are limits on what our numbers can be read as claiming, and we keep them visible rather than absorbing them into a

headline: the 19-row audit is **single-rater** (no second coder, so no inter-rater statistic exists and we do not claim one); the third-party log sample is a **convenience sample** found by repository search, so it corroborates that the component exists elsewhere and is not a rate; the GPU memory-pressure effect is **bounded, not estimated** (**≤0.05 pp** where same-config pairs can be compared — fresh replicates are bit-identical — while a same-config control that had no such event is itself not reproducible across dates (a 0.440 pp drift), so cross-date comparisons cannot test the bound; inside the zero-event-control spread elsewhere, §10 defect 10 and Table S41); and the §8.3 replicates vary the **training file order**, not independent training seeds.

None of the four has an interval that could be folded into §8.6's rate — the first two have no numeric bound at all without a second coder — so they are stated here as scope, not as error bars.

(The Chinese draft numbers these rows 1–3, 3, 4–7 with a duplicate "3"; renumbered here without changing content.) The four rule gaps it had to fix — `n_a`'s applicability, which unit absorbs ground-truth withholding, the two precedence questions (§1 verdict vs §4 caveat; §3 enumerations vs row-level caveats), and version anchoring for multi-version rows — are exactly the points a human second rater would have to rule on before any mark could be mechanical, and they are recorded as findings in the manual and the summary, not smoothed over.

Table S35 — the estimand and scope map

Table S35. *Added by this revision; not moved from the article, and therefore not listed in the §S8 index (which indexes tables moved from the article).* One row per quantity the article measures, with the assumption that makes it identifiable and the scope that bounds it. The last column collects the "we do not claim" statements that the article's prose carries at §5.1, §5.4, §8.6 and §12, so that each quantity's boundary is stated once, beside the quantity rather than scattered across sections.

Quantity	Estimator	What has to hold	What it identifies	What we do not claim
selection premium	`best − last` **within one run**, mAP50-95 pp, on the	that `best` is the argmax of the same curve the reported split	the checkpoint-selection term of run-to-run	not that the convention *understates* the effect; not a single

	monitored split; paired across arms	scores (the definitional part)	variation under the published convention	bias direction across corpora
Δgap	the paired difference of the premium between the two arms, per corpus	that the two arms see equal split difficulty — an assumption we state, not one this design demonstrates	a split-dependent **arm contrast**	not an isolated selection effect; the selection contribution proper is `Δprem_val − Δprem_test`
realization rate	`prem_test / prem_val`, run-weighted, over the five arm-cells at n = 10	that the readout split is disjoint from the selection set and untouched by any selection rule	the share of the `val`-selected premium that survives on a disjoint split, **for these five cells**	not a constant of the convention; not significant read at the level of the claim; the aggregate is a ratio of small means
κ (transfer slope)	through-origin slope of ΔT on ΔV over a run's epochs; run-clustered inference with runs as clusters	that "epoch" is the unit of the curve and that runs are the clusters	how much of a run's movement on `val` appears on `test`	κ is relative to each run's own final epoch and is **not** causal; κ is not constant across arms
peak-agreement predictor	whether the `val` and `test` peaks fall on the same 5-epoch checkpoint	the shared 5-epoch grid, and that agreement is an observable, not an intervention	a **partial** predictor of Δgap's sign, via a lower median Δgap when the peaks agree	not an explanation of the sign; the agreement group's mean is carried by one run; not refined beyond the 5-epoch grid

Table S36 — the analysis-status ledger

Table S36. *Added by this revision; not moved from the article, and therefore not listed in the §S8 index.* Every quantity and procedural step this paper reports,

classed as **pre-registered** (frozen before the runs existed), **pre-planned** (fixed in advance of the analysis but not registered) or
post-hoc (decided after the runs were seen). The class each item carries is the one the article's own prose asserts; the ledger collects them so that no post-hoc reading can be mistaken for a registered one. **Any p-value this paper quotes for a post-hoc row is uncorrected and descriptive**: the multiplicity corrections here are **three and not interchangeable**: a Benjamini–Hochberg step over the **declared, descriptive** 13-cell family (with **Holm** across the two targets of the escalation plan), and the **confirmatory** family, which is the **four escalated cells**; only the pre-planned rows belong to the first.

Item	Status	What fixes the class
the registered multi-target replication	**pre-registered**	its own frozen registration stamp, with zero runs existing at freeze time; reported by the companion
the four-budget epoch series' criteria C1–C3	**pre-registered**	the criteria are recorded as fixed before the runs
the n = 10 escalation criterion for the Tier-2 cells	**pre-planned**	the criterion was fixed in advance of the escalations
the endpoint convention `best − last`	**pre-planned**	frozen in writing before the re-analysis
the 13-cell family and its Benjamini–Hochberg correction	**pre-planned**	the family was declared before the correction was computed; its **thirteen members and their raw p-values are Table S8**, and the Benjamini–Hochberg step over that family is stated in this section rather than tabulated, and the two cells escalated to n = 10 after that scan are named in Table 2's note
the σ√(2 ln E) scaling	**post-hoc**	tested, then **withdrawn** — the form does not survive the archive
the complete-curve κ and its run-clustered interval	**post-hoc**	the sweep was completed after the five-epoch reading
the per-epoch Δgap sign reading	**post-hoc**	read off the completed curve, correcting a

		four-point sample
the second coding of the 19-row table	**post-hoc**	reliability evidence gathered after the printed coding; it does not replace it
the peak-agreement predictor	**post-hoc**	a reading of our own 40 runs
the mAP50 / mAP75 re-scoring batches	**post-hoc**	evaluation-side robustness checks run for this revision
plateau selection	**post-hoc**	offered as a candidate remedy, not as a validated rule

Detail relocated from the article's single-rater audit

Two clauses. Which reported number the COCO row judges, and this table's first-version correction (`yolo_dist` is **4/19**, not 2/19).

Algorithm self-check. On the **literal two-key subset** the table **reproduces the audit's 2/19**, so the difference comes from **the unit**, not a transcription error; the full `yolo_dist` column and the four subtotal equalities are in **Supplementary S1**.

One row belongs to a different unit. Mendeley face-mask's `train == val` comes from a **generator** (a VOC→YOLO script writing two keys from one `images_dir`), not a distribution package: not counted in the 2/19, described separately.

A number we deliberately do not use: the audit's "63–79 %" bound (15/19) has **no enumeration**.

"Same directory" has four levels: the levels, the rows each admits, and the resulting shift in the headline number are in **Supplementary S12** and Table S12.

The literal-search check, displaced from the article's section 2.1 (#31, 2026-09-28)

Non-collision, checked by literal search. Three exact queries return **0** hits — `"same directory" train val test yaml`, `configuration-level leakage "data.yaml"` and `"split aliasing" benchmark` — so the configuration-level phrasing is not in the indexed literature; the channel is **metadata-only**, which establishes that these strings are unused, **not that the idea is new**. The comparison that carries the claim is the **estimand** contrast with [10], which stays in the article: [10] reports a raw `val`-minus-`test` gap that mixes the two splits' difficulty, whereas this paper's paired difference removes

the difficulty term under the stated equal-difficulty assumption and still shows a residual shift.

Coding separation: structure versus selection (added 2026-10-01)

The five-value structural coding in §S1's table (`independent_test`, `n_a`, `test_gated`, `alias`, `no_test_key`, with `contradictory` / `unknown` as evidence states) records **what the release's configuration binds**, and nothing more. Two readings must not be folded into it, or a structural label gets read as a claim about *selection*:

*`test_gated` is not "the reported number was selected on `test`."** It says the official protocol does not let a holder evaluate an independent test locally (a licence or a server gate). Whether the paper reporting the number selected its checkpoint on that split is a separate question, recorded per row in the `reported` column and in the second-coding record.

*`no_test_key` / `absent` is not "the literature reported a `val`-selected endpoint."** It says the release ships no `test` key at all. Whether the cited paper's headline is a `val`-selected endpoint is again the `reported` column's question, and where the release gives no evidence that column reads `unknown` rather than being inferred from the absence.

Neither label decides `clean` or `alias` on its own: those two words appear only where the release's own artifact settles the binding, and every row's evidence class is printed beside it in §S1's table.

Known discrepancies inside the frozen §S4 (recorded, not edited; added 2026-10-01)

The frozen region (§S0–§S7, byte-frozen at 50,600 characters) cannot be edited without breaking the cross-draft contract, so two internal discrepancies found by the 2026-10-01 independent read are **recorded here** rather than repaired in place:

*the same cell is printed as -0.158 in its per-run table and -0.157** in its summary line;

*that summary reports +0.205 pp / 23.3 %, while its own five rows average 0.2034 pp, i.e. 23.1 %**.

Neither changes a printed headline number: the article's realization rates come from **Table S20** (the released 50-run aggregate), and the frozen dump's role is verbatim provenance for the two escalated cells. A later revision that unfreezes

§S0–§S7 should correct both lines at source; until then this note is the pointer a reader should follow.

Known discrepancy inside the frozen §S1 (recorded, not edited; added 2026-10-10)

The same byte-frozen region (§S0–§S7) contains a **heading/value mismatch** in §S1's embedded block: its sub-heading reads `## The three units (the main text must state which one it uses)` while the table beneath it lists **four** — `release`, `protocol`, `yolo_dist`, `reported`. The word is a leftover from an earlier three-unit framing; the table itself, the recomputation beneath it and the article's §3 definition are all four-unit and mutually consistent, so **no count moves**. It is **recorded here rather than repaired in place**, for the same reason as the §S4 note above: editing the frozen region breaks the cross-draft contract. A later revision that unfreezes §S0–§S7 should change `three` to `four` at source.

Author-side names that are not in the released package (added 2026-10-01)

Several names appear in this supplement and are **not** shipped, because they are working-tree artefacts rather than released ones; each is marked at its point of use where that point lies outside the frozen section, and listed here once for the two that lie inside it:

*`PROVENANCE.md` (the log-mirror directory's own provenance record) -- author-side; the released copy of that directory carries `published_logs/SHA256SUMS.txt` instead — which is also author-side** (that directory is not redistributed; only its hashes and source pointers are).

*`work/build_provenance.py` -- author-side** (the script that wrote the record above).

*`work/r102_cluster_20260927.py` -- author-side** (a working-tree analysis driver whose outputs are shipped under `02_release_data/`).

*`published_logs/SHA256SUMS.txt`, `nonyolo_settle_20260925/`, `settle_all.json` and `nonyolo_n6_aggregate.json` -- author-side** (nothing under these names is in the released package; their points of use in this supplement say so).

*`A_results_20260916_t1d/` (the `T1-d` batch's archive, 32 files) -- author-side: it is that batch's register and test-side ledger. Its `val` side is no longer a non-recomputable exception** (2026-10-02): the three seeds the archive lacked were

still in the training machine's run directories, and their curves now ship under `02_release_data/t1d_valext_20261002/`, taking the registered pair's `val`-side reading to `n = 10` (note under Tables S24-S27).

*`x1_tier2_20260918/runs/` (the forty per-run records behind Table S29, 15 MB) -- author-side; its file list ships** (`02_release_data/x1_tier2_file_list.txt`) together with the eighty test evaluations it is read against (`teval.csv`, shipped under `02_release_data/x1_tier2_20260918/`), so Table S29's two sources are one shipped file plus one declared directory.

The rule this follows is the package's own: a name is either shipped or labelled author-side, never left looking shipped when it is not.

Table S42 — the prefix-n view of the two clean-protocol cells (n = 3…10; paired *t* and exact sign-flip permutation; both endpoints)

Table S42. *7.2* — requested in review: the two registered cells at every prefix n, under **both endpoint conventions**. Source: `release_selection-transfer_perseed.csv` (**40 runs** = 2 cells × 2 arms × 10 seeds, seeds 42–51). The permutation p is **exact** (all 2^n sign flips enumerated, no sampling), so it has a **floor of $2/2^n$** — at n = 3 that is **0.25**, which is why the permutation column, not the *t* column, is the binding constraint for the n = 3 cells.

> **Sign convention — this column is not Table 2's Δgap.** It is the **within-seed arm contrast**

> `Δgap(seed) = prem(strategy) − prem(baseline)` (definition from the generating script,

> `04_verification/x5_prefix_n_table_20261004.py`), i.e. the **difference in selection premium between the two arms**,

> not the **split-level** difference that Table 2 reports as Δgap. The two need not share a sign, and here they do not:

> the `oracle` row at n = 10 is **−0.7025 pp** for `shwd2sf` while Table 2's split-level Δgap is **+0.779** — the two are

> related through the selection term, `0.779 − 1.481 = −0.702`. Read this column as "how the premium differs between

> the arms", never as Table 2's Δgap.

shwd2sf · endpoint = `oracle`

n	seeds	paired Δgap	SD	paired *t*	*t* p	exact sign-flip p

		contrast (pp)				
---	---	---:	---:	---:	---:	---:
3	42..44	-0.7146	0.3364	-3.6796	0.06657	**0.25**
4	42..45	-0.6632	0.2932	-4.5231	0.02021	**0.125**
5	42..46	-0.6517	0.2553	-5.7086	0.004656	**0.0625**
6	42..47	-0.6158	0.2446	-6.1673	0.001632	**0.03125**
7	42..48	-0.6556	0.2468	-7.0270	0.0004147	**0.01562**
8	42..49	-0.7197	0.2917	-6.9783	0.0002157	**0.007812**
9	42..50	-0.7456	0.2837	-7.8845	4.848e-05	**0.003906**
10	42..51	-0.7025	0.3001	-7.4030	4.09e-05	**0.001953**

shwd2sf · endpoint = `selected`

n	seeds	paired Δgap contrast (pp)	SD	paired *t*	*t* p	exact sign-flip p
---	---	---:	---:	---:	---:	---:
3	42..44	-0.3933	0.4456	-1.5287	0.2659	**0.25**
4	42..45	-0.4000	0.3641	-2.1973	0.1155	**0.125**
5	42..46	-0.5194	0.4132	-2.8109	0.04827	**0.0625**
6	42..47	-0.5410	0.3733	-3.5494	0.0164	**0.03125**
7	42..48	-0.6185	0.3978	-4.1136	0.006261	**0.01562**
8	42..49	-0.7589	0.5416	-3.9635	0.005437	**0.007812**
9	42..50	-0.7796	0.5104	-4.5825	0.001796	**0.003906**
10	42..51	-0.6993	0.5441	-4.0645	0.002823	**0.005859**

smoke2sf · endpoint = `oracle`

n	seeds	paired Δgap contrast (pp)	SD	paired *t*	*t* p	exact sign-flip p
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3	42..44	-0.7938	0.1080	-12.7284	0.006116	**0.25**
4	42..45	-0.8020	0.0897	-17.8789	0.0003816	**0.125**
5	42..46	-0.7702	0.1054	-16.3356	8.219e-05	**0.0625**
6	42..47	-0.8390	0.1933	-10.6340	0.0001272	**0.03125**
7	42..48	-0.8622	0.1867	-12.2149	1.832e-05	**0.01562**
8	42..49	-0.8187	0.2122	-10.9147	1.198e-05	**0.007812**
9	42..50	-0.7417	0.3046	-7.3035	8.36e-05	**0.003906**
10	42..51	-0.6934	0.3252	-6.7419	8.442e-05	**0.001953**

smoke2sf · endpoint = `selected`

n	seeds	paired Δgap contrast (pp)	SD	paired *t*	*t* p	exact sign-flip p

| --- | --- | ---: | ---: | ---: | ---: | ---: |
3	42..44	-0.5631	1.3339	-0.7311	0.5408	**0.5**
4	42..45	-0.5253	1.0917	-0.9623	0.4069	**0.375**
5	42..46	-0.7462	1.0668	-1.5641	0.1928	**0.1875**
6	42..47	-0.9124	1.0374	-2.1544	0.08377	**0.09375**
7	42..48	-0.9372	0.9493	-2.6122	0.04	**0.04688**
8	42..49	-1.0691	0.9547	-3.1672	0.01577	**0.02344**
9	42..50	-0.9684	0.9427	-3.0818	0.01508	**0.01953**
10	42..51	-0.8716	0.9401	-2.9318	0.01671	**0.01953**

Table S43 — the transfer slope κ: per-run distribution and the free-intercept control

Table S43. *8.7* — requested in review (a free-intercept control was asked for explicitly; a per-run distribution was asked for as "the cheapest statistical gain"). Definition follows §8.7: `ΔV(e) = V(e) − V(final)`, `ΔT(e) = T(e) − T(final)` **within a run**, over `val` per epoch (`xeval_20260916/runs/*_results.csv`) intersected with `test` per epoch (`xeval_perepoch_20260918/matrix_perepoch.csv`).

Reproduction of the printed estimate. This independent recomputation resolves **765 (run, epoch) points over 41 runs** — the same count the printed estimate states — and gives **κ = 0.710** and **R^2 = 0.928** (R^2 taken relative to zero, the convention of the printed fit), against the printed **κ = 0.708, R^2 = 0.937**. The point count agrees **exactly**; the slope agrees to **0.003** and R^2 to **0.009**, i.e. the printed estimate is reproducible to that tolerance, and the remaining gap is in the fit details rather than the data.

Free-intercept control (this is the new information): fitting `ΔT = a + κ_free·ΔV` over the same points gives **κ_free = 0.701** with **intercept = -0.121 pp**. **The intercept's interval** (added 2026-10-10, run-clustered): sandwich SE **0.0643**, 95 % CI **[−0.2509, +0.0089]**; run-level bootstrap over the 41 runs (2,000 draws) **[−0.2403, +0.0024]**. Both contain zero, so the through-origin constraint is **supported rather than assumed** — the point estimate is reproduced by `work/x8_intercept_clustered_ci_20261010.py`, which re-uses the same pairing as `work/x6_arm_paired_kappa_20261004.py` and asserts `−0.1210` before reporting.

Per-run κ (n = 41 runs): mean **0.737**, SD **0.152** — spread across runs is genuine, and the printed run-clustered CI **[0.671, 0.745]** lies inside it.